

Experiment – 2 Analysing Android App Permissions and Mobile Traffic

1. Aim

To analyse Android application network traffic and permissions by intercepting HTTP and HTTPS requests using an Android Emulator and Burp Suite.

2. Objective

- To create and run an Android Virtual Device (AVD).
- To configure a proxy between the Android Emulator and Burp Suite.
- To capture HTTP traffic from the Android device.
- To intercept and analyse HTTPS traffic by installing the Burp Suite CA certificate.

3. Required Tools / Software

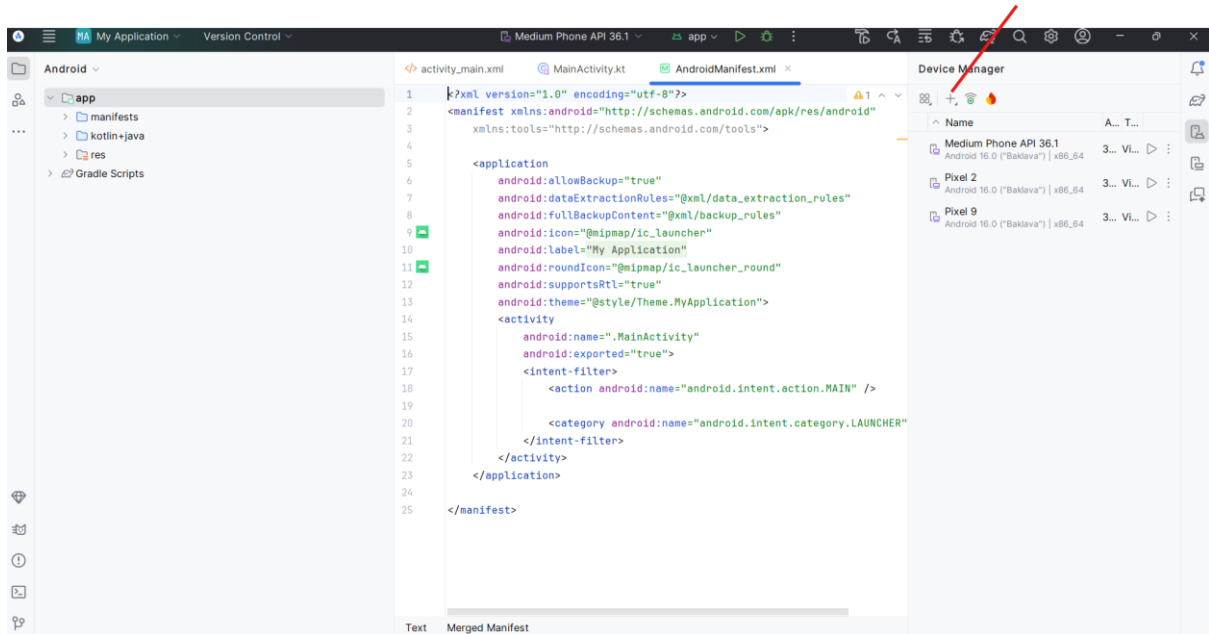
- Android Studio
- Android Emulator (AVD Manager)
- Burp Suite
- ADB (Android Debug Bridge)
- Google Chrome (inside emulator)
- Internet connection

4. Procedure

Step 1: Create Android Virtual Device

1. Install **Android Studio**.
2. Open **AVD Manager**.
3. Create a new Virtual Device.

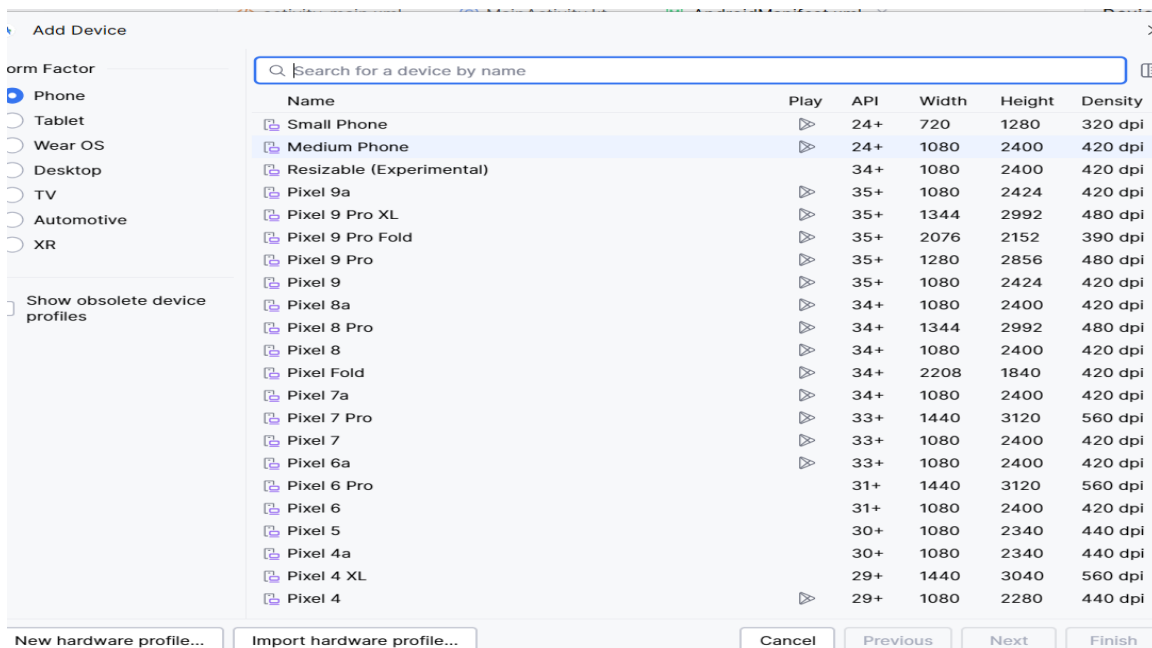
Step-1: Download Android Studio->Create Virtual Device



Step 2: Select Device

Choose any device such as:

- Pixel 2
- Pixel 9



Step-3: Start AVD from Command Prompt1

```
C:\Users\Admin>cd C:\Users\Admin\AppData\Local\Android\Sdk\emulator
```

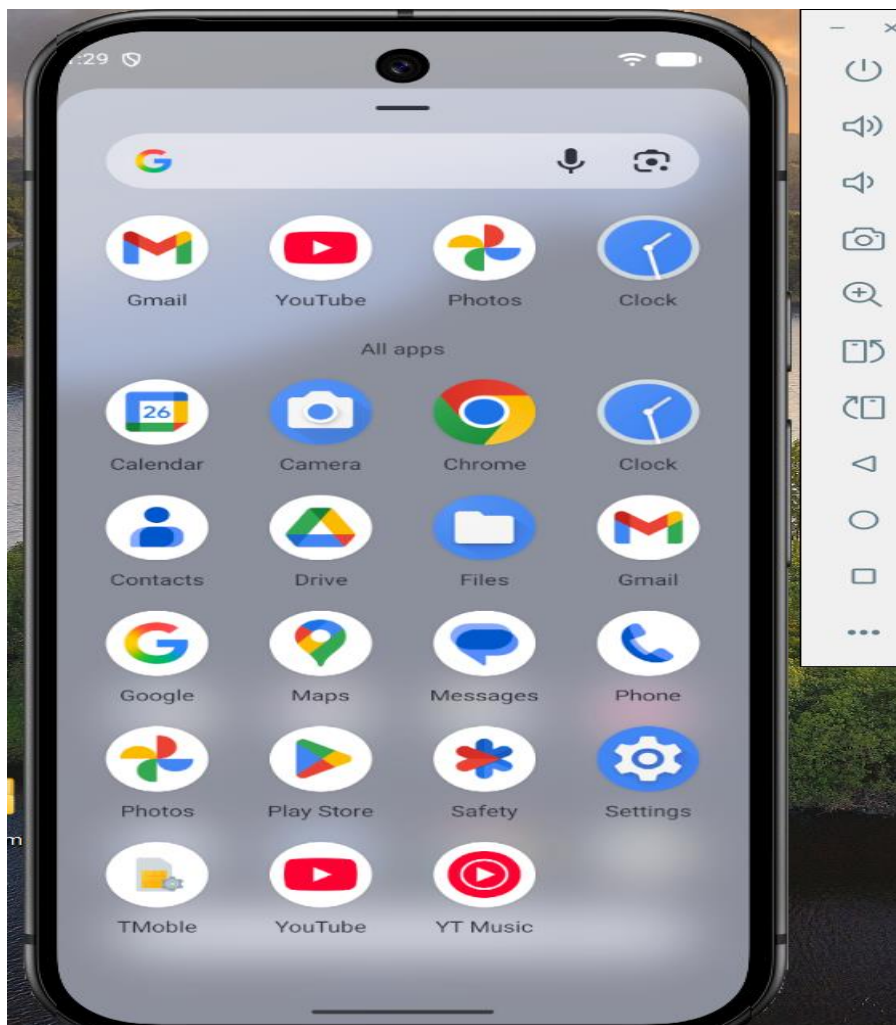
```
C:\Users\Admin\AppData\Local\Android\Sdk\emulator>emulator -list-avds
```

Medium_Phone_API_36.1

Pixel_2

Pixel_9

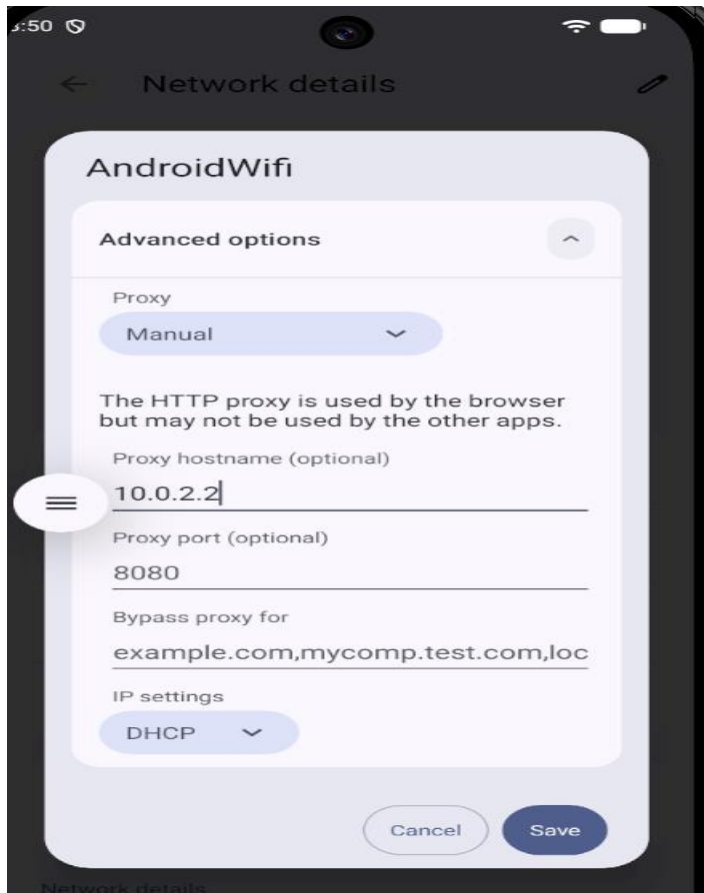
```
C:\Users\Admin\AppData\Local\Android\Sdk\emulator>emulator -avd Pixel_9 -http-proxy  
http://10.0.2.2:8080
```



Step-4: Setup Proxy:

In AVD

Settings-> Network and Internet-> internet->select icon  -> top right corner there is edit->advanced options -> proxy->Change proxy None->Manual->Hostname as 10.0.2.2 ->save



Note:if save option is not working then follow the steps in command prompt

Command Prompt2:

```
C:\Users\Admin>cd C:\Users\Admin\AppData\Local\Android\Sdk\platform-tools
```

```
C:\Users\Admin\AppData\Local\Android\Sdk\platform-tools>adb devices
```

List of devices attached

```
emulator-5554 device
```

```
C:\Users\Admin\AppData\Local\Android\Sdk\platform-tools>adb shell settings put global http_proxy 10.0.2.2:8080
```

```
C:\Users\Admin\AppData\Local\Android\Sdk\platform-tools>adb shell settings get global http_proxy
```

```
10.0.2.2:8080
```



```

C:\Users\Admin>cd C:\Users\Admin\AppData\Local\Android\Sdk\platform-tools
C:\Users\Admin\AppData\Local\Android\Sdk\platform-tools>adb devices
List of devices attached
emulator-5554    device

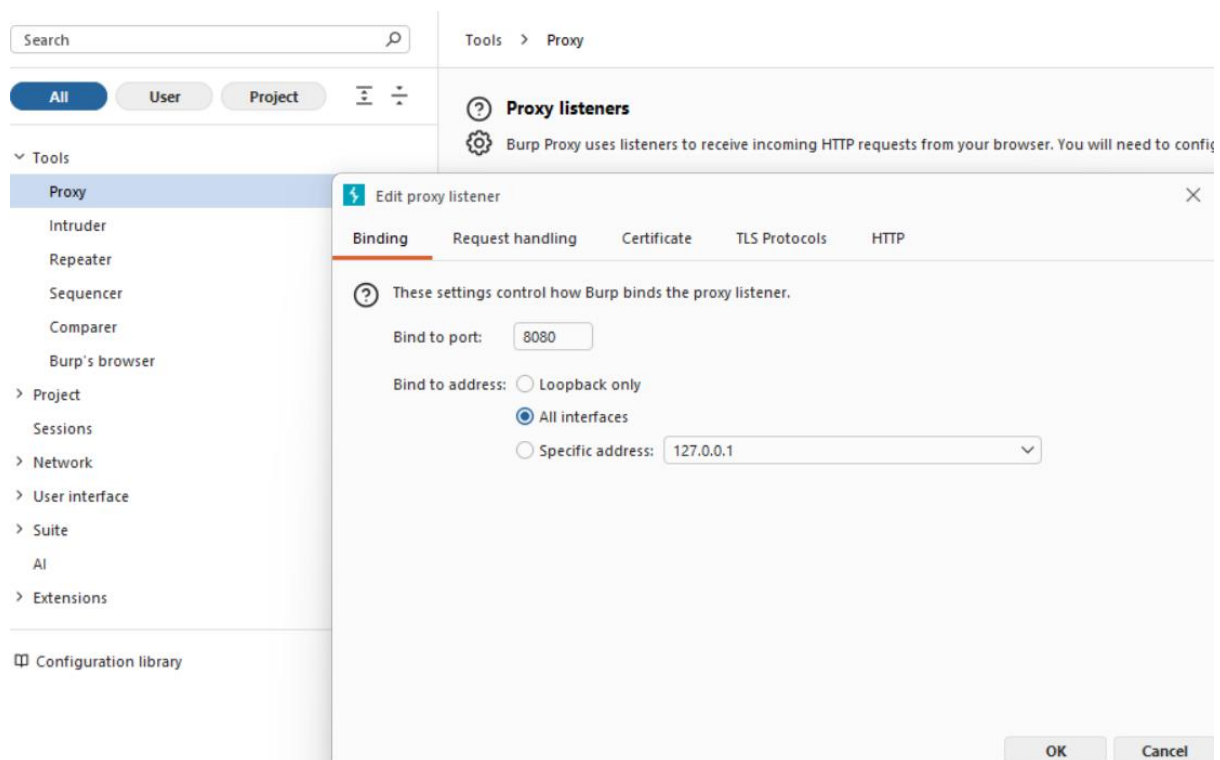
C:\Users\Admin\AppData\Local\Android\Sdk\platform-tools>adb shell settings put global http_proxy 10.0.2.2:8080
C:\Users\Admin\AppData\Local\Android\Sdk\platform-tools>adb shell settings get global http_proxy
10.0.2.2:8080
C:\Users\Admin\AppData\Local\Android\Sdk\platform-tools>|

```

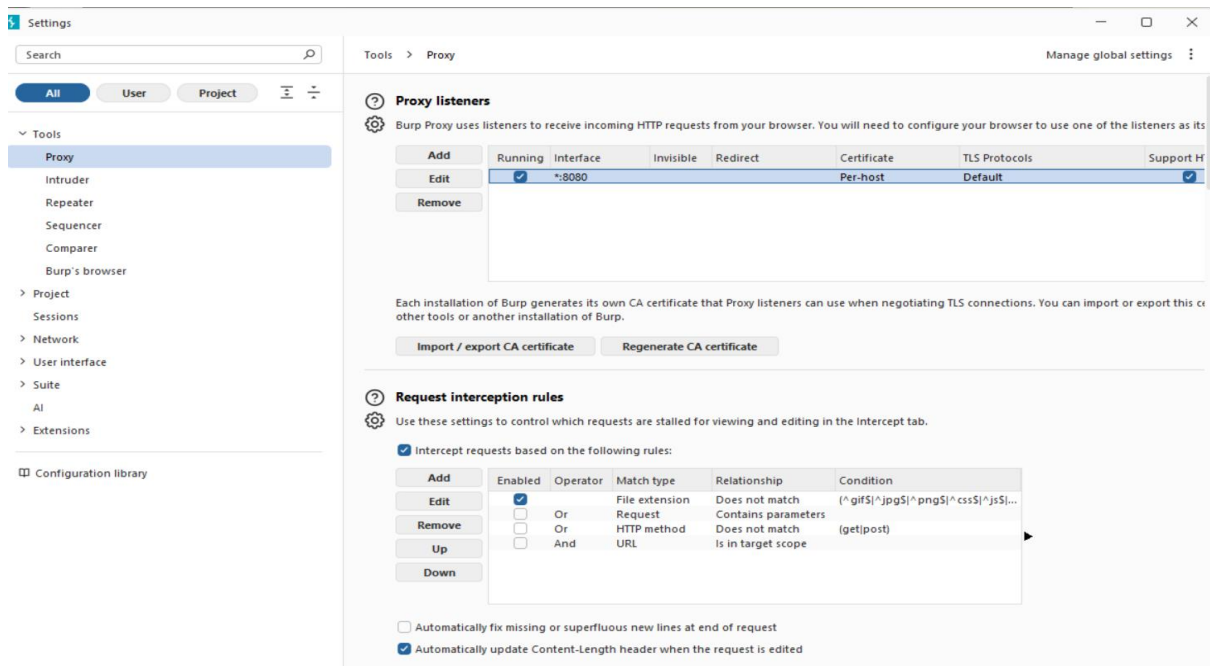
Step 5: In BurpSuit

Go proxy->options/proxy settings->under Proxy Listeners->Edit->select All interfaces->ok

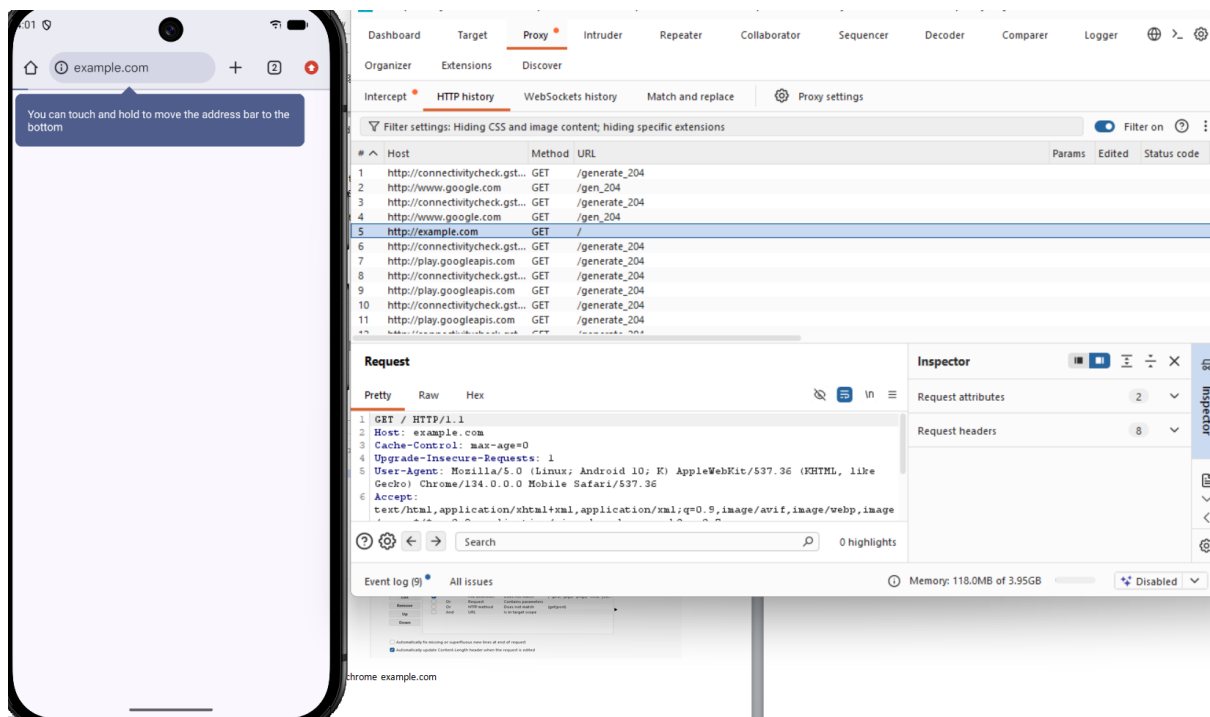
Go proxy->intercept on



You can see BurpSuit is accepting all interfaces traffic

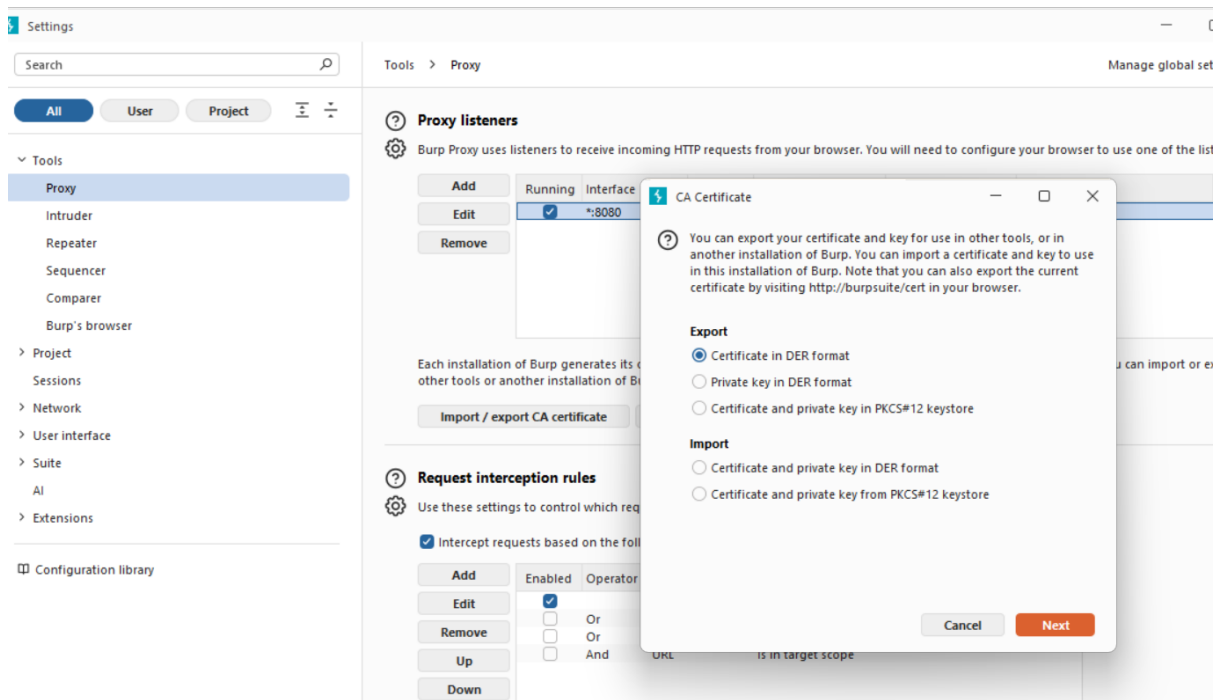


Step 6: in AVD type in chrome example.com and you can see burpsuit HTTP history



Step 7: in Step 5 we only able to http request and response only but if you want to analyse https request and response follow the Procedure to download CA certificate from Bunrpsuit.

Go proxy->options/proxy settings->import/export CA certificate->Certificate in DER format->select file-> Downloads->give name as burpcer.der



Step 8: Sending CA certificate to AVD

Go to command Prompt2

```
C:\Users\Admin\AppData\Local\Android\Sdk\platform-tools>adb push
C:\Users\Admin\Downloads\burpcer.cer /sdcard/Download/
```

```
C:\Users\Admin\AppData\Local\Android\Sdk\platform-tools>adb push C:\Users\Admin\Downloads\burpcer.cer /sdcard/Download/
C:\Users\Admin\Downloads\burpcer.cer: 1 file pushed, 0 skipped. 0.0 MB/s (939 bytes in 0.022s)
C:\Users\Admin\AppData\Local\Android\Sdk\platform-tools>
```

To install Certificate in AVD

Settings

→ Security & privacy

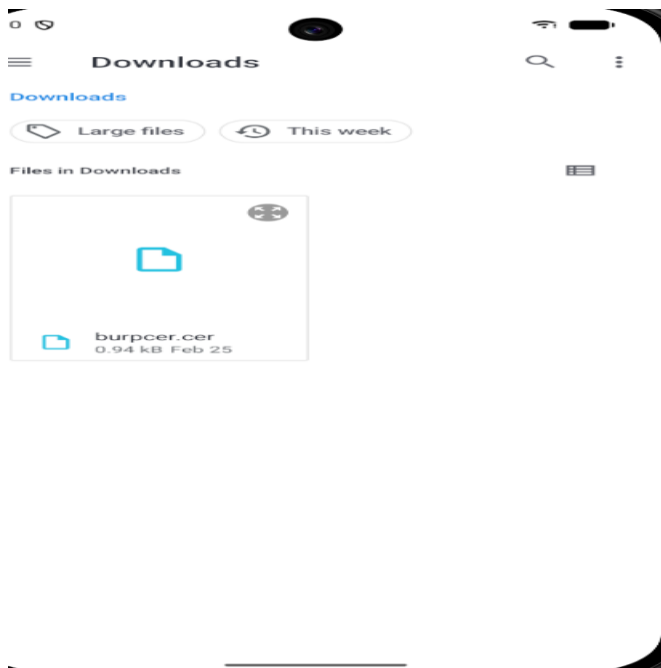
→ More security & privacy

→ Encryption & credentials

→ Install a certificate

→ CA certificate

In AVD we can see burpcer.der certificate



Step 9:verify HTTPS Interception

1. Open browser in emulator.
2. Visit any HTTPS website.

Example:

`https://example.com`

3. In Burp Suite, check **Proxy** → **HTTP History**.

Now HTTPS requests and responses will be visible.

5. Result

Android application network traffic was successfully intercepted and analysed using Burp Suite by configuring proxy settings in the Android emulator and installing the Burp Suite CA certificate.

6. Precautions

- Ensure Burp Suite proxy port (8080) is correctly configured.
- Verify the emulator is running before executing ADB commands.
- Install the CA certificate properly to capture HTTPS traffic.

- Disable intercept when not required to allow traffic to pass normally.

Conducting a Security Audit and Risk Assessment

1. Aim

To perform a security audit on a Windows system and identify potential vulnerabilities by conducting a risk assessment and suggesting mitigation strategies.

2. Tools Required

- Windows Operating System (Windows 10/11)
- Windows Security (Defender / Antivirus)
- Command Prompt
- Task Manager
- Web Browser (Google Chrome / Microsoft Edge)

3. Introduction

A security audit is the process of evaluating a system's security measures to identify vulnerabilities. Risk assessment involves analyzing potential threats, vulnerabilities, and their impact on the system. It helps in implementing appropriate security controls to protect data and resources from cyber threats.

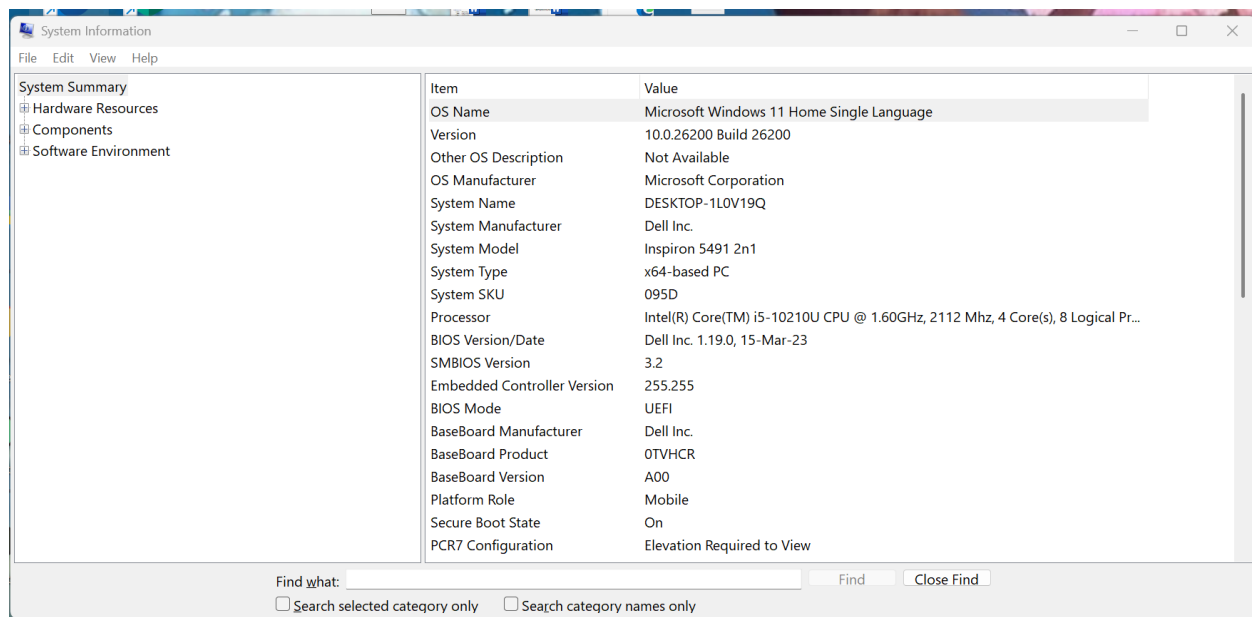
4. Procedure

STEP 1: System Information

- Press **Windows + R**
- Type: msinfo32

Note:

- OS Version
- System Type



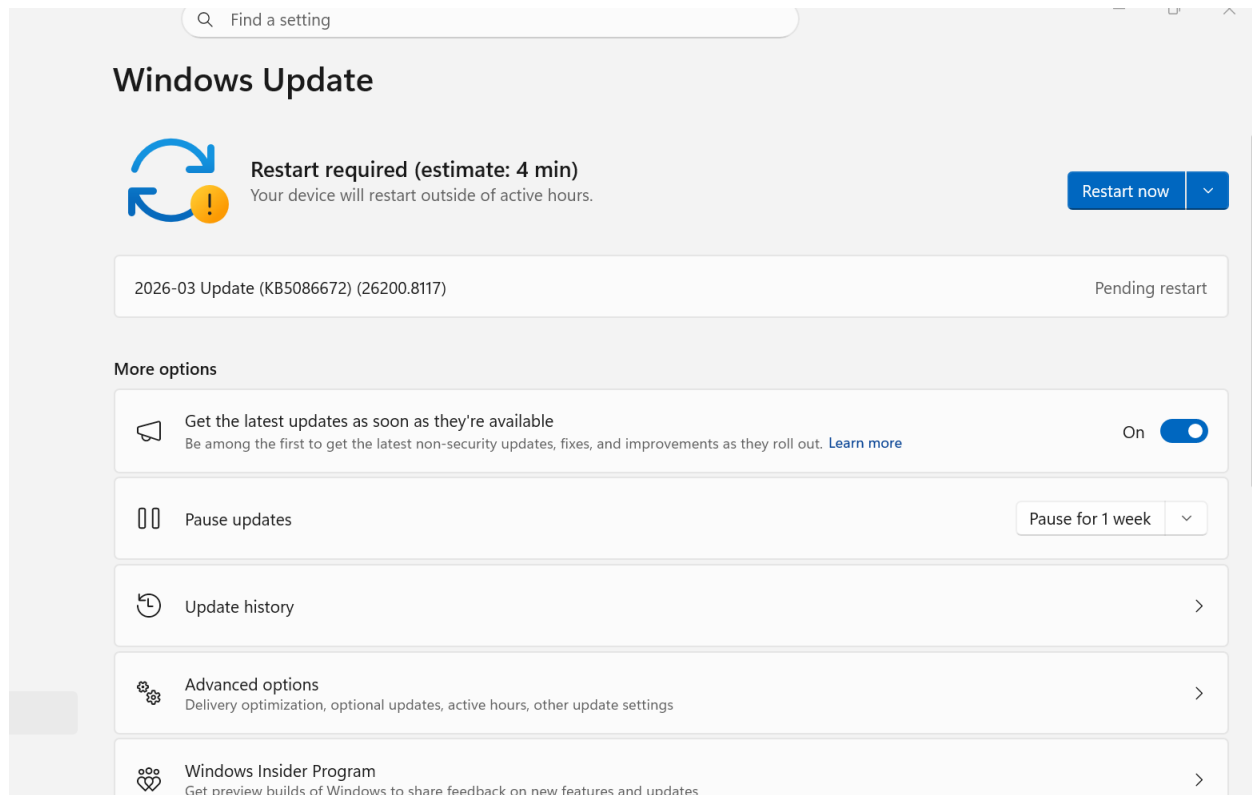
Observation:

The system information was obtained using the *msinfo32* tool. The system is running on Microsoft Windows 11 Home Single Language, 64-bit operating system, with Intel i5 processor and UEFI-based architecture. Secure Boot is enabled, indicating a secure boot configuration. Secure Boot being enabled enhances system security by preventing unauthorized boot loaders.

STEP 2: Check Windows Updates

Steps:

- Open **Settings**
- Go to **Windows Update**
- Click **Check for updates**



Observation:

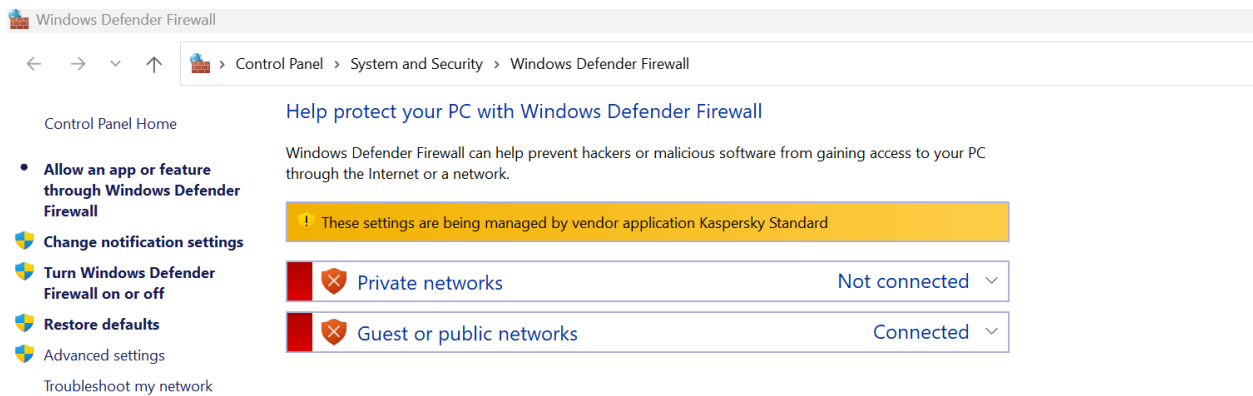
The Windows Update section was checked, and it was found that updates are available and pending installation. The system requires a restart to complete the update process.

Pending updates may expose the system to known vulnerabilities. Regular updates are essential to maintain system security and protect against threats.

STEP 3: Check Firewall Status

Steps:

- Search: **Windows Defender Firewall**
- Click **Turn Windows Defender Firewall on or off**



Observation:

The firewall settings were checked using Windows Defender Firewall. It was observed that firewall settings are managed by Kaspersky antivirus, indicating that a third-party firewall is active on the system.

The presence of a third-party firewall ensures network protection; however, misconfiguration may still expose the system to risks.

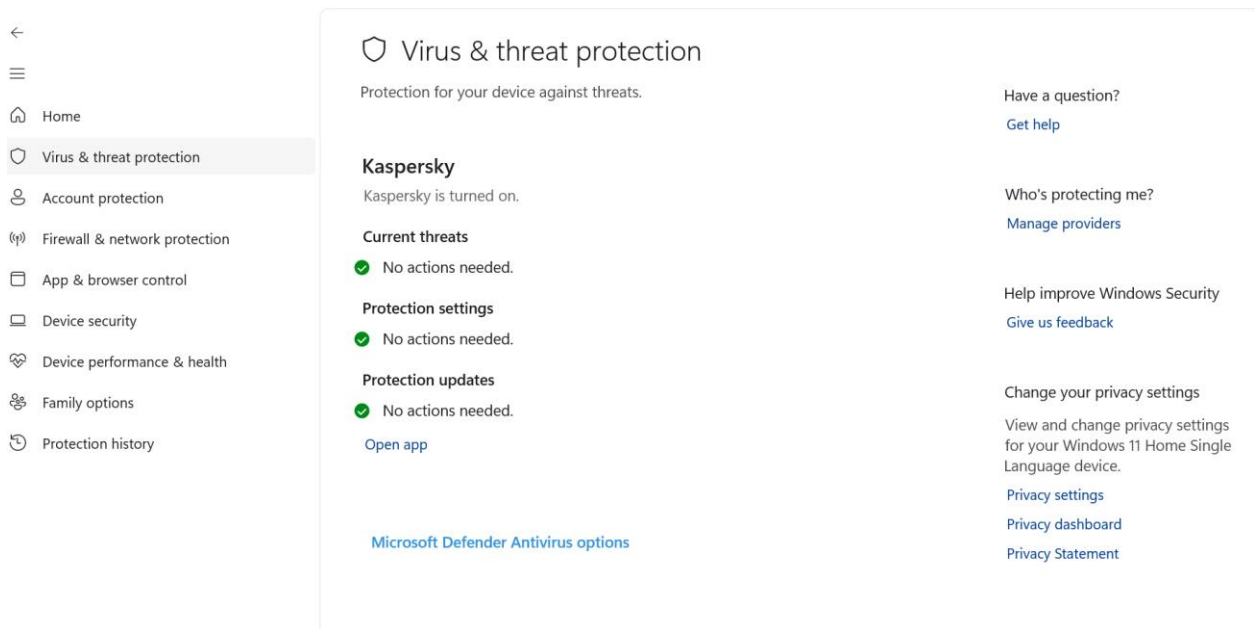
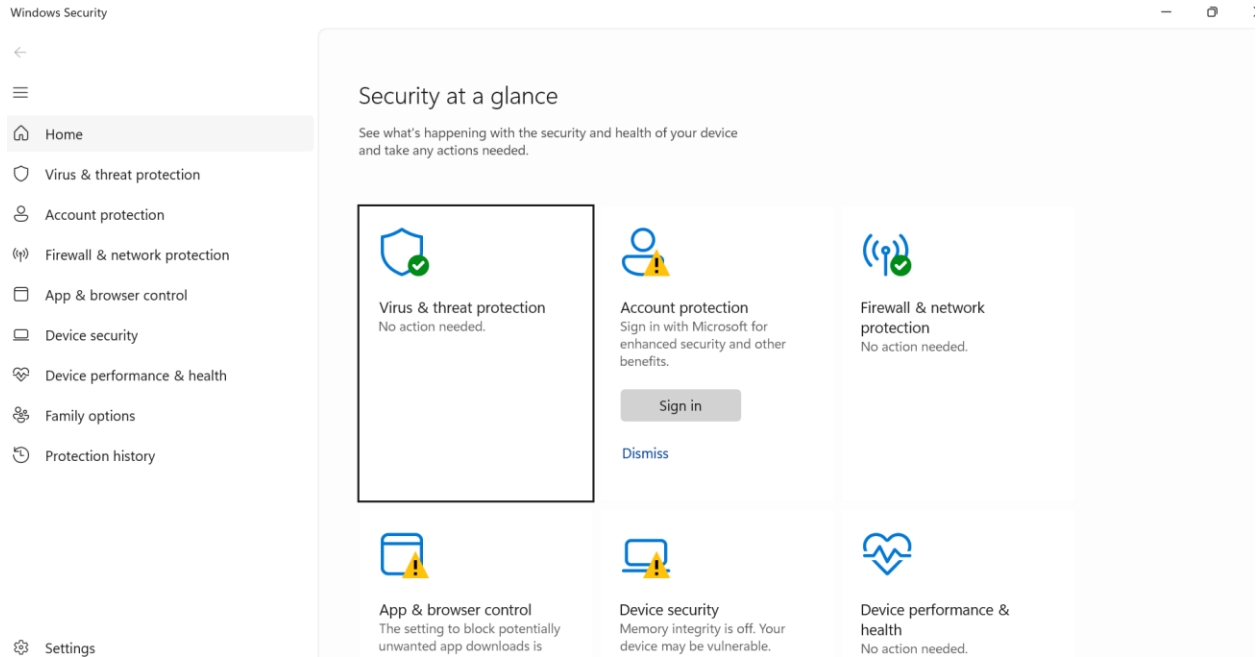
STEP 4: Check Antivirus (Windows Security)

Steps:

- Open **Windows Security**
- Go to **Virus & Threat Protection**

Check:

- Real-time protection → ON
- Last scan → Recent



Observation:

The antivirus status was verified under Virus and Threat Protection. It was observed that Kaspersky antivirus is active, and no current threats were detected. Protection settings and updates are functioning properly. A third-party antivirus like Kaspersky provides advanced protection against malware, phishing, and other cyber threats.

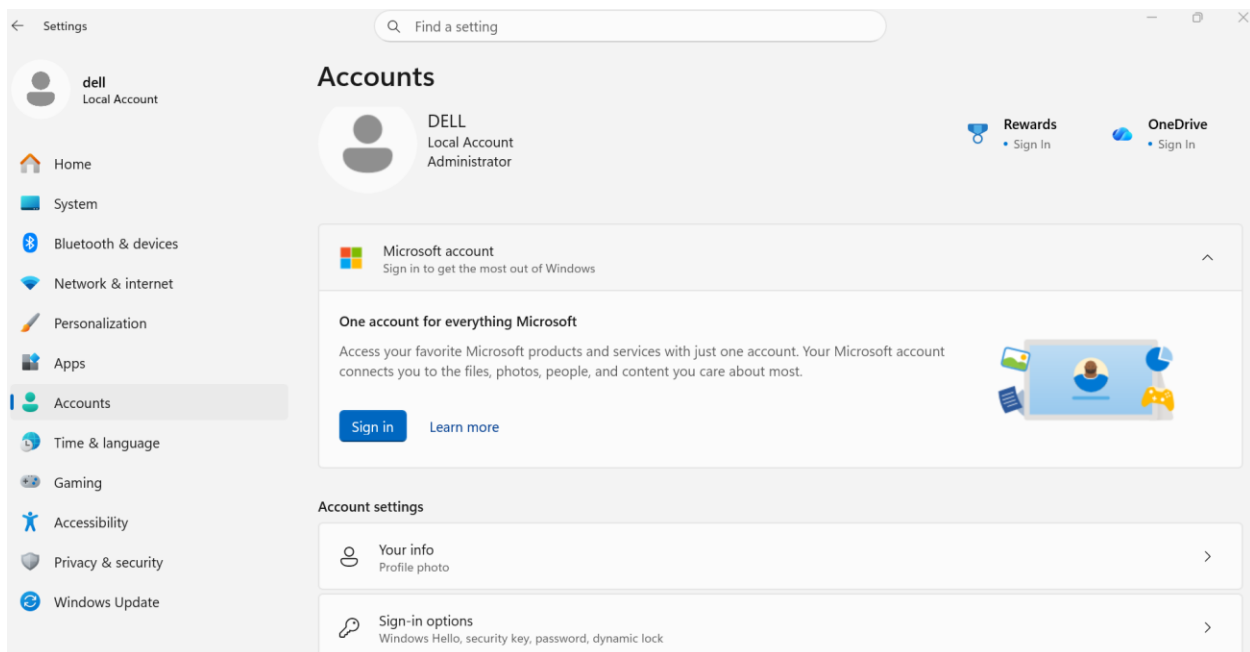
STEP 5: Check Password & Account Security

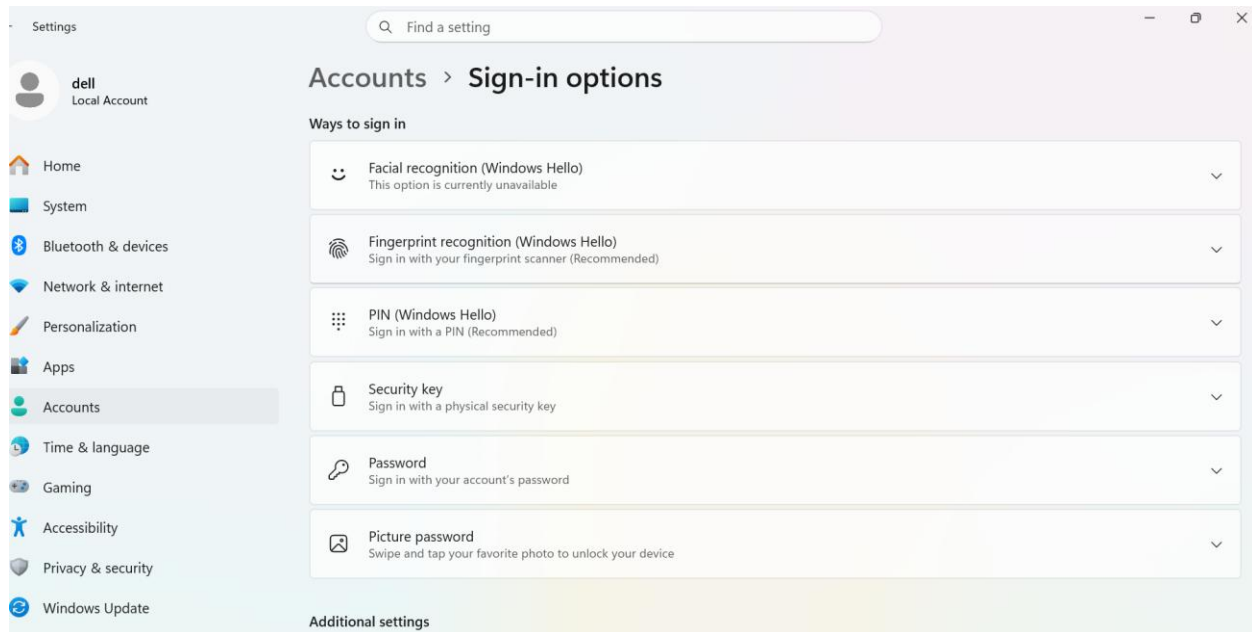
Steps:

- Settings → **Accounts** → **Sign-in options**

Check:

- Password / PIN enabled
- Windows Hello (optional)





Observation:

The account settings and sign-in options were checked. The system uses a local administrator account with multiple authentication methods such as password and PIN (Windows Hello), ensuring secure access.

Strong authentication mechanisms like PIN and biometric options enhance system security and prevent unauthorized access.

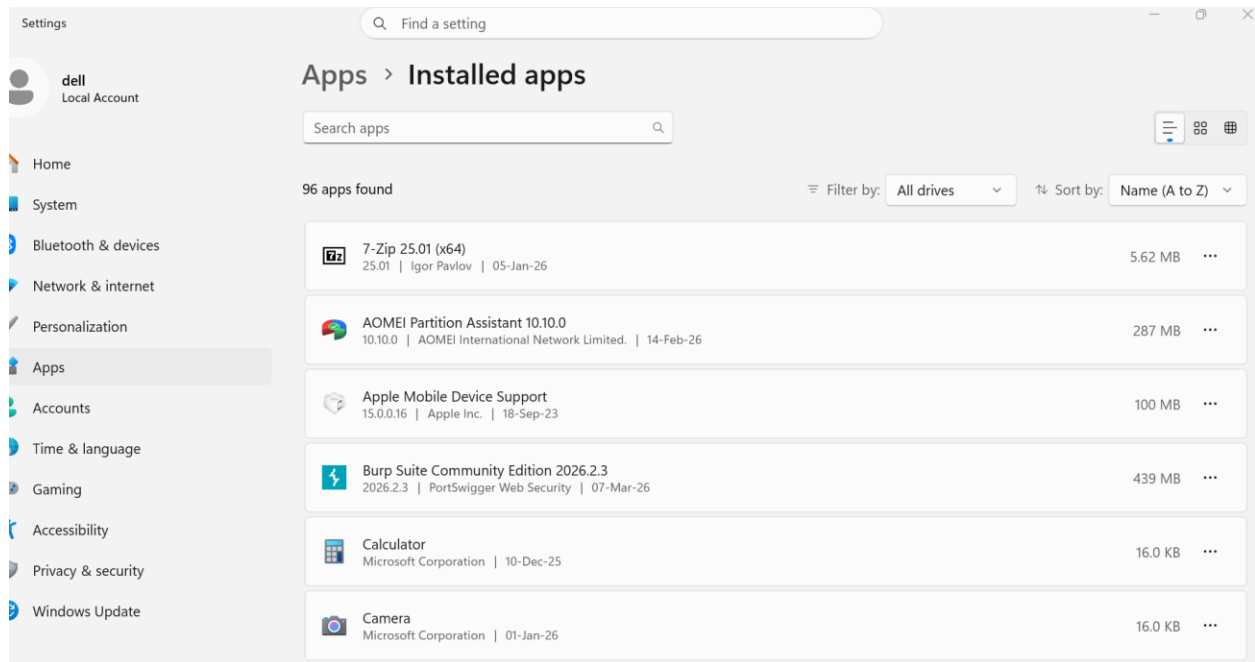
STEP 6: Check Installed Applications

Steps:

- Settings → **Apps** → **Installed apps**

Look for:

- Unknown software
- Cracked tools



Observation:

The list of installed applications was reviewed. No unauthorized or suspicious software was identified in the system.

Unverified or cracked software may introduce malware and compromise system security.

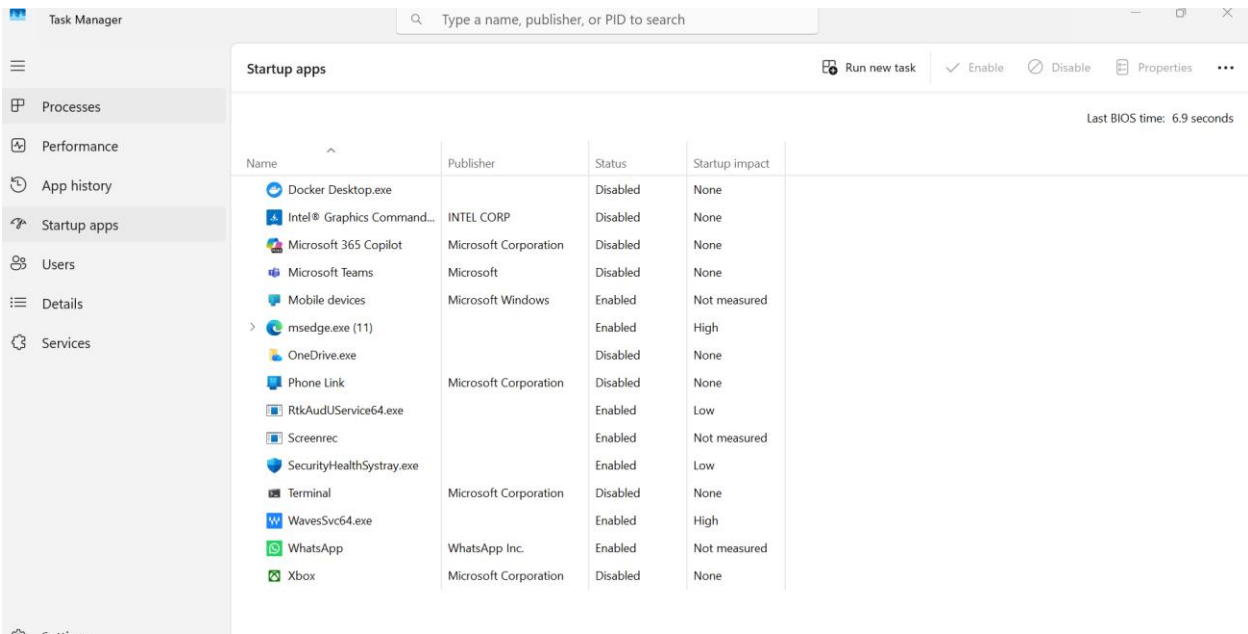
STEP 7: Check Startup Programs

Steps:

- Press **Ctrl + Shift + Esc**
- Go to **Startup tab**

Disable:

- Unknown apps



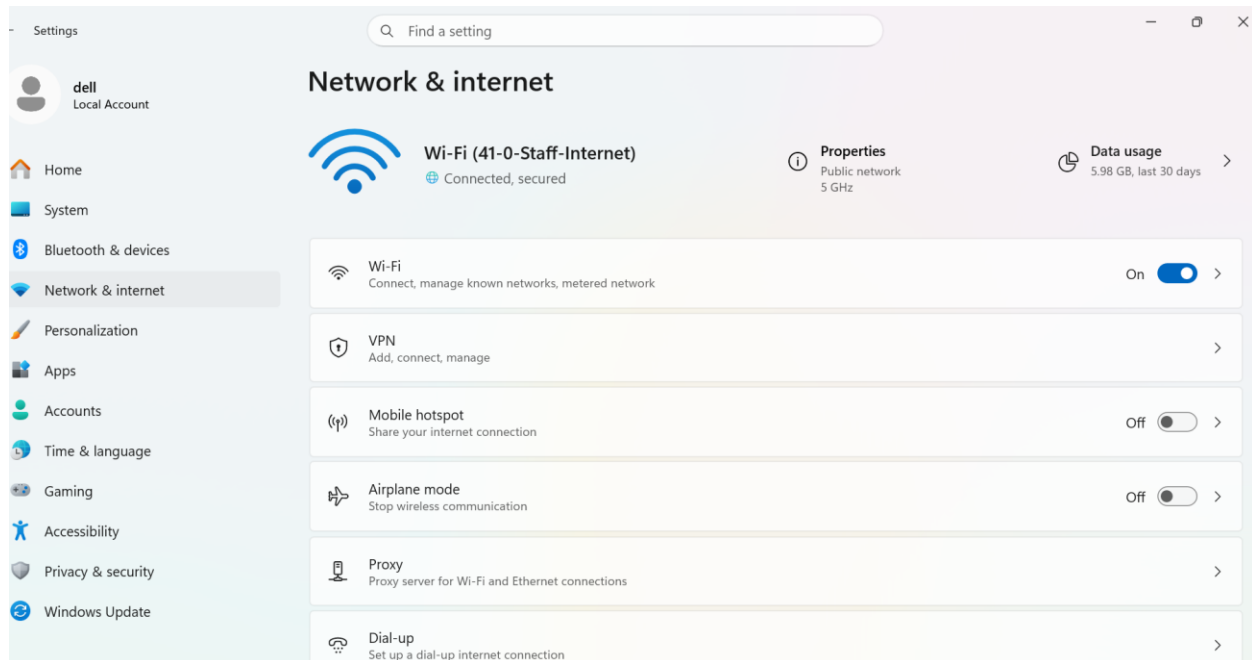
Observation:

Startup applications were analyzed using Task Manager. Some applications were enabled at startup, which may impact system performance and security. Disabling unnecessary startup applications reduces attack surface and improves system performance.

STEP 8: Check Network Security

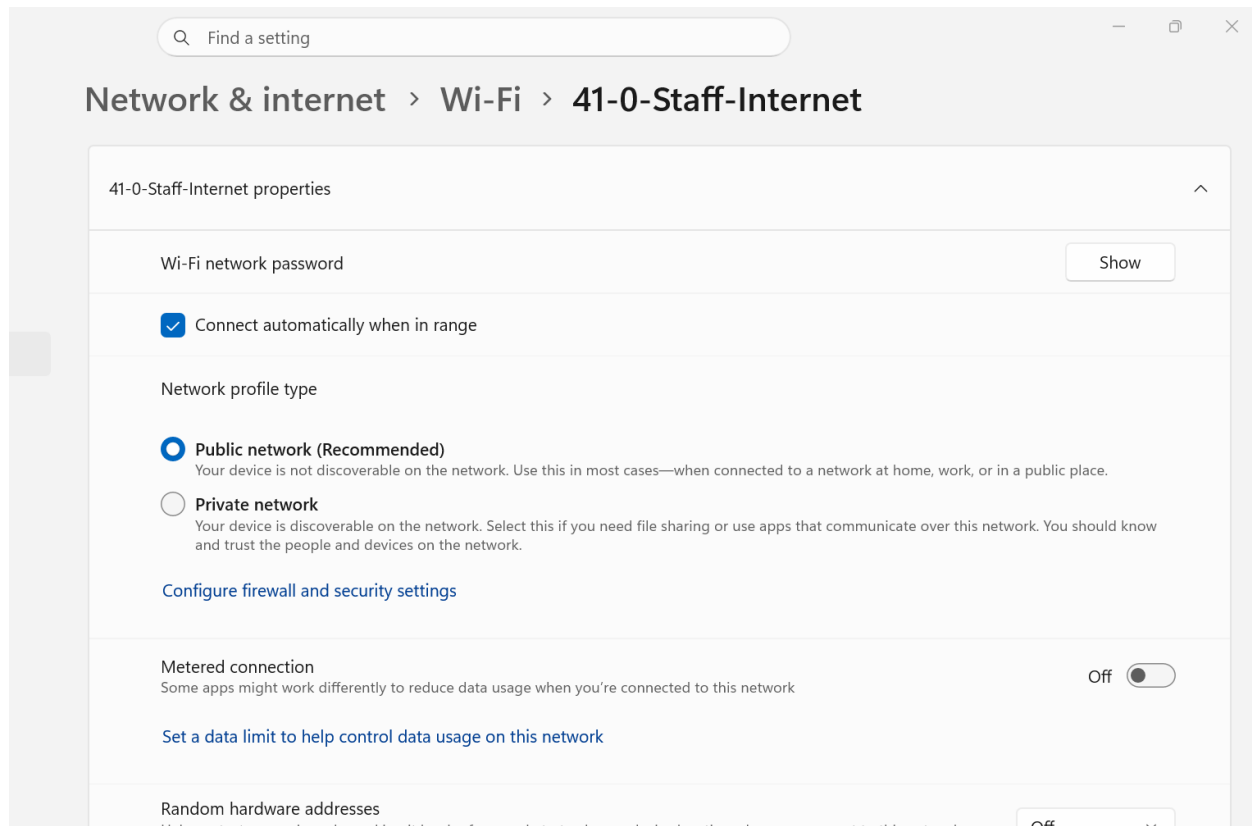
Steps:

- Settings → **Network & Internet**
- Check:
 - WiFi connection
 - Network type → Private/Public



Network Overview

- Wi-Fi connected → **41-0-Staff-Internet**
- Status → **Connected, secured**
- Shows general network info



Network Type

- Network profile → **Public network (selected)**
- Private option available

Observation:

The network settings were analyzed, and the system is connected to a Wi-Fi network with a public network profile. This restricts device visibility and enhances security.

Using a public network profile improves security by limiting unauthorized access, especially in untrusted environments.

STEP 9: Check Browser Security

Steps (Chrome/Edge):

- Go to **Settings** → **Privacy & Security**
- Check:
 - Safe browsing ON
 - Remove unknown extensions

Settings

chrome://settings

Ask Gemini

Verify it's you

Settings

You and Google

Autofill and passwords

Privacy and security

Performance

Appearance

Search engine

Default browser

On startup

Languages

Downloads

Accessibility

System

Search settings

You and Google

JUHI JUWAIYYAH

Verify it's you to use and save Chrome data in your account, dr.juhiwairiyah@gmail.com

Remove account

Verify it's you

Sync and Google services

Manage your Google Account

Customize your Chrome profile

Import bookmarks and settings

Settings - Privacy and security

chrome://settings/privacy

Ask Gemini

Verify it's you

Settings

You and Google

Autofill and passwords

Privacy and security

Performance

Appearance

Search engine

Default browser

On startup

Languages

Downloads

Accessibility

System

Search settings

Take the Privacy Guide

Review key privacy and security controls in Chrome

Get started

No thanks



Safety Check

Chrome found some safety recommendations for your review

Passwords

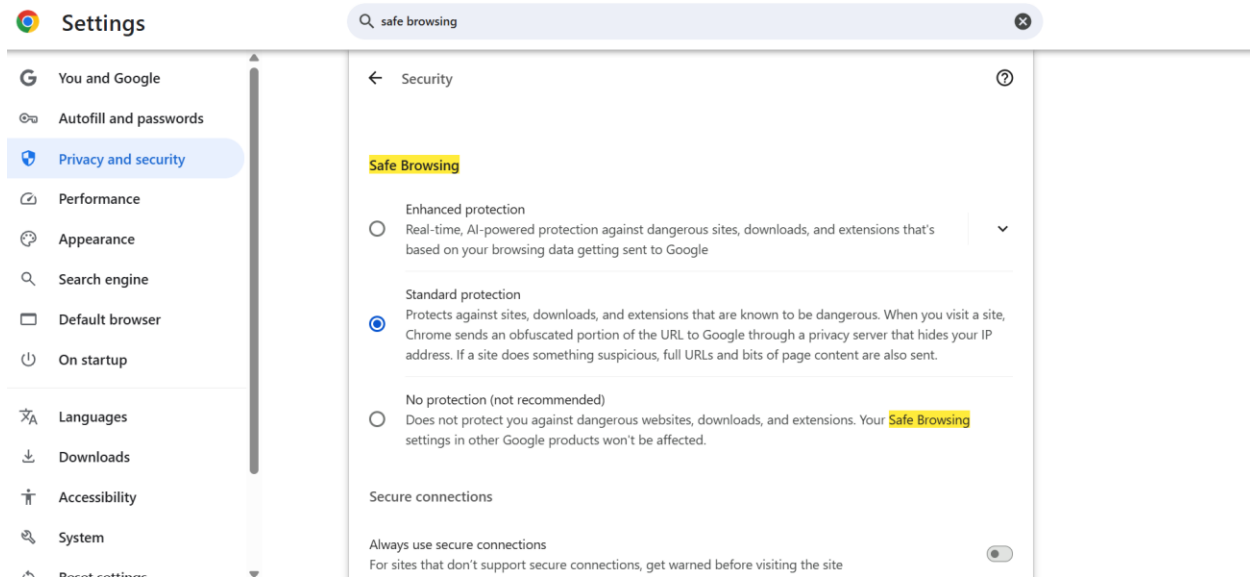
Go to Safety Check

Privacy and security

Delete browsing data

Delete history, cookies, cache, and more

Privacy Guide



Observation:

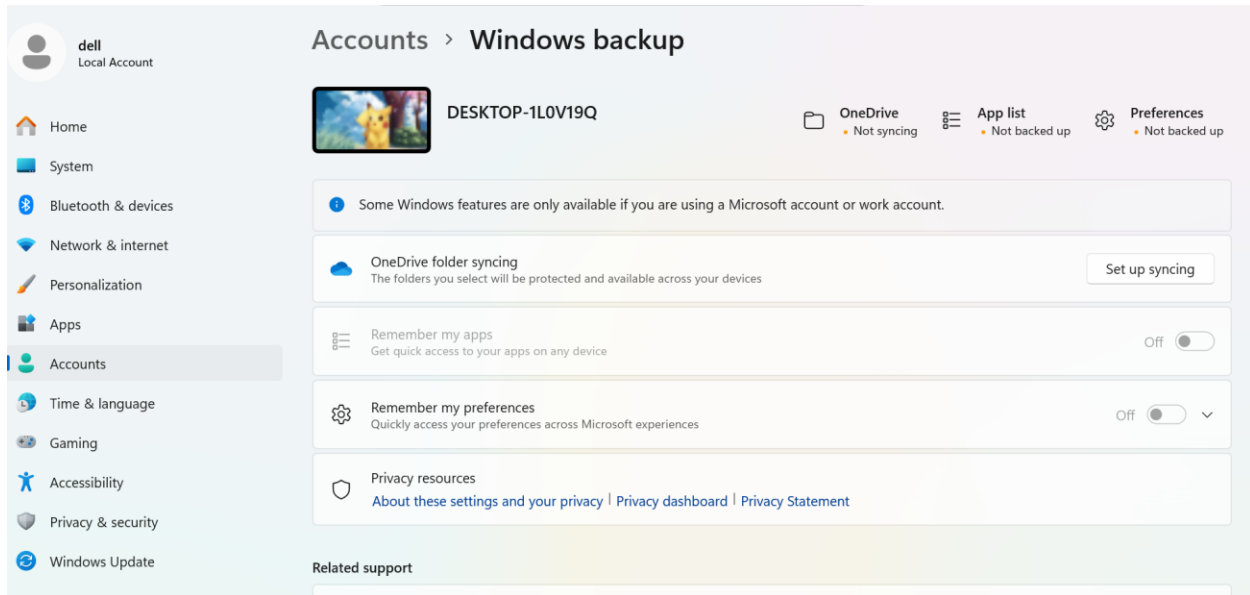
The browser security settings were examined under the Privacy and Security section. Safe Browsing was found to be enabled with standard protection, ensuring protection against malicious websites and unsafe downloads.

Safe Browsing helps prevent phishing attacks, malware infections, and unsafe browsing activities.

STEP 10: Check Data Backup

Steps:

- Search: **Backup settings**
- Check:
 - OneDrive / External backup



Observation:

The backup settings were checked, and it was observed that OneDrive backup is not configured and system data is not being backed up.

Lack of proper backup mechanisms may lead to permanent data loss in case of system failure or cyber attacks such as ransomware.

STEP 11: Risk Assessment Table

Asset	Threat	Vulnerability	Risk Level	Solution
Personal Files	Data Loss	No Backup	High	Enable OneDrive
System	Malware	Unknown Apps	High	Remove apps
Network	Hacking	Public Network	Medium	Use Private
Account	Unauthorized Access	Weak Password	High	Strong password

STEP 12: Security Recommendations

✓ Technical Controls

- Enable firewall

- Enable antivirus
- Regular updates

✓ **Administrative Controls**

- Strong passwords
- Avoid unknown downloads

✓ **User Awareness**

- Avoid phishing links
- Safe browsing

STEP 13: Result

The system security audit identified potential vulnerabilities such as lack of backup and configuration risks. Suitable mitigation strategies were suggested to improve system security.

Experiment

Creating and Analyzing Disk Images Using dc3dd and Autopsy (Alternative to FTK Imager)

1. Aim

To create a forensic disk image and analyze digital evidence using **dc3dd** and **Autopsy** in **Kali Linux**.

2. Objectives

After completing this experiment, students will be able to:

- Understand the concept of **digital forensics and disk imaging**
- Create a **bit-by-bit forensic copy of a storage device**
- Maintain **data integrity using hashing**
- Analyze disk images to identify **files, deleted files, and system artifacts**

3. Introduction / Theory

Digital Forensics

Digital forensics is the process of **identifying, collecting, preserving, analyzing, and presenting digital evidence** from computers or storage devices.

Digital evidence may include:

- Documents
- Emails
- Browser history
- Logs
- Deleted files

To preserve evidence, investigators create a **forensic disk image**.

Disk Imaging

Disk imaging is the process of creating an **exact replica of a storage device**.

Characteristics of forensic disk images:

- Bit-by-bit copy of storage
- Preserves deleted files
- Maintains data integrity
- Allows investigation without altering original evidence

Hashing in Forensics

Hashing is used to verify the **integrity of digital evidence**.

Common hash algorithms:

Algorithm	Purpose
MD5	evidence integrity
SHA1	verification
SHA256	stronger security

If the hash value of the original disk and the disk image match, the **evidence is considered intact**.

4. Tools Used

Tool	Purpose
Kali Linux	forensic analysis environment
dc3dd	disk image acquisition
Autopsy	forensic investigation and analysis

5. System Requirements

Component	Specification
Processor	Intel i3 or higher
RAM	Minimum 4 GB
Disk	20 GB free space
Software	Kali Linux VM

6. Algorithm

1. Start the Kali Linux environment.
2. Generate activity on the system to create forensic artifacts.
3. Identify the disk partition using system commands.
4. Create a forensic disk image using dc3dd.
5. Generate hash values for integrity verification.
6. Start the Autopsy forensic analysis tool.
7. Import the disk image into Autopsy.
8. Analyze files, logs, and deleted artifacts.
9. Extract relevant digital evidence.

7. Procedure

Step 1: Open Terminal in Kali Linux

Press:

Ctrl + Alt + T

This opens the terminal.

Step 2: Create Sample Evidence

Generate a file that will later be discovered during forensic analysis.

```
echo "Cybersecurity Lab Evidence" > evidence.txt
```

Verify the file creation.

```
Ls
```

Step 3: Identify Disk Partition

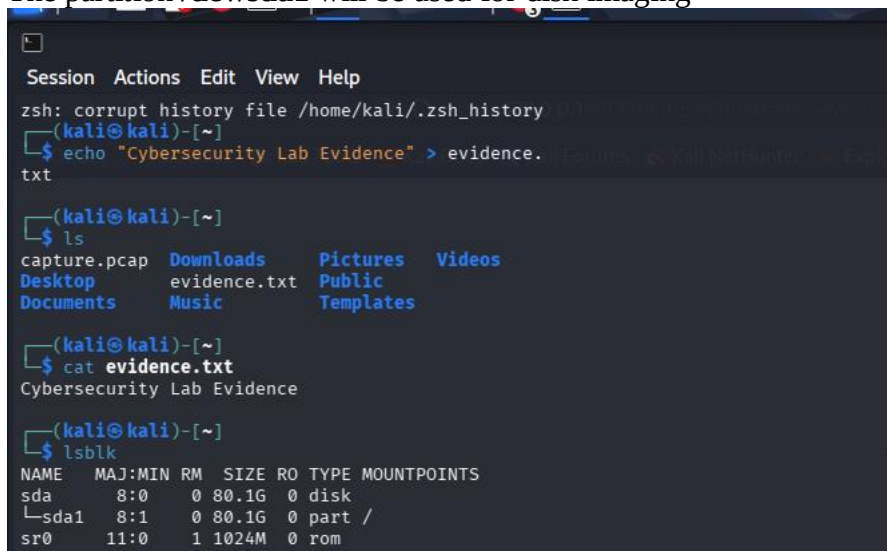
List storage devices.

```
lsblk
```

Example output:

```
sda
├─sda1
└─sda2
```

The partition **/dev/sda1** will be used for disk imaging



```
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history (1)
(kali@kali)-[~]
$ echo "Cybersecurity Lab Evidence" > evidence.txt
(kali@kali)-[~]
$ ls
capture.pcap  Downloads  Pictures  Videos
Desktop      evidence.txt  Public
Documents    Music       Templates
(kali@kali)-[~]
$ cat evidence.txt
Cybersecurity Lab Evidence
(kali@kali)-[~]
$ lsblk
NAME MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 80.1G  0 disk
├─sda1 8:1    0 80.1G  0 part /
└─sda2 8:1    0 80.1G  0 part /
sr0   11:0    1 1024M  0 rom
```

Install dc3dd

```
(kali㉿kali)-[~]
└─$ sudo apt install dc3dd
Installing:
  dc3dd

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not U
pgrading: 1757
  Download size: 117 kB
  Space needed: 489 kB / 60.2 GB available

Get:1 http://kali.download/kali kali-rolling/main
amd64 dc3dd amd64 7.3.1-4 [117 kB]
Fetched 117 kB in 5s (21.9 kB/s)
Selecting previously unselected package dc3dd.
(Reading database ... 454289 files and directorie
s currently installed.)
Preparing to unpack .../dc3dd_7.3.1-4_amd64.deb .
..
Unpacking dc3dd (7.3.1-4) ...
Setting up dc3dd (7.3.1-4) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.4.3) ...

(kali㉿kali)-[~]
└─$ dc3dd --help
usage:
_____

        dc3dd [OPTION 1] [OPTION 2] ... [OPTION N
]

                *or*

        dc3dd [HELP OPTION]

        where each OPTION is selected from the ba
sic or advanced
        options listed below, or HELP OPTION is s
elected from the
        help options listed below.
```

Step 4: Create Disk Image

Step 1: Create a Small Disk File

`dd if=/dev/zero of=practice_disk.dd bs=1M count=100`

```
(kali㉿kali)-[~]
└─$ dd if=/dev/zero of=practice_disk.dd bs=1M cou
nt=100
100+0 records in
100+0 records out
104857600 bytes (105 MB, 100 MiB) copied, 0.28316
5 s, 370 MB/s
```

Explanation:

Parameter	Description
if	input disk
of	output image file
hash	generate MD5 hash
log	acquisition log file

Creates a **100 MB disk**.

Step 2: Mount the Disk

```
mkfs.ext4 practice_disk.dd
```

```
(kali㉿kali)-[~]  
$ mkfs.ext4 practice_disk.dd  
mke2fs 1.47.2 (1-Jan-2025)  
Discarding device blocks: done  
  
Creating filesystem with 102400 1k blocks and 255  
84 inodes  
Filesystem UUID: 514f4c47-066d-4a31-8af3-f13b33b7  
5b90  
Superblock backups stored on blocks:  
      8193, 24577, 40961, 57345, 73729  
  
Allocating group tables: done  
  
Writing inode tables: done  
  
Creating journal (4096 blocks): done  
Writing superblocks and filesystem accounting inf  
ormation: done
```

Step 5: Verify Image Creation

Check the created image.

```
ls -lh /home/kali/disk_image.dd
```

```
(kali㉿kali)-[~]  
$ ls -lh /home/kali/disk_image.dd  
-rw-r--r-- 1 root root 9.8G Mar 15 11:59 /home/ka  
li/disk_image.dd
```

Check log file.

```
cat /home/kali/acquisition.log
```

The log will contain:

- image size
- hash value
- acquisition details

```

(kali㉿kali)-[~]
$ cat /home/kali/acquisition.log

dc3dd 7.2.646 started at 2026-03-15 11:52:30 -0400
0
compiled options:
command line dc3dd if=/dev/sda1 of=/home/kali/disk_image.dd hash=md5 log=/home/kali/acquisition.log
device size: 167966702 sectors (probed), 85,998,951,424 bytes
sector size: 512 bytes (probed)
10503356416 bytes ( 9.8 G ) copied ( 12% ), 432.546 s, 23 M/s

input results for device `/dev/sda1':
20514368 sectors in
0 bad sectors replaced by zeros
ce827a581c4dfce5b976603df85bb5e0 (md5)

output results for file `/home/kali/disk_image.dd':
20514368 sectors out

dc3dd aborted at 2026-03-15 11:59:42 -0400

```

Step 6: Start Autopsy

Launch the forensic analysis tool.

Autopsy

```

(kali㉿kali)-[~]
$ autopsy

=====
Autopsy Forensic Browser
http://www.sleuthkit.org/autopsy/

ver 2.24

=====
Evidence Locker: /var/lib/autopsy
Start Time: Sun Mar 15 12:01:18 2026
Remote Host: localhost
Local Port: 9999

Open an HTML browser on the remote host and paste
this URL in it:

http://localhost:9999/autopsy

Keep this process running and use <ctrl-c> to exit

```

```
(kali㉿kali)-[~]
└─$ sudo autopsy
[sudo] password for kali:

Autopsy Forensic Browser
http://www.sleuthkit.org/autops
y/
ver 2.24

Evidence Locker: /var/lib/autopsy
Start Time: Sun Mar 15 12:16:57 2026
Remote Host: localhost
Local Port: 9999

Open an HTML browser on the remote host and paste
this URL in it:

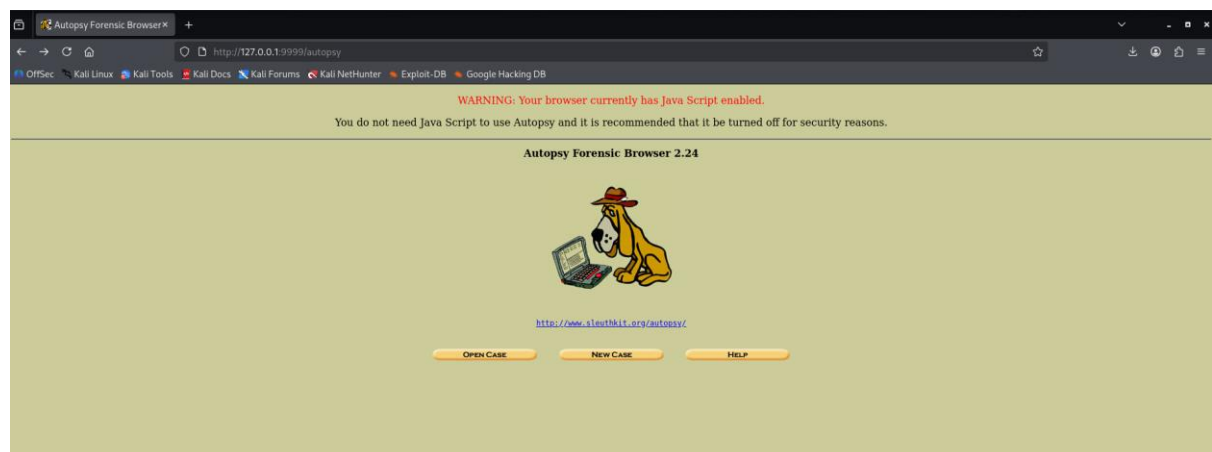
http://localhost:9999/autopsy

Keep this process running and use <ctrl-c> to exit
```

The terminal will display a URL:

`http://localhost:9999/autopsy`

Open this link in the browser.



Step 7: Create a New Case

In the Autopsy interface:

1. Click **Create New Case**
2. Enter case details:

The screenshot shows a web browser window with the title 'Create A New Case'. The address bar shows the URL `http://127.0.0.1:9999/autopsy/mod=0&view=1`. The browser's bookmark bar includes 'OffSec', 'Kali Linux', 'Kali Tools', 'Kali Docs', 'Kali Forums', 'Kali NetHunter', 'Exploit-DB', and 'Google Hacking DB'. The main content area has a light green background and a yellow box containing the form. The form is titled 'CREATE A NEW CASE' and has three sections: 1. 'Case Name' with a text input field containing 'CyberForensicsLab'; 2. 'Description' with a text input field containing 'Creating a new forensic investi'; 3. 'Investigator Names' with two columns of input fields labeled a. through j., where 'lavanya' is entered in field 'a.'. At the bottom of the yellow box are three buttons: 'New Case', 'CANCEL', and 'HELP'.

Step 8: Add Disk Image

Select:

Add Host → Add Image

Browse and select:

`/home/kali/disk_image.dd`

Autopsy will begin analyzing the disk image.

The screenshot shows a web browser window with the title 'Creating Case: cyberForensicsLab'. The address bar shows the URL `http://127.0.0.1:9999/autopsy?mod=0&view=2&case=CyberForensicsLab&desc=+Creating+a+new+fo`. The browser's bookmark bar is the same as in the previous screenshot. The main content area has a light green background. It displays the following text: 'Case directory (/var/lib/autopsy/CyberForensicsLab/) created' and 'Configuration file (/var/lib/autopsy/CyberForensicsLab/case.aut) created'. Below this, it says 'We must now create a host for this case.' and 'Please select your name from the list: lavanya'. At the bottom is a yellow button labeled 'ADD HOST'.

ADD A NEW HOST

1. **Host Name:** The name of the computer being investigated. It can contain only letters, numbers, and symbols.

KaliLabMachine

2. **Description:** An optional one-line description or note about this computer.

forensic disk image investigatio

3. **Time zone:** An optional timezone value (i.e. EST5EDT). If not given, it defaults to the local setting. A list of time zones can be found in the help files.

4. **Timeskew Adjustment:** An optional value to describe how many seconds this computer's clock was out of sync. For example, if the computer was 10 seconds fast, then enter -10 to compensate.

0

5. **Path of Alert Hash Database:** An optional hash database of known bad files.

6. **Path of Ignore Hash Database:** An optional hash database of known good files.

ADD HOST

CANCEL

HELP

Recommended Values

Host Name

KaliLabMachine

Description

Kali Linux virtual machine used for forensic disk image investigation

Time Zone

Leave Blank

Time Skew Adjustment

Leave Blank

Alert Hash Database

Leave Blank

Ignore Hash Database

Leave Blank

Then click:

Next

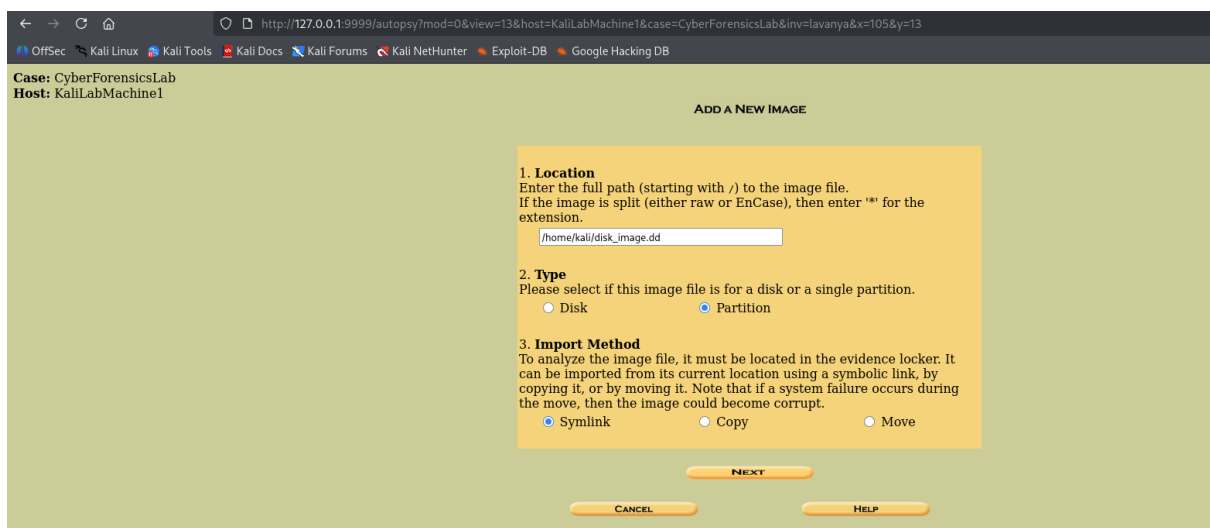
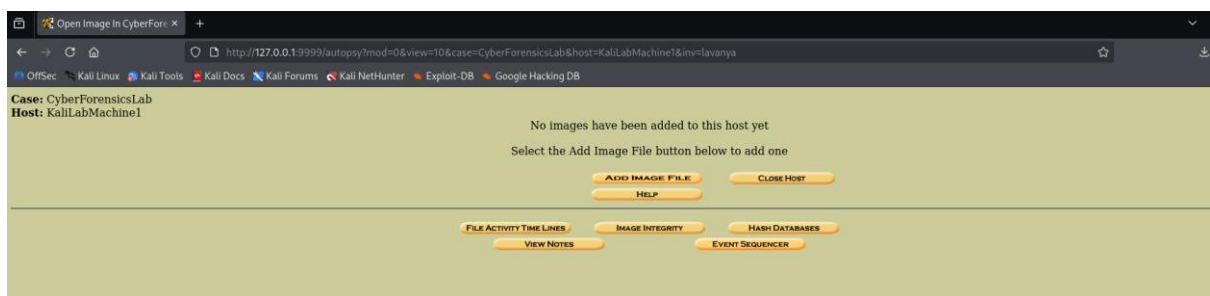
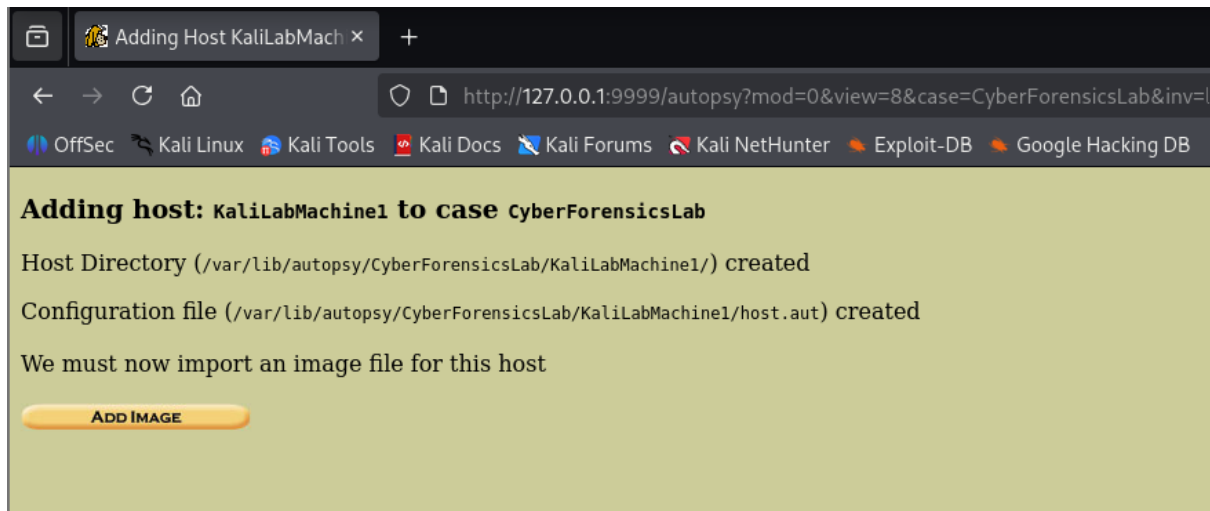


Image File Details

Local Name: images/disk_image.dd

Data Integrity: An MD5 hash can be used to verify the integrity of the image. (With split images, this hash is for the full image file)

- ☐ Ignore the hash value for this image.
- ☒ Calculate the hash value for this image.
- ☐ Add the following MD5 hash value for this image:

☒ Verify hash after importing?

File System Details

Analysis of the image file shows the following partitions:

Partition 1 (Type: ext4)

Mount Point:

File System Type:

ADD

CANCEL

HELP

Calculating MD5 (this could take a while)
Current MD5: ce827a581c4dfce5b976603df85b85e0
Testing partitions
Linking image(s) into evidence locker
Image file added with ID img1
Volume image (0 to 0 - ext - /1/) added with ID vol1

OK

ADD IMAGE

OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

Case: CyberForensicsLab
Host: KaliLabMachine1

Select a volume to analyze or add a new image file.

CASE GALLERY			HOST GALLERY	HOST MANAGER
mount	name	fs type		
/1/	disk_image.dd-0-0	ext	details	
ANALYZE			ADD IMAGE FILE	CLOSE HOST
			HELP	
FILE ACTIVITY TIME LINES			IMAGE INTEGRITY	HASH DATABASES
VIEW NOTES			EVENT SEQUENCER	

← → ↻ <http://127.0.0.1:9999/autopsy?mod=0&view=17&host=KaliLabMachine1&case=CyberForensicsLab&inv=lavanya&vol=vol1&x=78&y=16> ☆

OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

FILE ANALYSIS KEYWORD SEARCH FILE TYPE IMAGE DETAILS META DATA DATA UNIT HELP CLOSE
? X

To start analyzing this volume, choose an analysis mode from the tabs above.

OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB

FILE ANALYSIS
KEYWORD SEARCH
FILE TYPE
IMAGE DETAILS
META DATA
DATA UNIT
HELP
CLOSE

Directory Seek

Enter the name of a directory that you want to view.

VIEW

File Name Search

Enter a Perl regular expression for the file names you want to find.

SEARCH

ALL DELETED FILES

EXPAND DIRECTORIES

Current Directory: /

ADD NOTE
GENERATE MD5 LIST OF FILES

DEL	Type	NAME	WRITTEN	ACCESSED	CHANGED	Size	UID
	d /	d	2025-12-02 22:02:53 (EST)	2026-03-15 01:45:08 (EDT)	2025-12-02 22:02:53 (EST)	4096	0
	d /	d	2025-12-02 22:02:53 (EST)	2026-03-15 01:45:08 (EDT)	2025-12-02 22:02:53 (EST)	4096	0
	l /	l	2025-11-10 04:50:42 (EST)	2026-03-15 01:31:26 (EDT)	2025-12-02 22:02:40 (EST)	7	0
	d / -	boot	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0	0
	d / -	dev	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0	0
	d / -	etc	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0	0
	d / -	home	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0	0

Error Parsing File (Invalid Characters?):
V/V 5251073: \$OrphanFiles 0000-00-00 00:00:00 (UTC) 0000-00-00 00:00:00 (UTC) 0000-00-00 00:00:00 (UTC) 0000-00-00 00:00:00 (UTC) 0 0 0

File Browsing Mode

In this mode, you can view file and directory contents.
File contents will be shown in this window.
More file details can be found using the Metadata link at the end of the list (on the right).
You can also sort the files using the column headers

Find in page

Highlight All
Match Case
Match Diacritics
Whole Words

OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB

FILE ANALYSIS
KEYWORD SEARCH
FILE TYPE
IMAGE DETAILS
META DATA
DATA UNIT
HELP
CLOSE

Keyword Search of Allocated and Unallocated Space

Enter the keyword string or expression to search for:

☒ ASCII
☒ Unicode

☐ Case Insensitive
☐ grep Regular Expression

SEARCH

EXTRACT STRINGS
EXTRACT UNALLOCATED

Regular Expression Cheat Sheet

NOTE: The keyword search runs grep on the image.
A list of what will and what will not be found is available here.

Predefined Searches

SSN2
IP
CC
Date

SSN1

OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB

FILE ANALYSIS
KEYWORD SEARCH
FILE TYPE
IMAGE DETAILS
META DATA
DATA UNIT
HELP
CLOSE

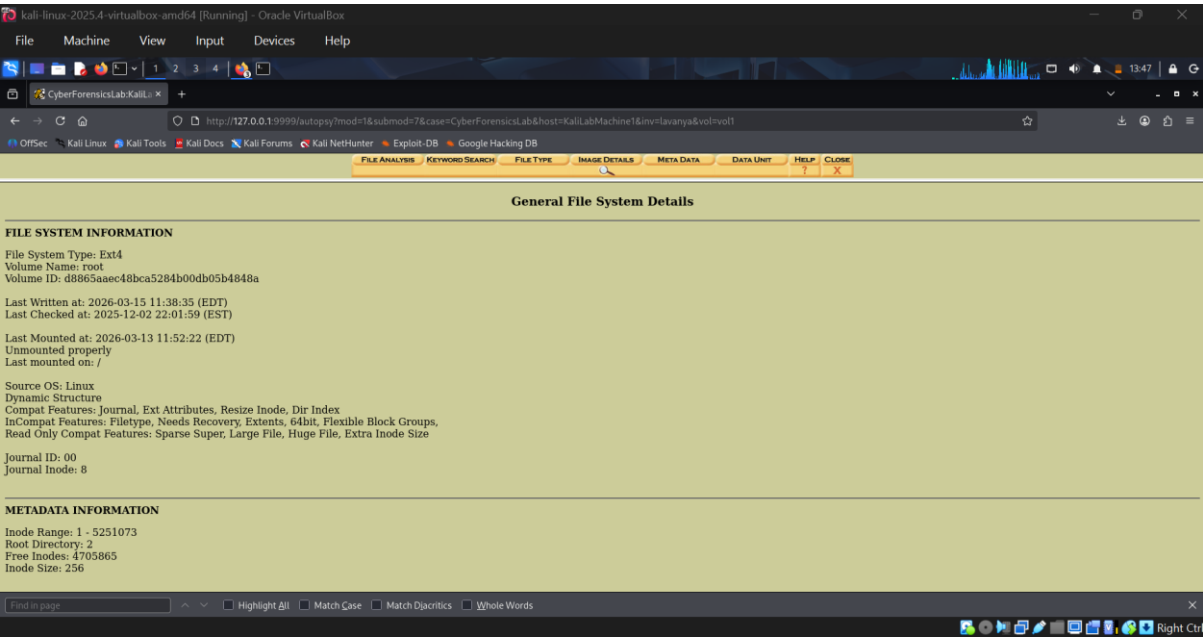
Sort Files by Type

View Sorted Files

File Type Sorting

In this mode, Autopsy will examine allocated and unallocated files and sort them into categories and verify the extension.
This allows you to find a file based on its type and find "hidden" files.
WARNING: This can be a time intensive process.

Step 9: Analyze Evidence



EXERCISE: LOG FILE ANALYSIS FOR INCIDENT DETECTION LAB

Aim

To analyze system log files and detect suspicious activities such as failed login attempts and identify suspicious IP addresses.

Objective

- Understand system log files
- Detect suspicious activities and incidents
- Analyze logs using Linux tools
- Identify attack patterns (failed logins, brute force, etc.)

Requirements

- Kali Linux / Ubuntu VM
- Basic Linux command knowledge
- Tools:
 - Kali Linux
 - journalctl
 - ssh
 - grep, awk, sort
 - grep
 - awk
 - less
 - cat
 - journalctl

Understanding Log Files

Navigate to Log Directory

cd /var/log

ls

The `/var/log` directory contains system and application logs such as Apache, Nginx, and login records used for security analysis.

```
kali@kali: /var/log/apache2
Session Actions Edit View Help
(kali@kali)-[~]
$ cd /var/log
(kali@kali)-[/var/log]
$ ls
alternatives.log  inetsim      openvpn      stunnel4
apache2           journal      postgresql   sysstat
apt              lastlog     private      wtmp
boot.log         lightdm     README       wtmp.db
btmtp            macchanger.log  redis       Xorg.0.log
dpkg.log         mosquitto    runit
fontconfig.log   nginx       samba
gvm             notus-scanner speech-dispatcher

(kali@kali)-[/var/log]
$ sudo less /var/log/auth.log
[sudo] password for kali:
/var/log/auth.log: No such file or directory

(kali@kali)-[/var/log]
$ sudo lastb
[sudo] password for kali:
sudo: lastb: command not found

(kali@kali)-[/var/log]
$ last
kali    tty7      :0                Tue Mar 24 06:25 - still logged in
lightdm tty7      :0                Tue Mar 24 06:24 - 06:25 (00:00)
```

What Your Output Shows

From your `ls`:

- ✓ `dpkg.log` → package activity
- ✓ `apache2` → web server logs
- ✓ `nginx` → web logs
- ✓ `journal` → system logs (IMPORTANT)
- ✓ `wtmp`, `btmtp` → login records

Successful Logins (`wtmp`)

Command:

`last`

Capture:

- Login history

This displays successful login sessions. It helps identify legitimate and suspicious user access.

System Logs using `journalctl`

Command:

```
journalctl | less
```

Capture:

- Logs output

Failed SSH Attempts

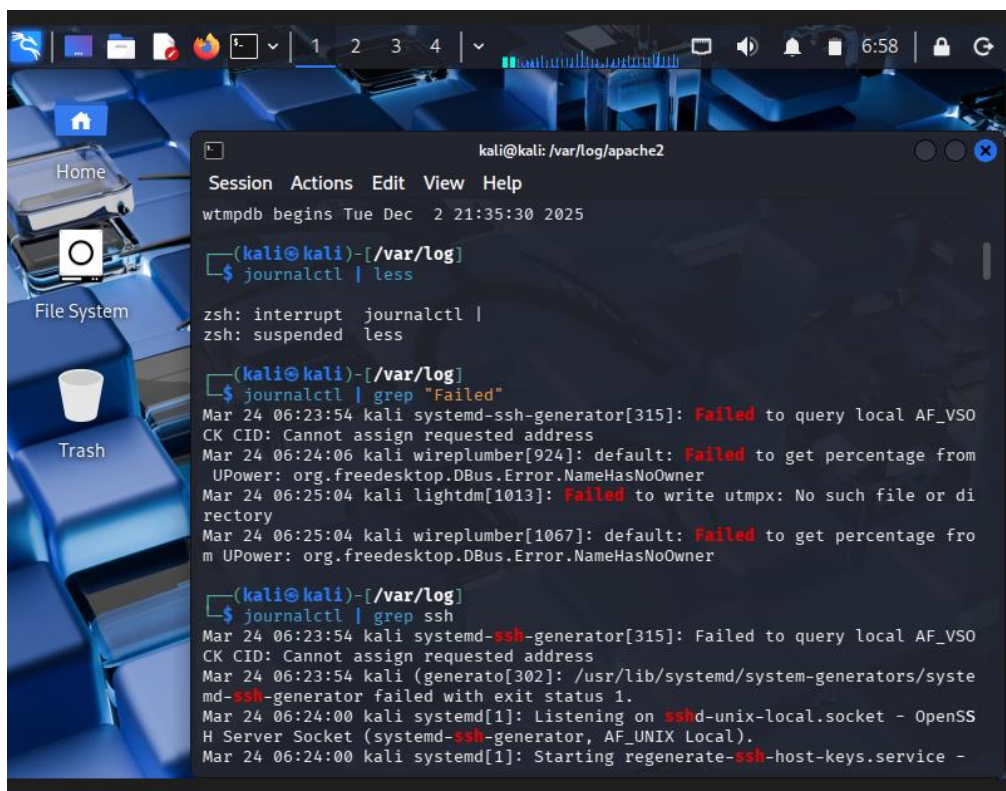
Command:

```
journalctl | grep "Failed"
```

SSH Login Activity

Command:

```
journalctl | grep ssh
```

A screenshot of a Kali Linux desktop environment. The desktop background is a blue and black keyboard. On the left side, there are icons for 'Home', 'File System', and 'Trash'. A terminal window is open in the center, titled 'kali@kali: /var/log/apache2'. The terminal shows the output of the 'journalctl | less' command, which displays system logs. The logs include messages from 'systemd-ssh-generator' and 'systemd-ssh-generator' indicating failures. The terminal also shows the output of the 'journalctl | grep "Failed"' command, which displays logs with the word 'Failed' in red. Finally, the terminal shows the output of the 'journalctl | grep ssh' command, which displays logs related to SSH, including 'Failed to query local AF_VSOCK CID: Cannot assign requested address' and 'Listening on sshd-unix-local.socket - OpenSSH Server Socket (systemd-ssh-generator, AF_UNIX Local)'.

```
kali@kali: /var/log/apache2
Session Actions Edit View Help
wtmddb begins Tue Dec 2 21:35:30 2025

(kali@kali)-[/var/log]
$ journalctl | less

zsh: interrupt journalctl |
zsh: suspended less

(kali@kali)-[/var/log]
$ journalctl | grep "Failed"
Mar 24 06:23:54 kali systemd-ssh-generator[315]: Failed to query local AF_VSO
CK CID: Cannot assign requested address
Mar 24 06:24:06 kali wireplumber[924]: default: Failed to get percentage from
UPower: org.freedesktop.DBus.Error.NameHasNoOwner
Mar 24 06:25:04 kali lightdm[1013]: Failed to write utmpx: No such file or di
rectory
Mar 24 06:25:04 kali wireplumber[1067]: default: Failed to get percentage fro
m UPower: org.freedesktop.DBus.Error.NameHasNoOwner

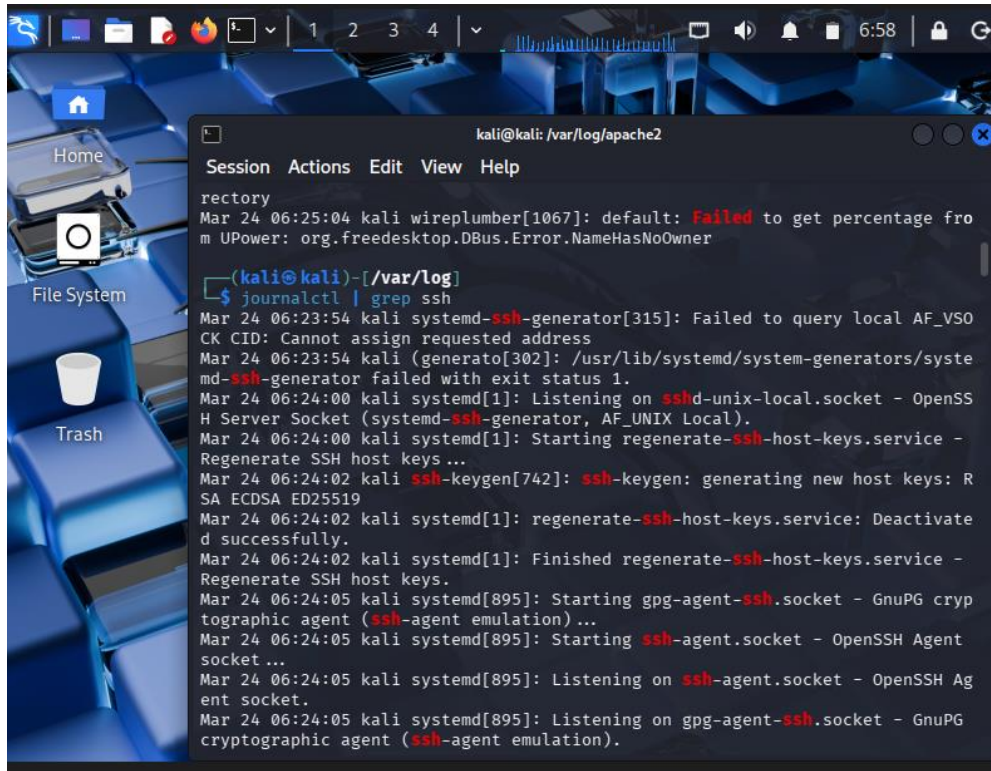
(kali@kali)-[/var/log]
$ journalctl | grep ssh
Mar 24 06:23:54 kali systemd-ssh-generator[315]: Failed to query local AF_VSO
CK CID: Cannot assign requested address
Mar 24 06:23:54 kali (generato[302]: /usr/lib/systemd/system-generators/syste
md-ssh-generator failed with exit status 1.
Mar 24 06:24:00 kali systemd[1]: Listening on sshd-unix-local.socket - OpenSS
H Server Socket (systemd-ssh-generator, AF_UNIX Local).
Mar 24 06:24:00 kali systemd[1]: Starting regenerate-ssh-host-keys.service -
```

Explanation:

The journalctl command provides centralized system logs including authentication, services, and system events.

This screenshot shows failed authentication attempts, which may indicate unauthorized access attempts.

This screenshot also displays SSH-related logs, including login attempts and connections.

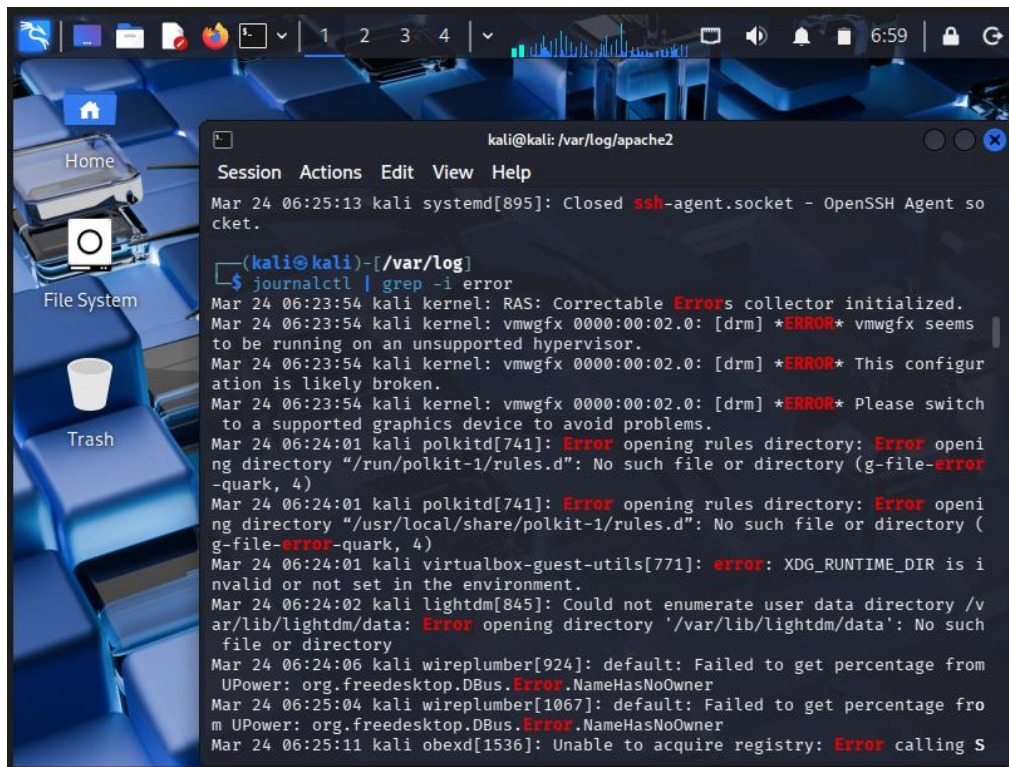
A screenshot of a Kali Linux desktop environment. The desktop background is a blue keyboard. On the left, there are icons for 'Home', 'File System', and 'Trash'. A terminal window is open in the center, titled 'kali@kali: /var/log/apache2'. The terminal shows the command 'journalctl | grep ssh' and its output, which includes several log entries related to SSH services, such as 'Failed to query local AF_VSOCK CID: Cannot assign requested address', 'Listening on sshd-unix-local.socket', and 'Starting regenerate-ssh-host-keys.service'.

```
kali@kali: /var/log/apache2
Session Actions Edit View Help
rectory
Mar 24 06:25:04 kali wireplumber[1067]: default: Failed to get percentage from
m UPower: org.freedesktop.DBus.Error.NameHasNoOwner

(kali@kali)-[/var/log]
$ journalctl | grep ssh
Mar 24 06:23:54 kali systemd-ssh-generator[315]: Failed to query local AF_VSO
CK CID: Cannot assign requested address
Mar 24 06:23:54 kali (generato[302]: /usr/lib/systemd/system-generators/syste
md-ssh-generator failed with exit status 1.
Mar 24 06:24:00 kali systemd[1]: Listening on sshd-unix-local.socket - OpenSS
H Server Socket (systemd-ssh-generator, AF_UNIX Local).
Mar 24 06:24:00 kali systemd[1]: Starting regenerate-ssh-host-keys.service -
Regenerate SSH host keys...
Mar 24 06:24:02 kali ssh-keygen[742]: ssh-keygen: generating new host keys: R
SA ECDSA ED25519
Mar 24 06:24:02 kali systemd[1]: regenerate-ssh-host-keys.service: Deactivate
d successfully.
Mar 24 06:24:02 kali systemd[1]: Finished regenerate-ssh-host-keys.service -
Regenerate SSH host keys.
Mar 24 06:24:05 kali systemd[895]: Starting gpg-agent-ssh.socket - GnuPG cryp
tographic agent (ssh-agent emulation)...
Mar 24 06:24:05 kali systemd[895]: Starting ssh-agent.socket - OpenSSH Agent
socket...
Mar 24 06:24:05 kali systemd[895]: Listening on ssh-agent.socket - OpenSSH Ag
ent socket.
Mar 24 06:24:05 kali systemd[895]: Listening on gpg-agent-ssh.socket - GnuPG
cryptographic agent (ssh-agent emulation).
```

Error Detection

Command: journalctl | grep -i error



The above screenshot highlights system errors, which may indicate abnormal behavior or system issues.

Apache Log Analysis

Command:

```
cd /var/log/apache2  
ls
```

Then:

```
sudo less access.log
```

Apache logs record web requests. These logs help identify suspicious web activity such as repeated requests or attacks.

Suspicious Requests

Command:

```
grep "404" /var/log/apache2/access.log
```

This shows failed web requests (404 errors), which may indicate scanning or probing attacks.

Package Activity

Command:

```
less /var/log/dpkg.log
```

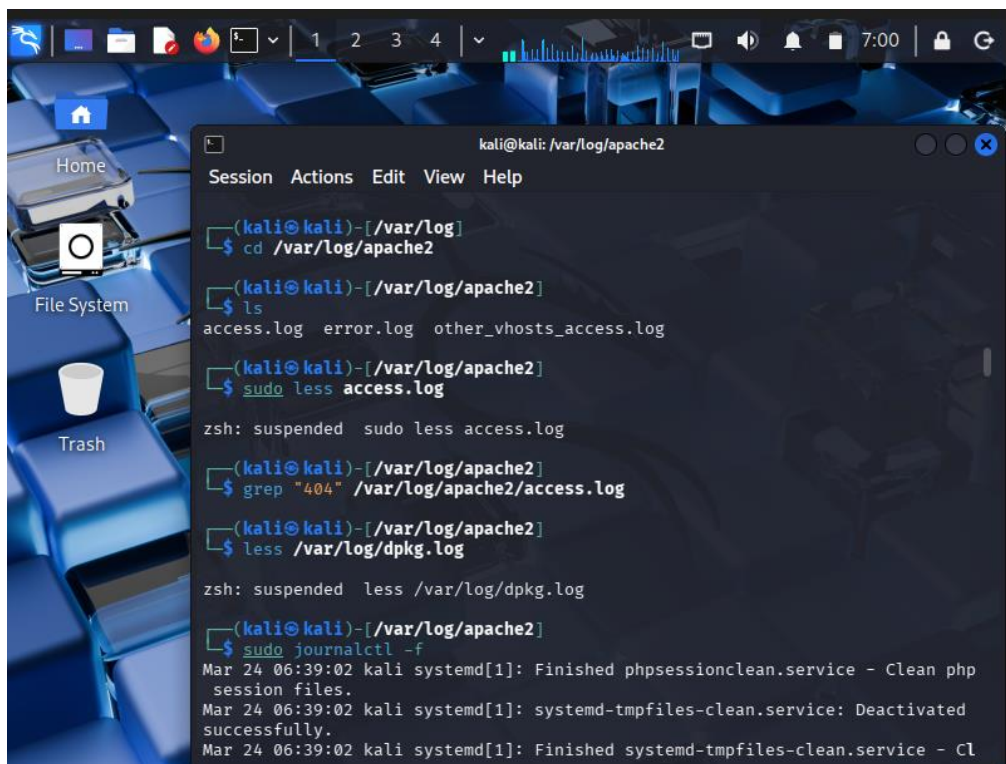
This shows software installation and update logs, useful for detecting unauthorized changes.

Real-Time Monitoring

Command:

```
sudo journalctl -f
```

This shows real-time system log monitoring, allowing detection of incidents as they occur.

A screenshot of a Kali Linux desktop environment with a terminal window open. The terminal shows a series of commands and their outputs for navigating and filtering logs. The background of the desktop is a blue keyboard-themed wallpaper. The terminal window has a title bar that reads 'kali@kali: /var/log/apache2'. The commands and outputs are as follows:

```
kali@kali: /var/log/apache2
Session Actions Edit View Help
(kali@kali)-[/var/log]
$ cd /var/log/apache2
(kali@kali)-[/var/log/apache2]
$ ls
access.log  error.log  other_vhosts_access.log
(kali@kali)-[/var/log/apache2]
$ sudo less access.log
zsh: suspended  sudo less access.log
(kali@kali)-[/var/log/apache2]
$ grep "404" /var/log/apache2/access.log
(kali@kali)-[/var/log/apache2]
$ less /var/log/dpkg.log
zsh: suspended  less /var/log/dpkg.log
(kali@kali)-[/var/log/apache2]
$ sudo journalctl -f
Mar 24 06:39:02 kali systemd[1]: Finished phpsessionclean.service - Clean php
session files.
Mar 24 06:39:02 kali systemd[1]: systemd-tmpfiles-clean.service: Deactivated
successfully.
Mar 24 06:39:02 kali systemd[1]: Finished systemd-tmpfiles-clean.service - Cl
```

QUICK CHECKLIST

Check	Command	Suspicious If
Failed logins	lastb	Many attempts
SSH activity	`journalctl	grep ssh`
Login history	last	Odd timing
Errors	`journalctl	grep error`
Web logs	grep 404	Many failed requests

STEP-BY-STEP ANALYSIS OF YOUR LOG

1. System Services (NORMAL)

```
Finished phpsessionclean.service
Finished systemd-tmpfiles-clean.service
```

- ✓ These are routine background tasks
- ✓ No issue

Regular system maintenance services were observed, indicating normal system operation.

2. CRON Jobs (NORMAL)

```
CRON: session opened for user root
(root) CMD (debian-sa1)
```

- ✓ Automated scheduled task
- ✓ Runs periodically

Scheduled cron jobs were observed, which are normal automated system processes.

3. Your Own SUDO Activity (NORMAL)

```
kali : COMMAND=/usr/bin/less access.log
kali : COMMAND=/usr/bin/journalctl -f
```

- ✓ This is YOU running commands
- ✓ Not suspicious

Sudo activity corresponds to legitimate administrative actions performed by the user.

4. XFCE / GUI Services (NORMAL)

```
xfconfd.service started
```

- ✓ Desktop environment service
- ✓ Normal

5. IMPORTANT PART

● Repeated Kernel Messages

```
dndHGCM: Received message HOST_DND_FN_GH_REQ_PENDING
dnd: No guest source window
```


These lines are:

Repeated MANY times

Coming from VirtualBox (VBoxDRM / DnD)

🔍 Is This Suspicious?

Technically:

- NOT an attack
- But **abnormal repetitive behavior**

! Why it matters:

- Repeated logs = unusual system activity
- Could indicate:
 - Misconfiguration
 - Resource issue
 - VirtualBox interaction problem

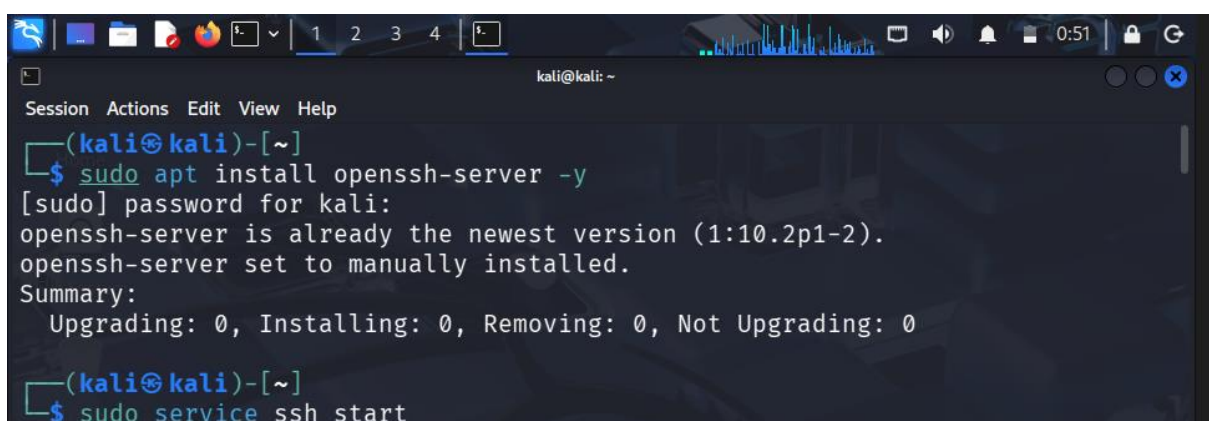
METHOD 1: SIMULATE FAILED SSH LOGIN

Step 1: Ensure SSH is Installed

```
sudo apt install openssh-server -y
```

Step 2: Start SSH Service

```
sudo service ssh start
```

A screenshot of a Kali Linux terminal window. The window title is 'kali@kali: ~'. The terminal shows the following commands and output:

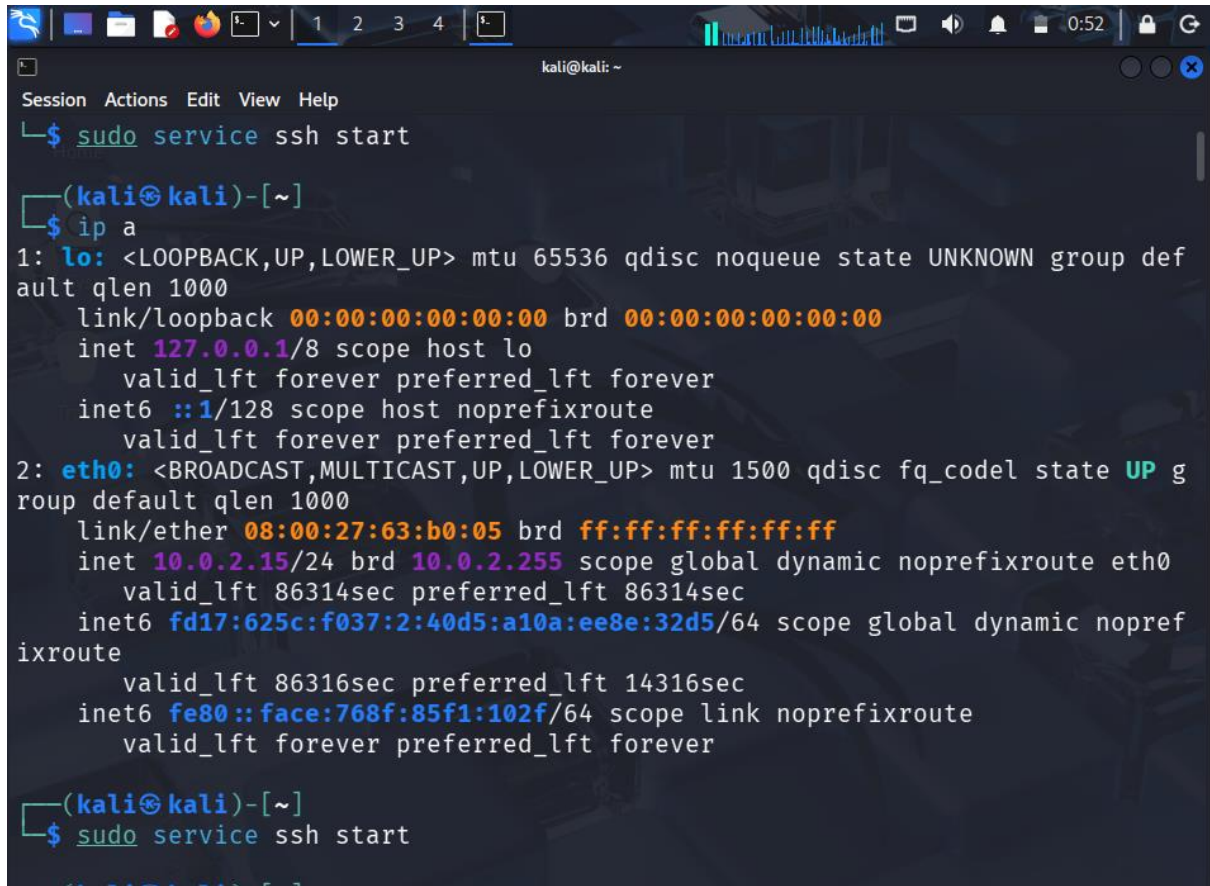
```
(kali@kali)-[~]  
$ sudo apt install openssh-server -y  
[sudo] password for kali:  
openssh-server is already the newest version (1:10.2p1-2).  
openssh-server set to manually installed.  
Summary:  
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0  
  
(kali@kali)-[~]  
$ sudo service ssh start
```

Step 3: Find Your IP Address

```
ip a
```

Look for something like:

inet 192.168.x.x

A screenshot of a Kali Linux terminal window. The window title is 'kali@kali: ~'. The terminal shows the command 'sudo service ssh start' being executed. Below that, the prompt is '(kali@kali)-[~]'. The user enters 'ip a', and the terminal displays the output for the loopback interface 'lo' and the ethernet interface 'eth0'. The 'lo' interface has the IP address 127.0.0.1. The 'eth0' interface has the IP address 10.0.2.15. The terminal also shows the command 'sudo service ssh start' being executed again.

```
kali@kali: ~  
Session Actions Edit View Help  
└─$ sudo service ssh start  
  
(kali@kali)-[~]  
└─$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def  
ault qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g  
roup default qlen 1000  
    link/ether 08:00:27:63:b0:05 brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0  
        valid_lft 86314sec preferred_lft 86314sec  
    inet6 fd17:625c:f037:2:40d5:a10a:ee8e:32d5/64 scope global dynamic nopref  
ixroute  
        valid_lft 86316sec preferred_lft 14316sec  
    inet6 fe80::face:768f:85f1:102f/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
  
(kali@kali)-[~]  
└─$ sudo service ssh start
```

Step 4: Attempt Wrong Login (GENERATE FAILURES)

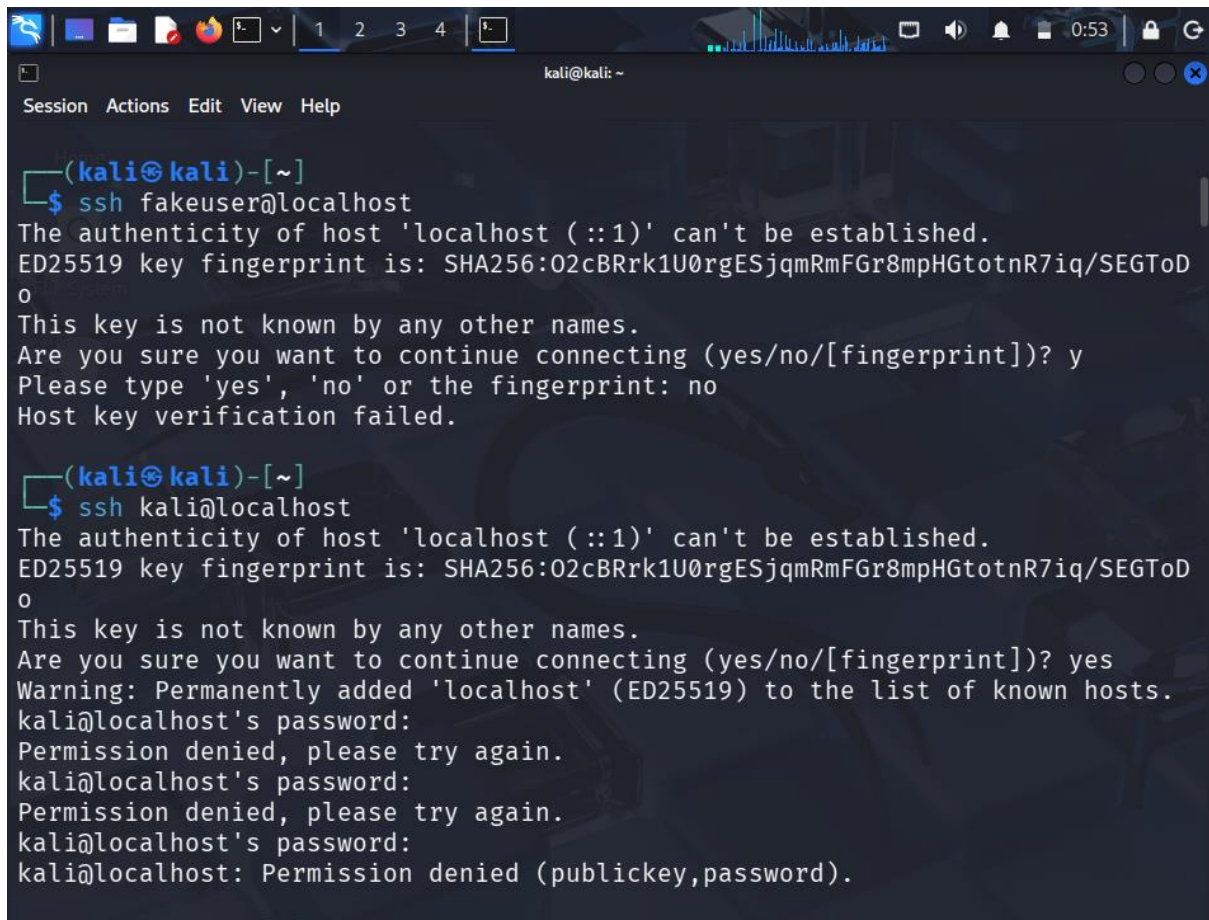
run:

ssh fakeuser@localhost

OR

ssh kali@localhost

Enter **wrong password multiple times (5–10 times)**



```
(kali@kali)-[~]
$ ssh fakeuser@localhost
The authenticity of host 'localhost (::1)' can't be established.
ED25519 key fingerprint is: SHA256:02cBRRk1U0rgESjqmRmFGr8mpHGtotnR7iq/SEGToD
0
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: no
Host key verification failed.

(kali@kali)-[~]
$ ssh kali@localhost
The authenticity of host 'localhost (::1)' can't be established.
ED25519 key fingerprint is: SHA256:02cBRRk1U0rgESjqmRmFGr8mpHGtotnR7iq/SEGToD
0
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
kali@localhost's password:
Permission denied, please try again.
kali@localhost's password:
Permission denied, please try again.
kali@localhost's password:
kali@localhost: Permission denied (publickey,password).
```

Now Analyze Logs

Step 5: Check Failed Logins

```
journalctl | grep "Failed password"
```

🔗 You should now see:

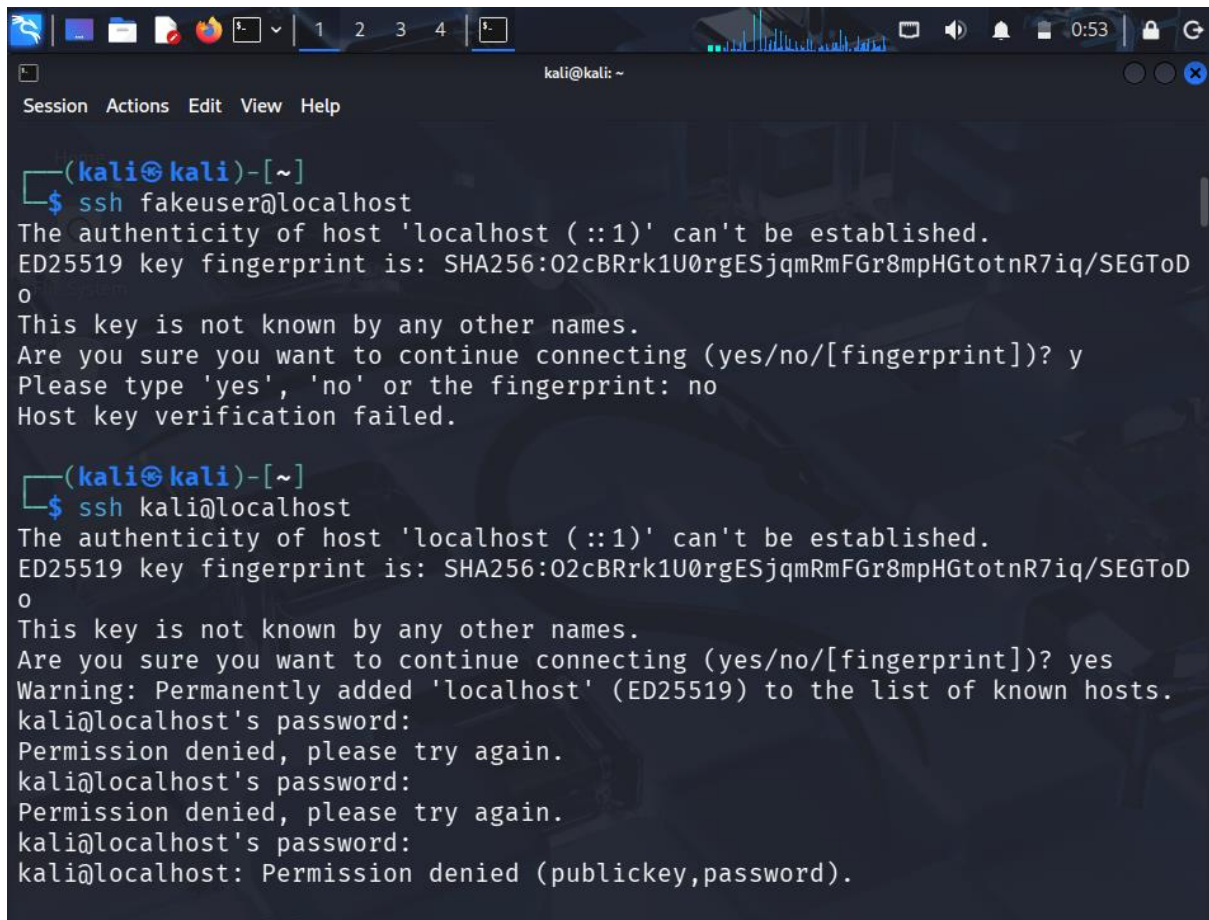
```
Failed password for kali from 127.0.0.1 port ...
```

Step 6: Extract Suspicious IP

```
journalctl | grep "Failed password" | awk '{print $11}'
```

Output:

```
127.0.0.1
```



```
(kali㉿kali)-[~]
$ ssh fakeuser@localhost
The authenticity of host 'localhost (::1)' can't be established.
ED25519 key fingerprint is: SHA256:02cBRRk1U0rgESjqmRmFGr8mpHGtotnR7iq/SEGToD
0
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: no
Host key verification failed.

(kali㉿kali)-[~]
$ ssh kali@localhost
The authenticity of host 'localhost (::1)' can't be established.
ED25519 key fingerprint is: SHA256:02cBRRk1U0rgESjqmRmFGr8mpHGtotnR7iq/SEGToD
0
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
kali@localhost's password:
Permission denied, please try again.
kali@localhost's password:
Permission denied, please try again.
kali@localhost's password:
kali@localhost: Permission denied (publickey,password).
```

Step 7: Count Attempts Per IP

```
journalctl | grep "Failed password" | awk '{print $11}' | sort
| uniq -c | sort -nr
```

Example:

```
10 127.0.0.1
```

```
kali@kali: ~  
Session Actions Edit View Help  
└─$ journalctl | grep "Failed password"  
Mar 26 00:42:21 kali sshd-session[3399]: Failed password for kali from ::1 port 36238 ssh2  
Mar 26 00:42:29 kali sshd-session[3399]: Failed password for kali from ::1 port 36238 ssh2  
Mar 26 00:42:37 kali sshd-session[3399]: Failed password for kali from ::1 port 36238 ssh2  
  
└─(kali@kali)-[~]  
└─$ journalctl | grep "Failed password" | awk '{print $11}'  
::1  
::1  
::1  
  
└─(kali@kali)-[~]  
└─$ journalctl | grep "Failed password" | awk '{print $11}' | sort | uniq -c | sort -nr  
3 ::1
```

INTERPRETATION (VERY IMPORTANT)

Multiple failed login attempts were detected from IP address 127.0.0.1, indicating a simulated brute force attack.

METHOD 2: Use lastb (Alternative)

sudo lastb

Shows failed login records

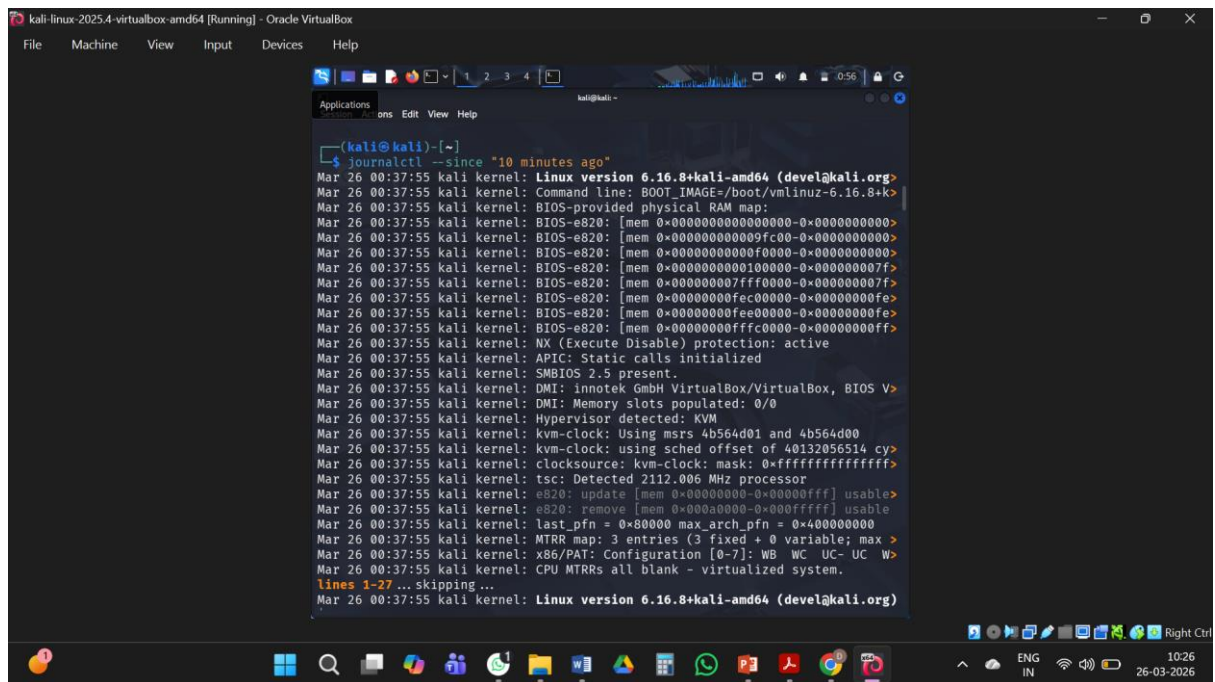
HOW TO IDENTIFY “SUSPICIOUS IP”

Simple rule:

Condition	Meaning
Same IP repeated many times	🚩 Suspicious
Unknown IP	🚩 Suspicious
High frequency attempts	🚩 Brute force

Run:

Shows recent attack activity



Lab Exp 3: NETWORK FORENSICS USING WIRESHARK

Aim

To capture and analyse network packets using Wireshark and understand network forensics techniques.

Objective

- To learn packet capturing using Wireshark
- To analyze different network protocols
- To apply filters for identifying specific traffic
- To understand how network forensics helps in detecting suspicious activities

Introduction

Network forensics is a branch of digital forensics that deals with monitoring, capturing, and analyzing network traffic to detect security breaches and malicious activities. Wireshark is a widely used open-source tool that helps in capturing real-time packets and analyzing them in detail.

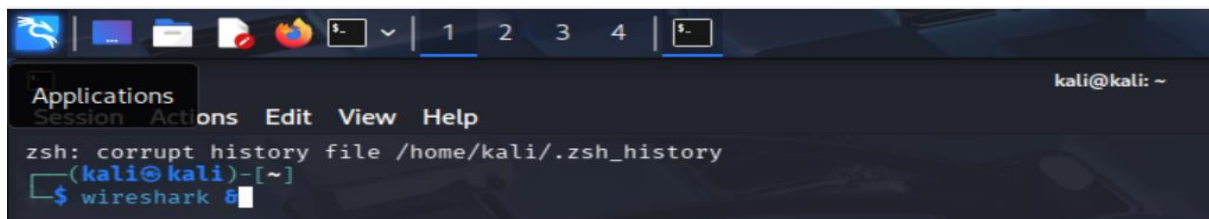
Tools Required:

1. Kali Linux
2. Wireshark

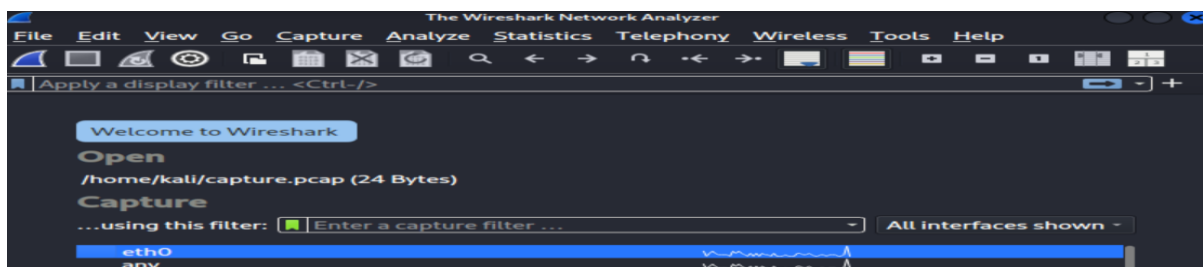
Procedure:

Step1:Open kali linux terminal

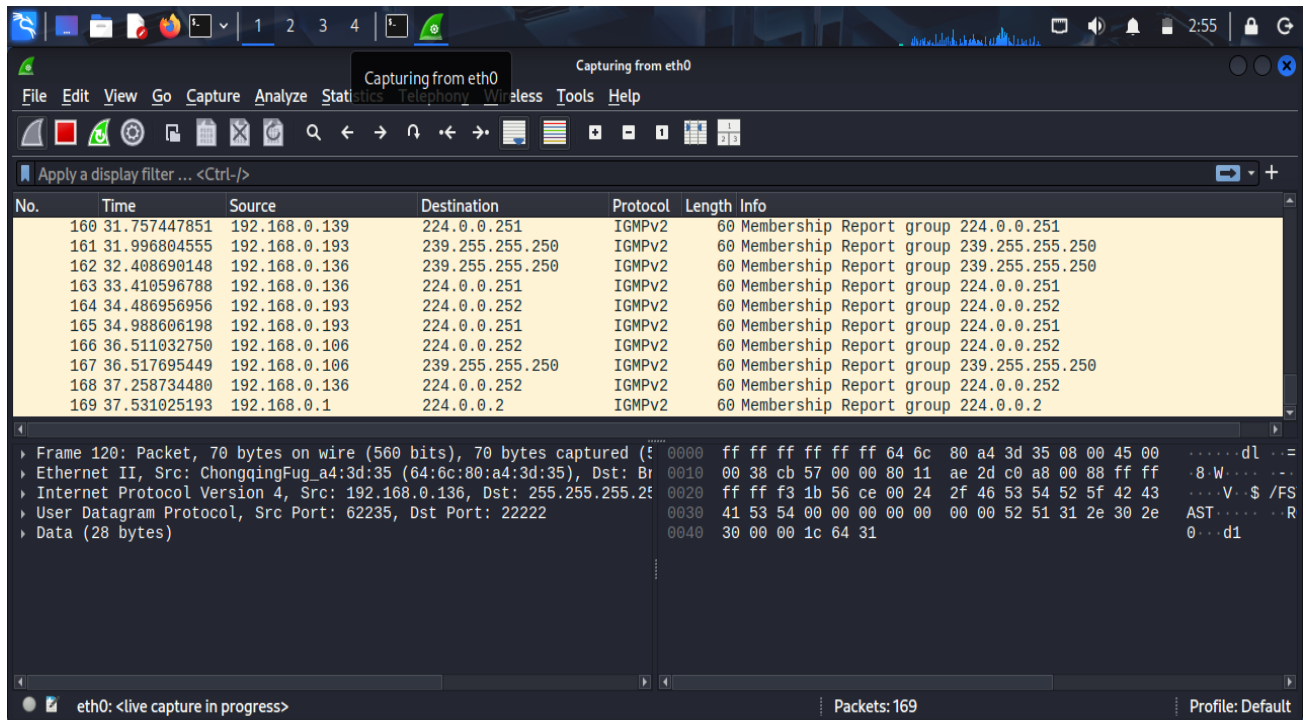
Type **wireshark &**



Step 2: Select eth0

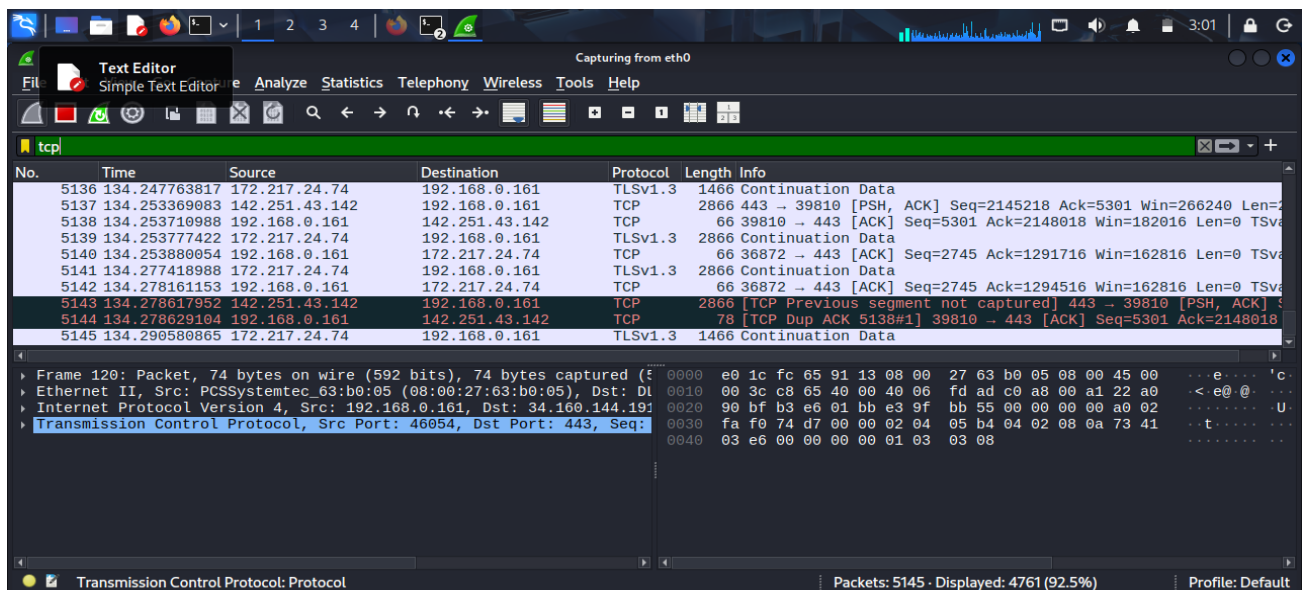


- Wireshark is launched to begin packet analysis. It provides a graphical interface to monitor all incoming and outgoing packets in a network. It supports multiple protocols and helps in detailed inspection.



Step 3: Protocol Analysis (TCP/UDP/HTTP): Wireshark supports deep inspection of protocols such as TCP, UDP, and HTTP. TCP ensures reliable communication, while UDP is faster but connectionless. HTTP packets help analyze web traffic and identify requests and responses.

- Apply filter as tcp...now you can able to see the only tcp packets. You can also try with UDP,HTTP

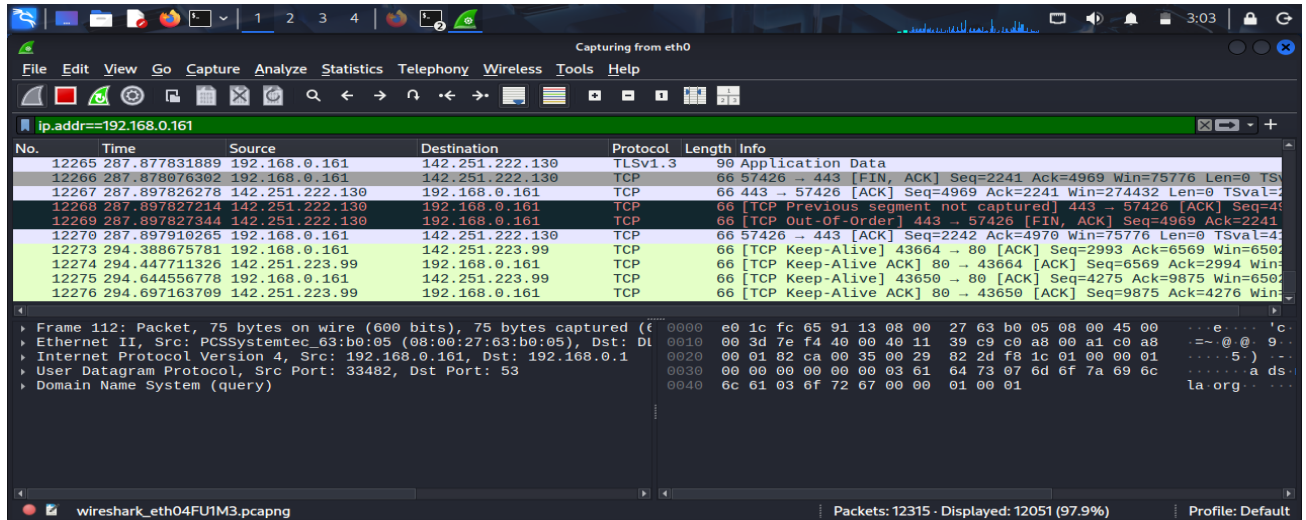


Step 4: Applying Filters

Filters are used to narrow down specific packets. For example:

- `ip.addr == 192.168.0.161` → filters packets of a specific IP

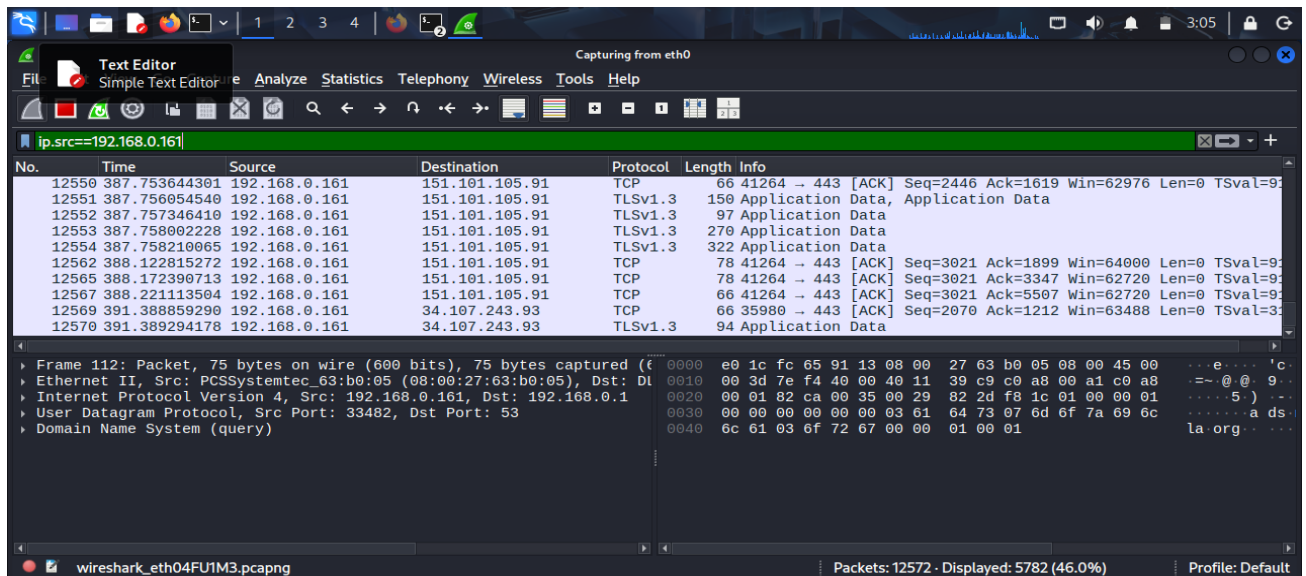
This simplifies analysis by focusing only on relevant data.



Step 5 :for particular Source address use

`Ip.src==<IP_Address>` like

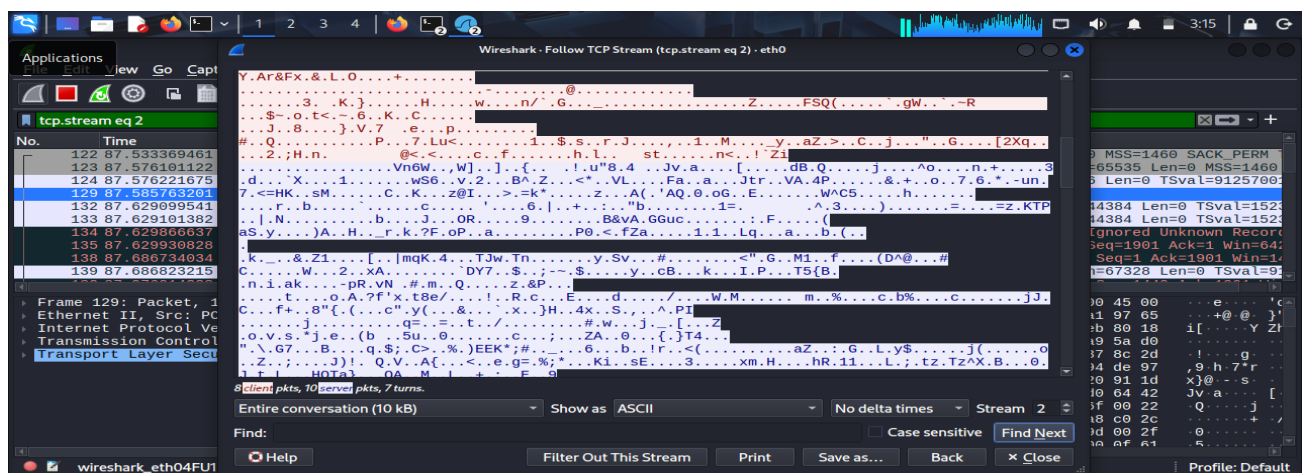
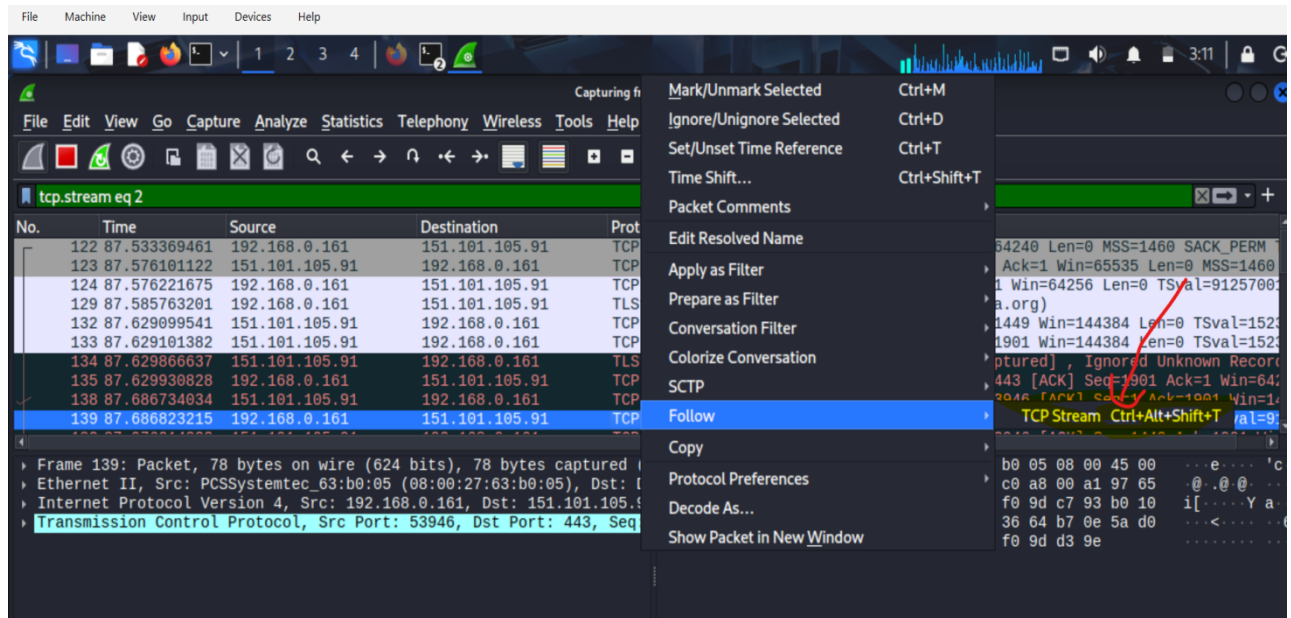
`Ip.src== 192.168.0.161`



Step 6: Following TCP Stream

The "Follow TCP Stream" feature reconstructs communication between two endpoints. It helps in analyzing complete conversations, such as HTTP requests and responses, making it useful in forensic investigations.

- Right click on any TCP packet->Follow->TCP Stream



- Observation on above tcp stream:

Complete conversation between two devices over TCP

- Red text → Data sent from one device
- Blue text → Data sent from the other device

It reconstructs the **entire communication stream**

Mixed Content (Binary + Text)

- You see:
 - Random characters → **binary data**

- Some readable text → **application data**

This means:

- It is **not plain HTTP**
- Likely:
 - Encrypted traffic (HTTPS)
 - Or application-level data

TCP Flags Visible

You can see:

[SYN], [ACK], [PSH], [FIN]

Meaning:

- **SYN** → Connection start
- **ACK** → Acknowledgement
- **PSH** → Data transfer
- **FIN** → Connection close

So this stream shows:

✓ Full TCP session lifecycle

Step 7: Identifying Suspicious Traffic

Open kali linux terminal

Type **nmap -sS <target_IP>**

Eg: **nmap -sS 192.168.0.156** (use your windows ip)

Open wireshark

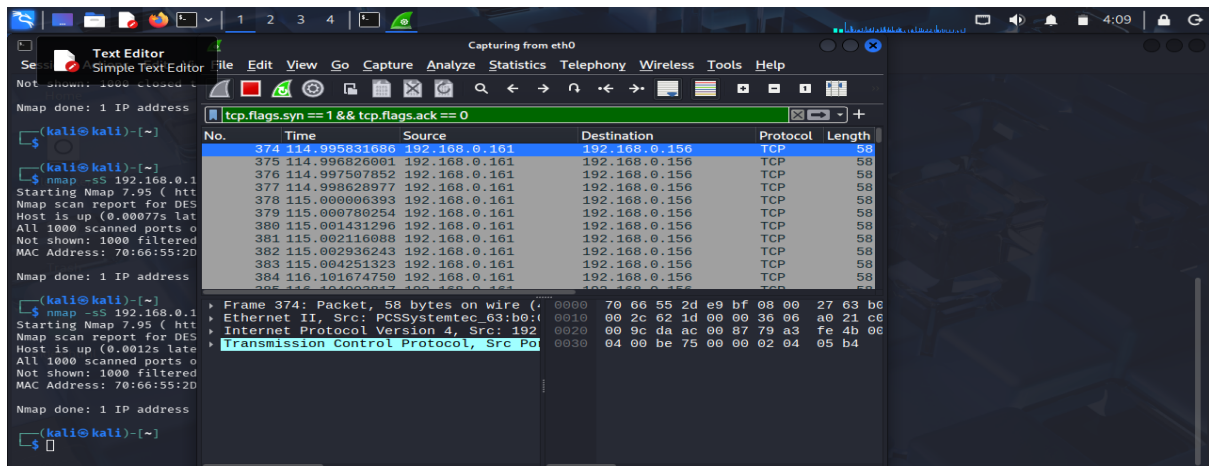
Use this filter:

`tcp.flags.syn == 1 && tcp.flags.ack == 0`

Now you will observe:

- Multiple SYN packets
- Different ports
- Incomplete connections
- Source address is Kali linux and Destination Address is Windows

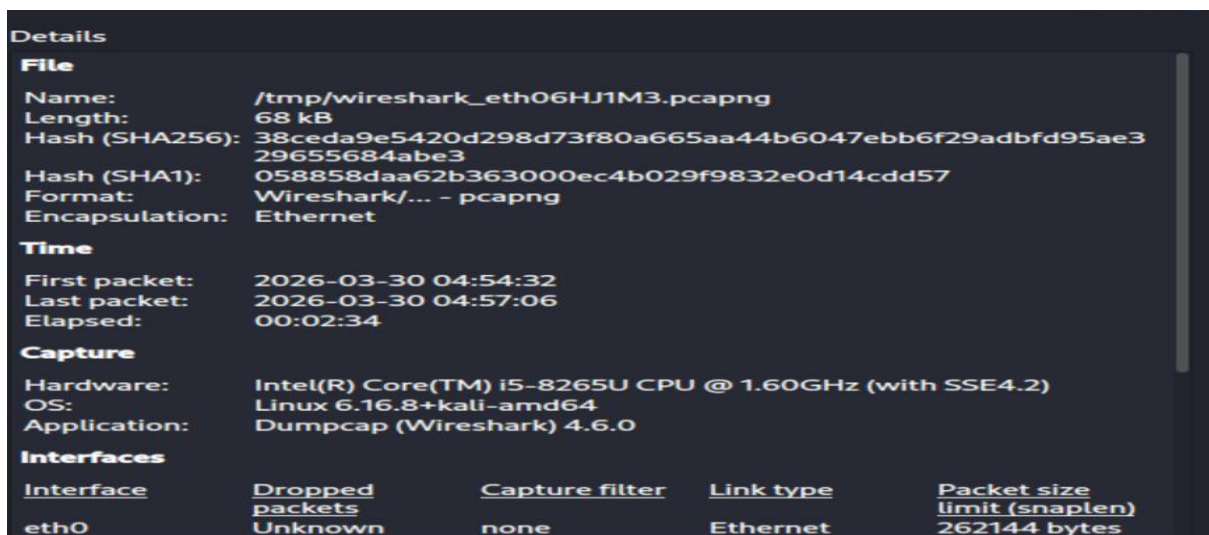
This is **port scanning behavior (suspicious)**. Any way our system firewall stops this.



Step 8:

In wireshark

Go to Statistics->Capture File Properties



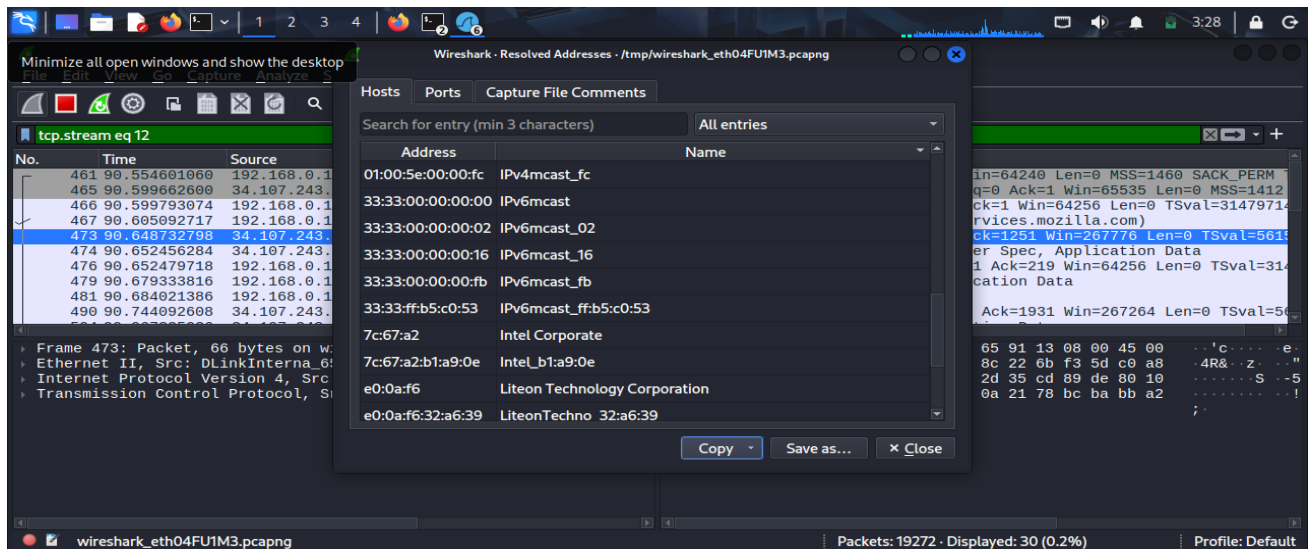
- It provides a quick overview of network traffic and helps detect suspicious activities without analyzing each packet individually.

Step 9:

In wireshark **Go To**

Statistics->Resolved Address

Resolve Address in Wireshark converts numerical network addresses (like IP or MAC) into **human-readable names** such as domain names or device names.



Step 10: Go to

Statistics-> Protocol Hierarchy Statistics

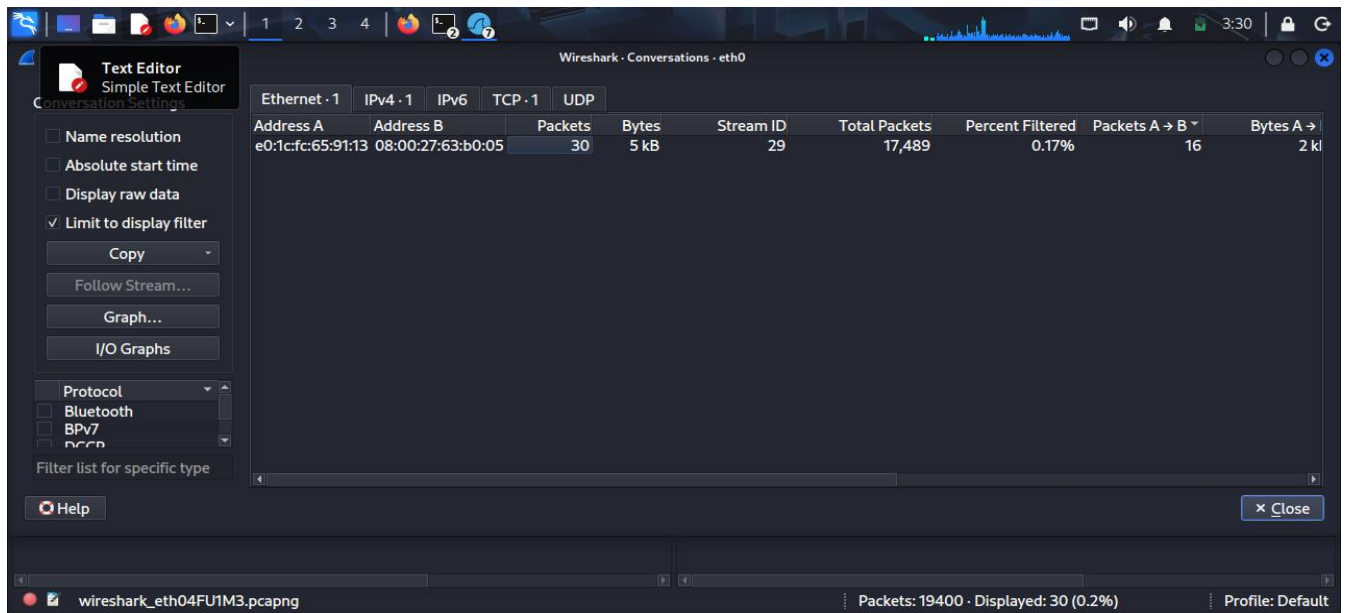
Protocol Hierarchy Statistics shows the breakdown of all network protocols in the captured traffic in a hierarchical (layer-wise) format along with their percentage and packet count.



Step 11: Go to

Statistics->Conversations

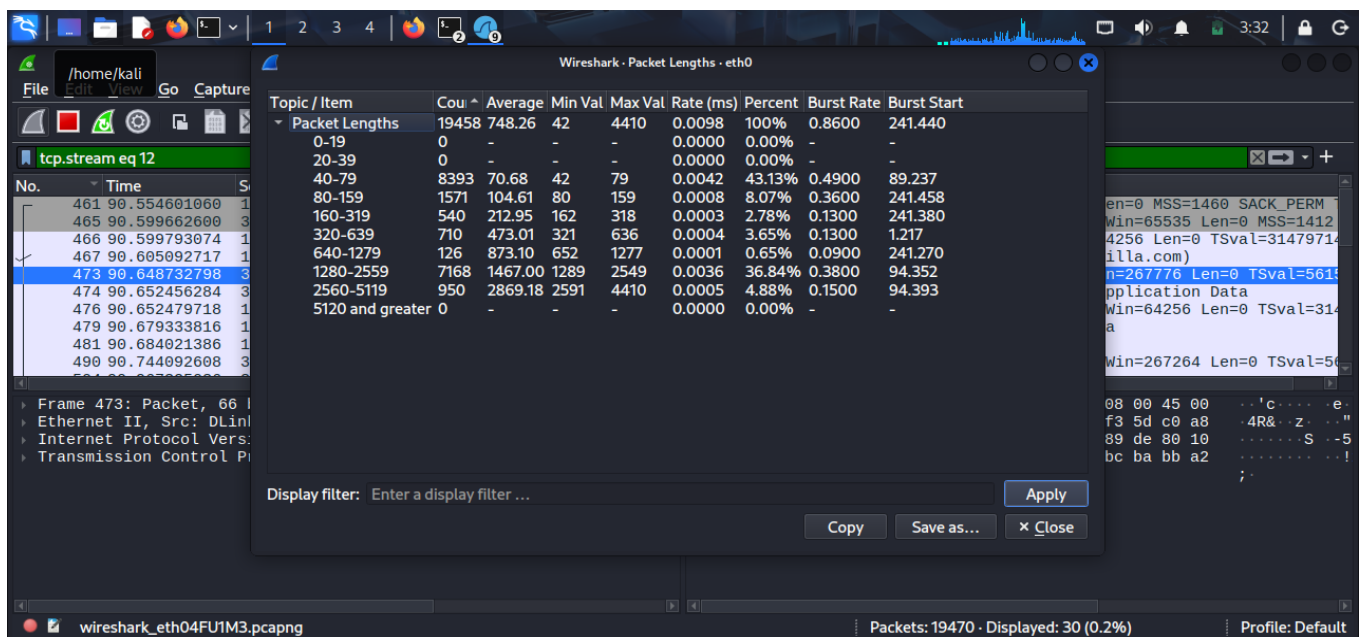
Conversations in Wireshark shows communication between two endpoints (devices), including source and destination addresses, number of packets, and data transferred.



Step 12:

Statistics-> Packet Length

Packet Length refers to the **size of a network packet** measured in bytes, as captured by Wireshark.



Step 13:

Statistics-> End Points

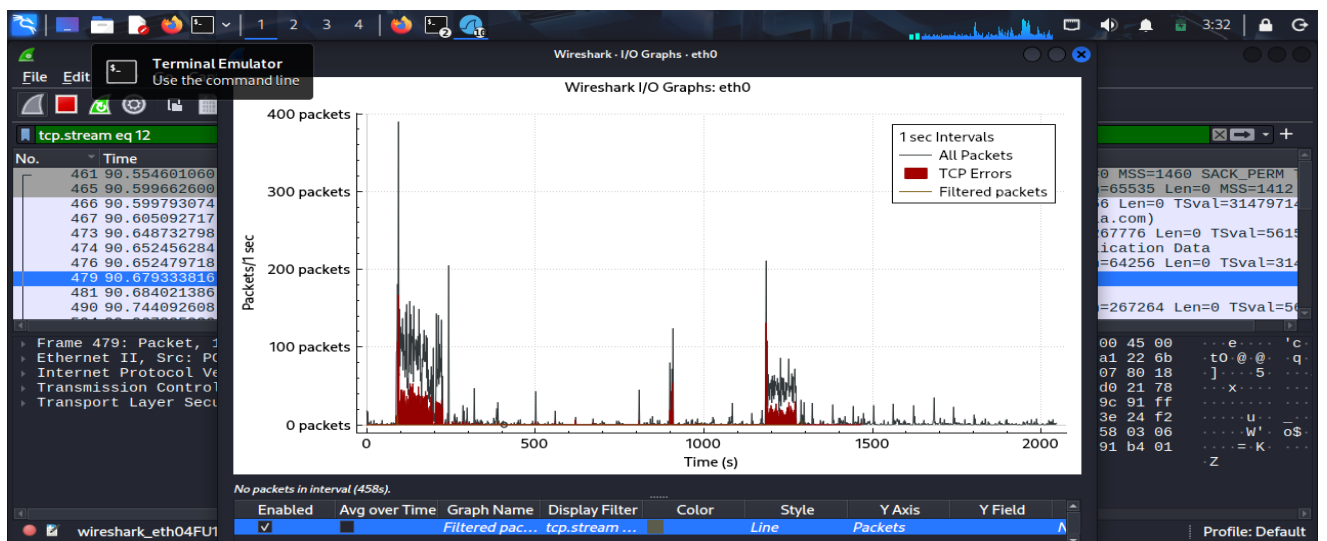
Endpoints in Wireshark represent individual network devices (hosts) involved in communication. It shows each device separately along with the number of packets sent/received and data usage.



Step 14:

Statistics-> I/O Graphs

I/O Graphs (Input/Output Graphs) in Wireshark are used to display network traffic over time in the form of a graph, showing the number of packets or bytes captured per second.



Result

Network packets were successfully captured and analyzed using Wireshark. Different protocols and communication patterns were observed and studied.

Conclusion

Wireshark is an effective tool for network forensics. It helps in capturing and analyzing network traffic, identifying suspicious activities, and understanding communication between systems. It plays a vital role in cybersecurity and incident investigation.

Viva Points

- What is packet sniffing?
- Difference between TCP and UDP
- What is a protocol analyzer?
- What is a filter in Wireshark?

CYBER SECURITY LAB EXPERIMENT

Title: Privacy Audit of Popular Apps (Desktop **WhatsApp**) and Websites (**Facebook**) & Data Breach Case Study Analysis

Aim

To analyze privacy, network behavior, and security risks of applications and websites using Kali Linux tools.

Tools Required

- **Wireshark**
- **Burp Suite**
- Kali Linux (VirtualBox)
- Desktop WhatsApp (Nativefier)
- Web Browser (Firefox)

Step 1: Install Node.js & npm

```
sudo apt update
```

```

kali@kali:~$ sudo systemctl status snmpd
Loaded: loaded (/usr/lib/systemd/system/snmpd.service; enabled; preset:~
Active: inactive (dead) since Wed 2026-04-31 00:57:45 EDT; 1min 0s ago
Duration: 0.851s
Invocation: 0267453dd20aa51894c1150bb8c9075
TriggeredBy: ● snmpd.socket
Docs: man:snmpd.service
Process: 9697 ExecStart=/usr/lib/snmpd/snmpd (code=exited, status=42)
Main PID: 9697 (code=exited, status=42)
Mem peak: 18.2M
CPU: 1.575s

Apr 01 00:57:39 kali snmpd[9697]: snmpmgr.go:1659: performing periodic snap
Apr 01 00:57:39 kali snmpd[9697]: snmpmgr.go:1669: cannot clean store downlo
Apr 01 00:57:42 kali snmpd[9697]: standby.go:101: standby conditions met, in
Apr 01 00:57:42 kali snmpd[9697]: daemon.go:556: gracefully waiting for runn
Apr 01 00:57:42 kali snmpd[9697]: daemon.go:558: done waiting for running h
Apr 01 00:57:43 kali snmpd[9697]: standby.go:121: standby monitoring stop re
Apr 01 00:57:45 kali snmpd[9697]: overlord.go:543: Released state lock file
Apr 01 00:57:45 kali snmpd[9697]: daemon stop requested to wait for socket >
Apr 01 00:57:45 kali systemd[1]: snmpd.service: Deactivated successfully.
Apr 01 00:57:45 kali systemd[1]: snmpd.service: Consumed 1.575s CPU time ove>

kali@kali:~$ sudo snmp install whatsapp-for-linux
error: snmp 'whatsapp-for-linux' not found

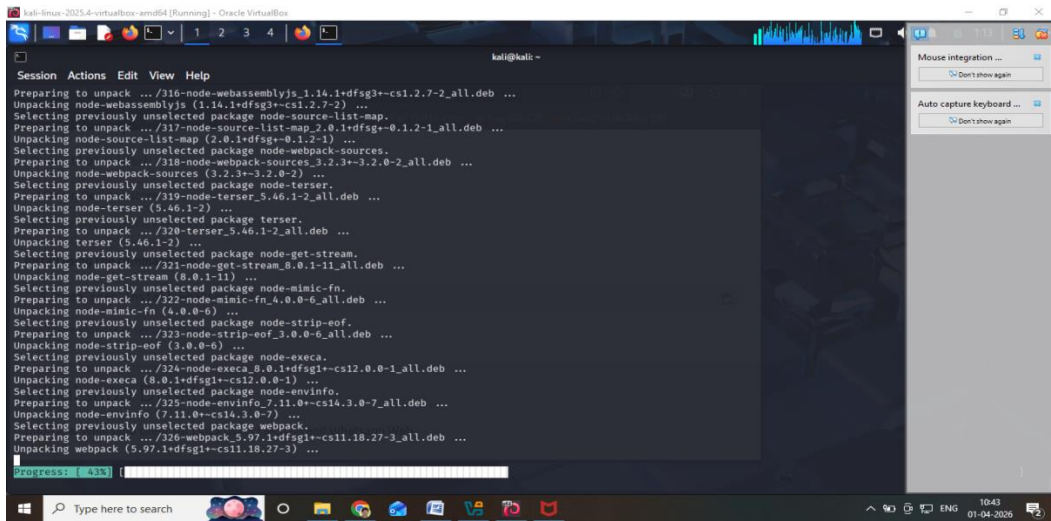
kali@kali:~$ sudo apt update
Hit:1 http://http.kali.org/kali kali-rolling InRelease
1614 packages can be upgraded. Run 'apt list --upgradable' to see them.

kali@kali:~$

```

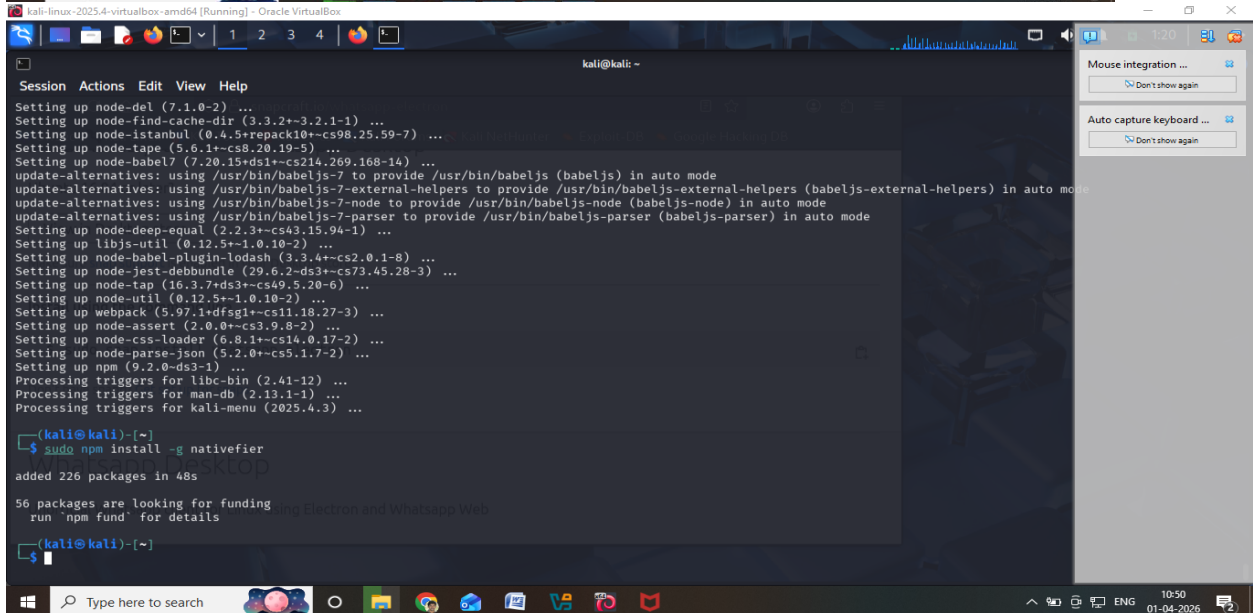
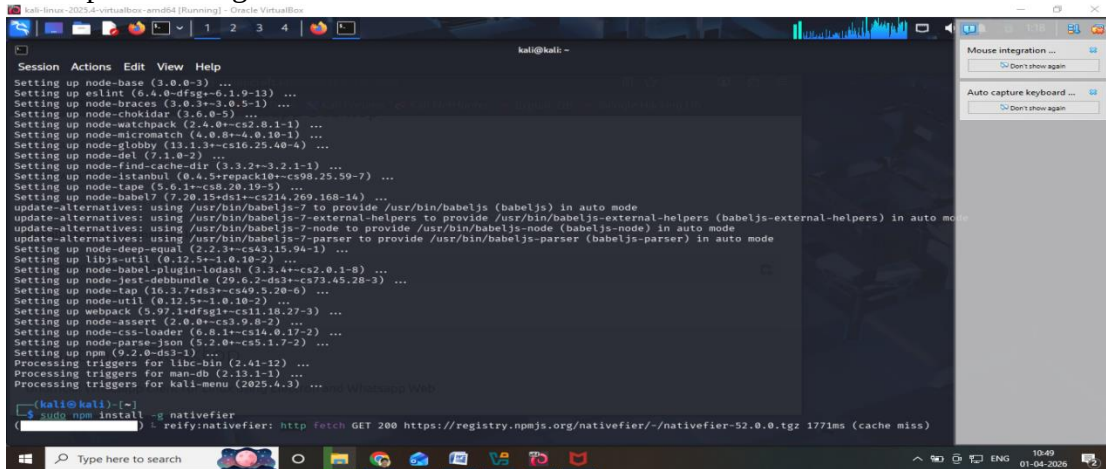
```
sudo apt install nodejs npm -y
```

[illegible]



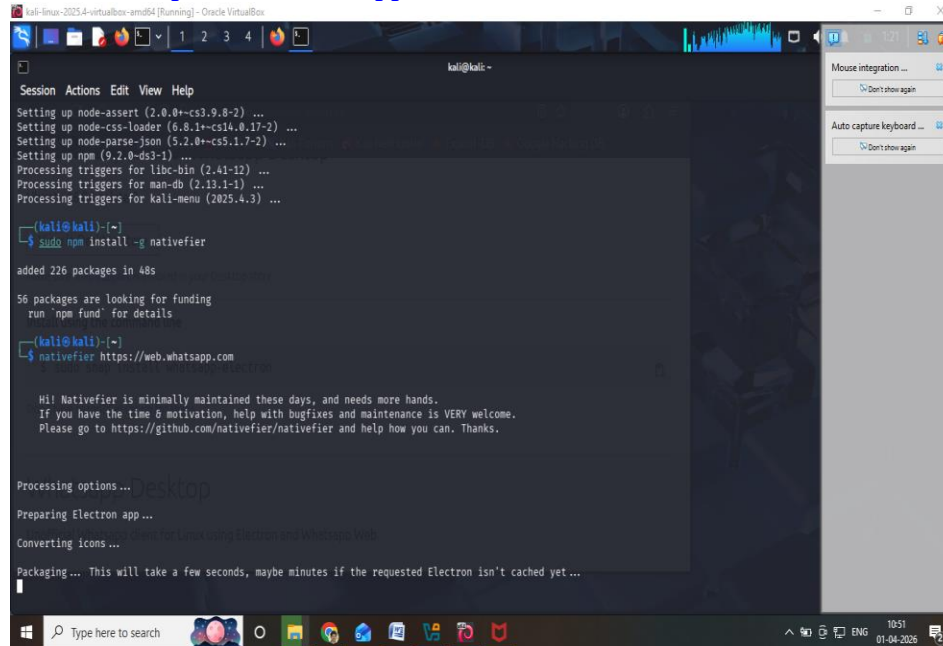
Step 2: Install Nativiefier

sudo npm install -g nativiefier



Step 3: Create WhatsApp Desktop App

nativefier <https://web.whatsapp.com>



```
kali@kali:~$ sudo npm install -g nativefier
added 226 packages in 48s
56 packages are looking for funding
  run `npm fund` for details

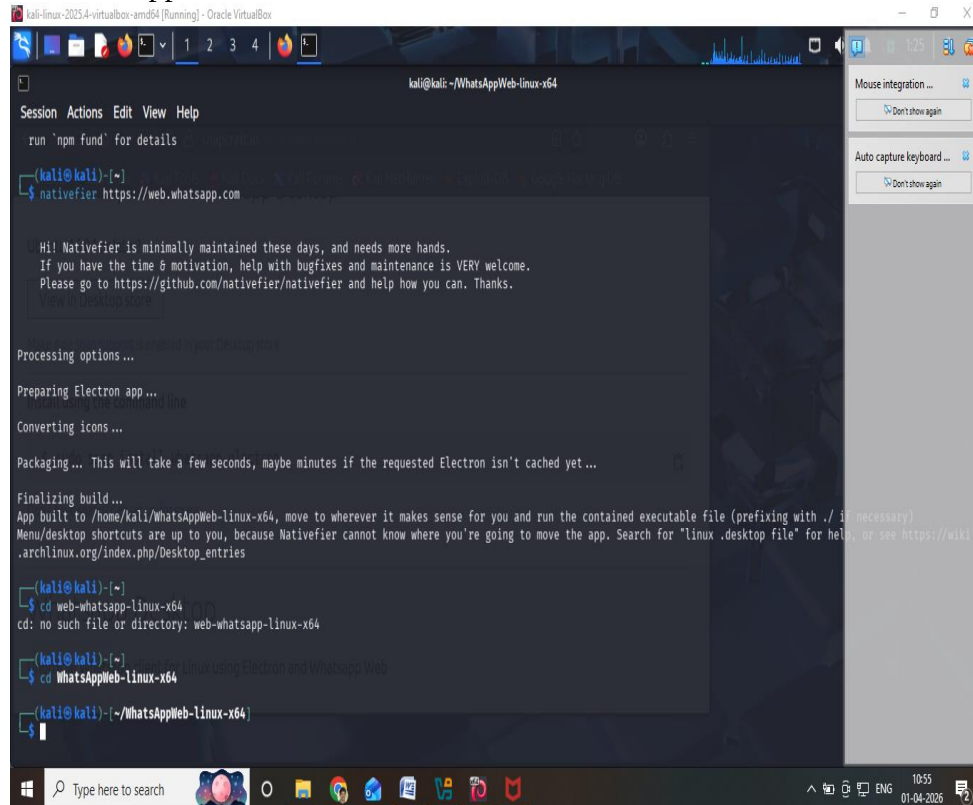
(kali@kali)~$ nativefier https://web.whatsapp.com
Processing options... Desktop
Preparing Electron app...
Converting icons...
Packaging... This will take a few seconds, maybe minutes if the requested Electron isn't cached yet...
```

It will create a folder like:

web-whatsapp-linux-x64

Step 4: Run WhatsApp

cd WhatsAppWeb-linux-x64



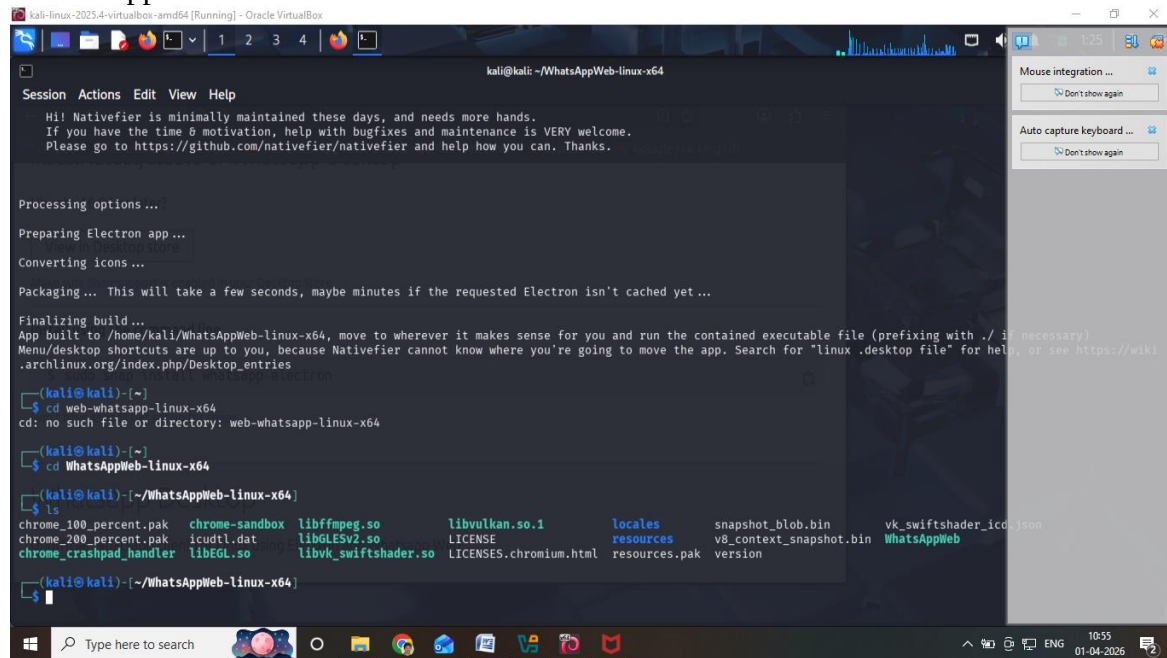
```
kali@kali:~/WhatsAppWeb-linux-x64$ nativefier https://web.whatsapp.com
Processing options...
Preparing Electron app...
Converting icons...
Packaging... This will take a few seconds, maybe minutes if the requested Electron isn't cached yet...

Finalizing build...
App built to /home/kali/WhatsAppWeb-linux-x64, move to wherever it makes sense for you and run the contained executable file (prefixed with ./ if necessary)
Menu/desktop shortcuts are up to you, because Nativefier cannot know where you're going to move the app. Search for "linux .desktop file" for help, or see https://wiki.archlinux.org/index.php/Desktop_entries

(kali@kali)~$ cd web-whatsapp-linux-x64
cd: no such file or directory: web-whatsapp-linux-x64

(kali@kali)~$ cd WhatsAppWeb-linux-x64
(kali@kali)~/WhatsAppWeb-linux-x64$
```

./WhatsAppWeb



```
kali@kali: ~/WhatsAppWeb-linux-x64
Session Actions Edit View Help
Hi! Nativefier is minimally maintained these days, and needs more hands.
If you have the time & motivation, help with bugfixes and maintenance is VERY welcome.
Please go to https://github.com/nativefier/nativefier and help how you can. Thanks.

Processing options...
Preparing Electron app...
Converting icons...
Packaging... This will take a few seconds, maybe minutes if the requested Electron isn't cached yet...
Finalizing build...
App built to /home/kali/WhatsAppWeb-linux-x64, move to wherever it makes sense for you and run the contained executable file (prefixing with ./ if necessary)
Menu/desktop shortcuts are up to you, because Nativefier cannot know where you're going to move the app. Search for "linux .desktop file" for help, or see https://wiki.archlinux.org/index.php/Desktop_entries

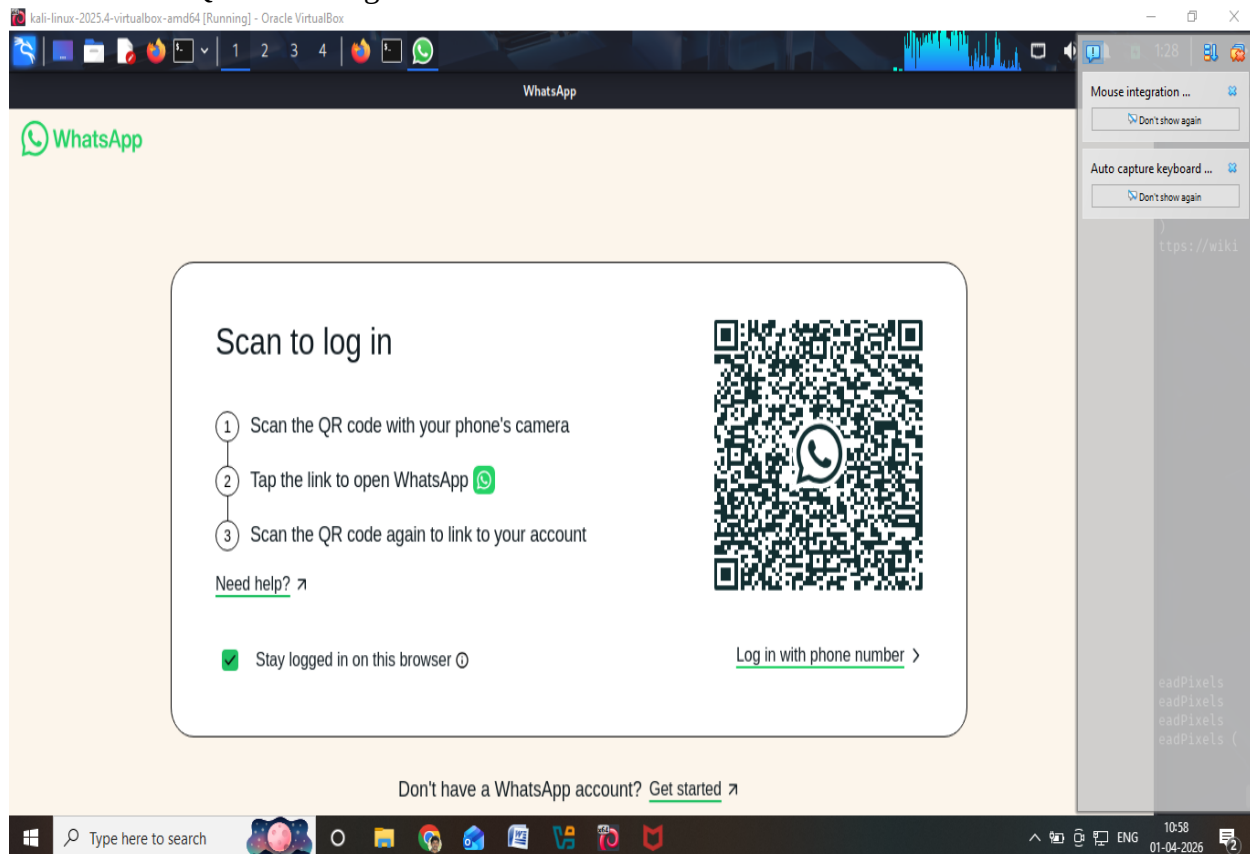
(kali@kali)~$ cd web-whatsapp-linux-x64
cd: no such file or directory: web-whatsapp-linux-x64

(kali@kali)~$ cd WhatsAppWeb-linux-x64

(kali@kali)~/WhatsAppWeb-linux-x64$ ls
chrome_100_percent.pak  chrome-sandbox  libffmpeg.so  libvulkan.so.1  locales  snapshot_blob.bin  vk_swiftshader_icd.json
chrome_200_percent.pak  icudtl.dat      libGLESv2.so  LICENSE         resources  v8_context_snapshot.bin  WhatsAppWeb
chrome_crashpad_handler  libEGL.so       libvk_swiftshader.so  LICENSES.chromium.html  resources.pak  version
```

✓ Step 5: Login

- Scan QR code using mobile



- Done

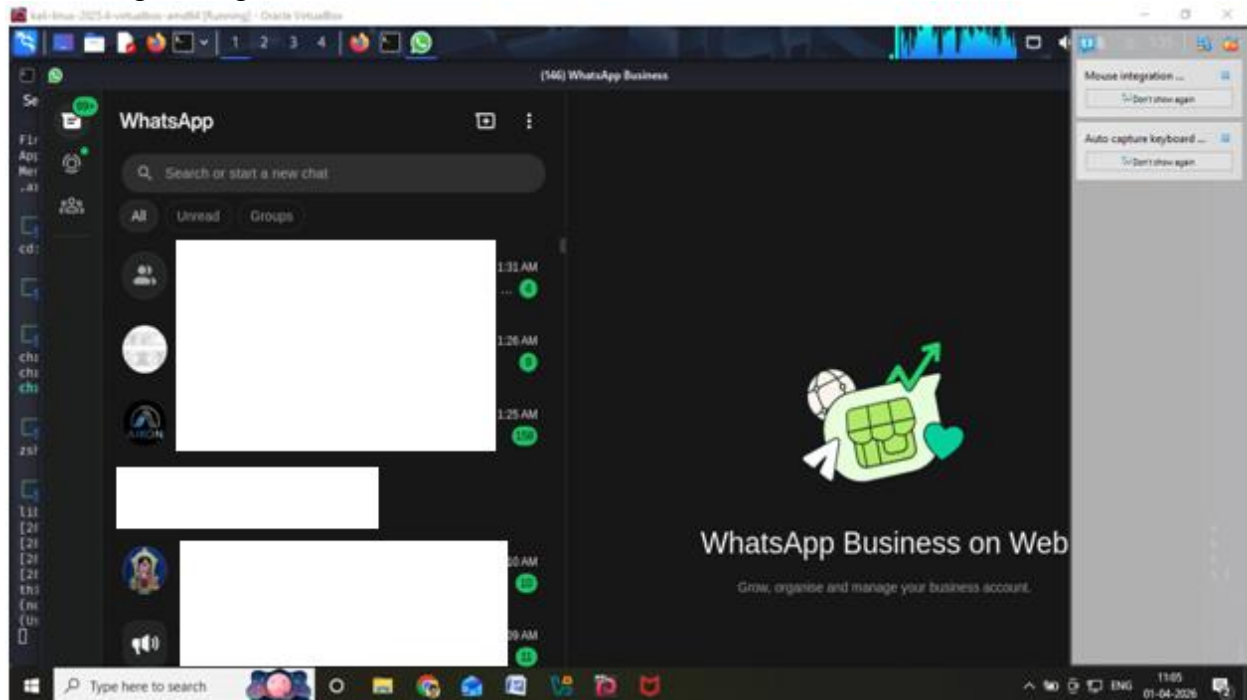
PART 1: Privacy Audit of Desktop WhatsApp

Step 1: Launch WhatsApp Desktop

cd WhatsAppWeb-linux-x64

./WhatsAppWeb

- QR Code appears
- Login using mobile → Linked Devices

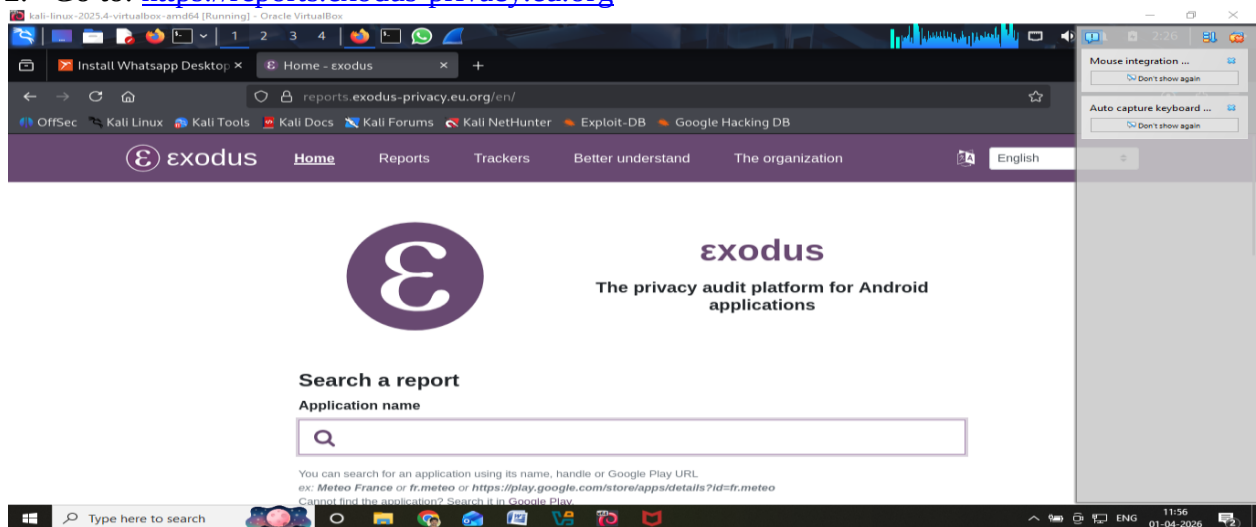


Step 1: Install WhatsApp APK (Optional in Emulator)

- Download APK from trusted source or use your mobile

Step 2: Analyze Trackers using Exodus Privacy

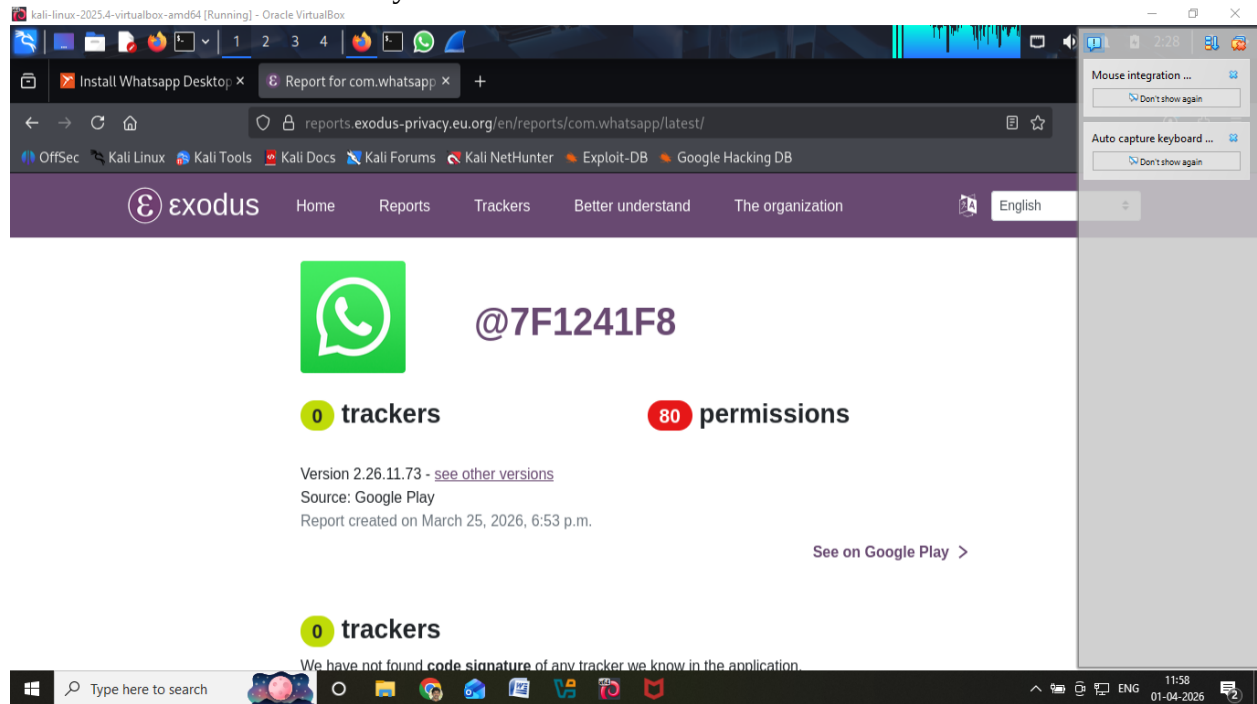
1. Open browser in Kali
2. Go to: <https://reports.exodus-privacy.eu.org>



3. Search for **WhatsApp**

Observation:

- Detects trackers like:
 - Google Analytics
 - Crashlytics
 - Facebook Analytics

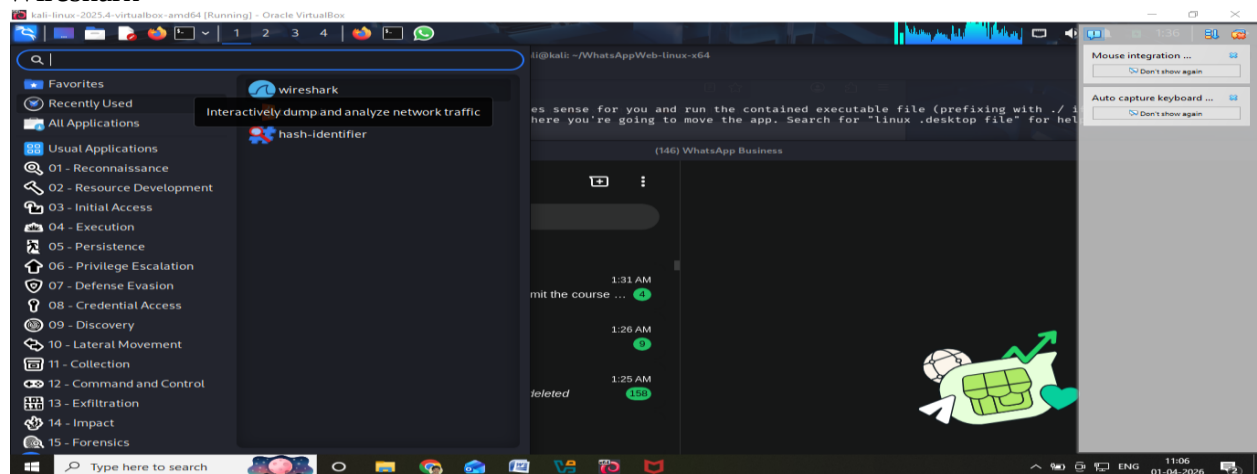


Record in lab:

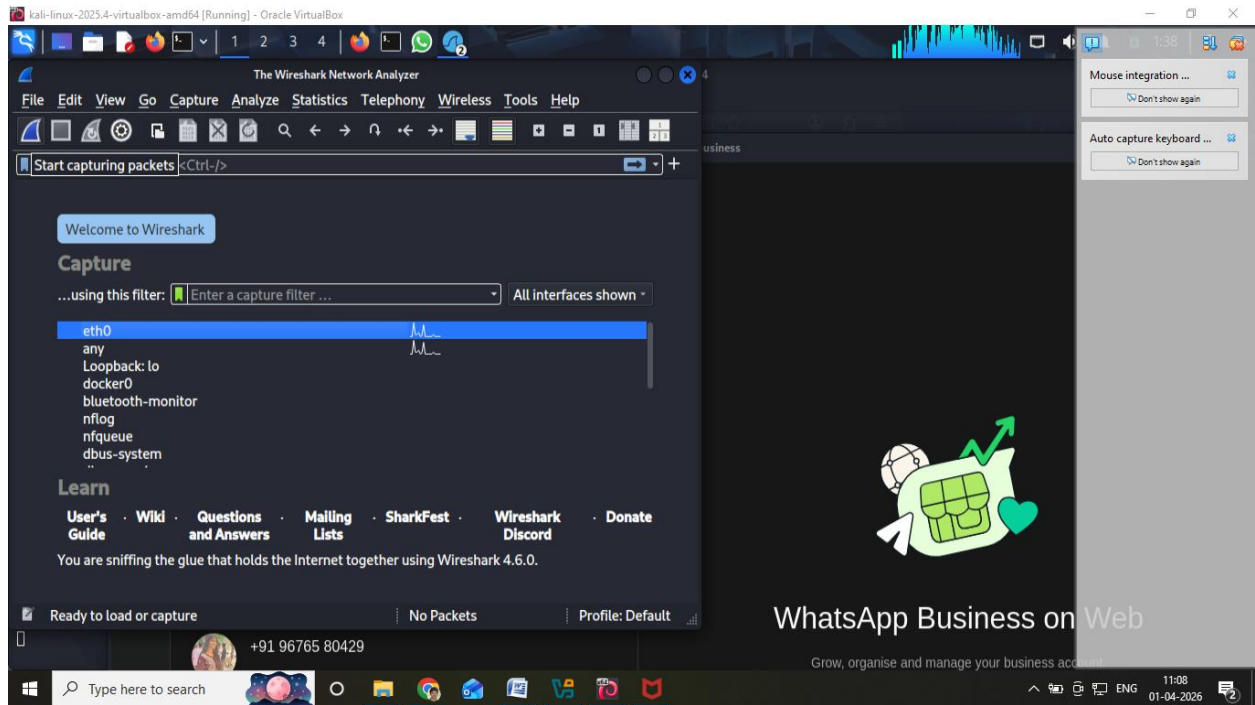
- Number of trackers
- Permissions used

Step 2: Start Wireshark

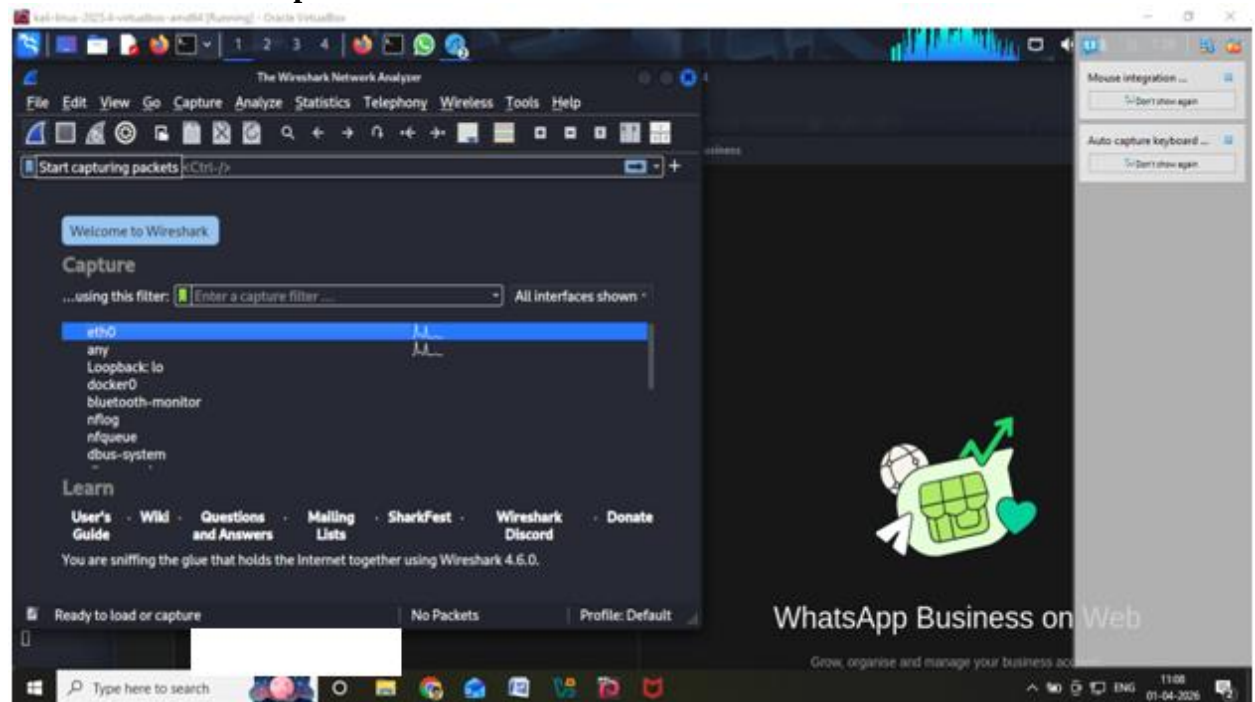
Wireshark



- Select interface (eth0 / wlan0)



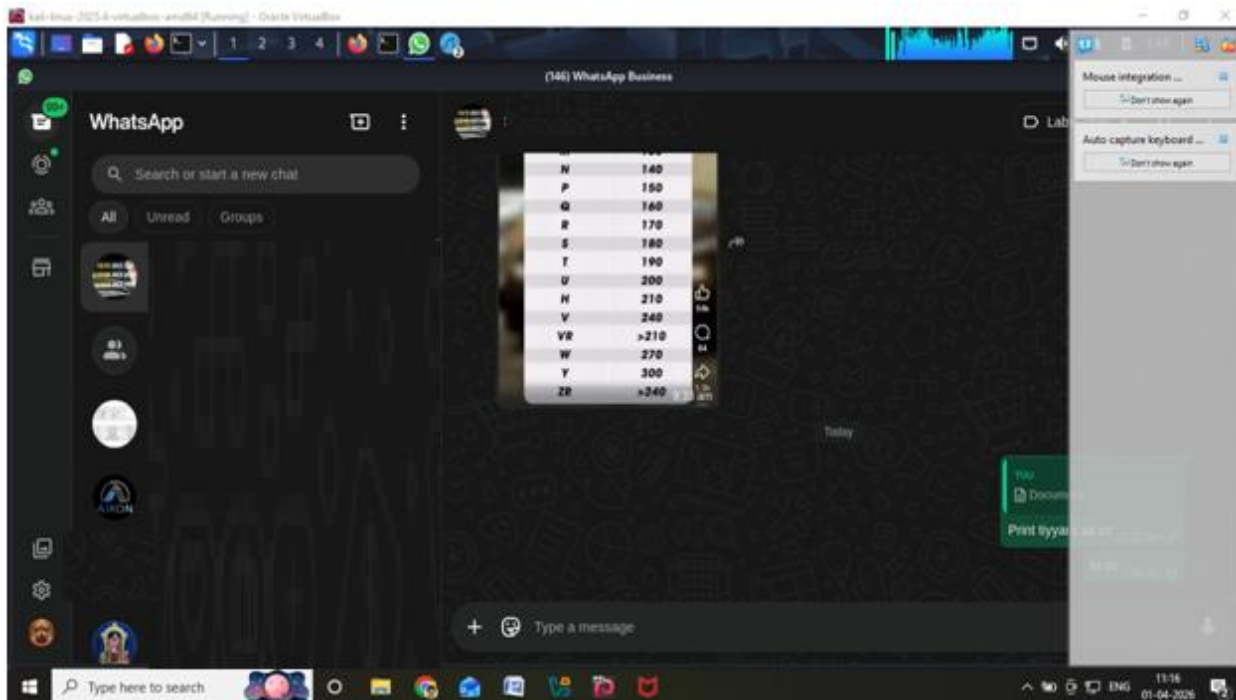
-
- Click Start Capture



Step 3: Generate Traffic

In WhatsApp Desktop:

- Send messages
- Send images
- Open chats

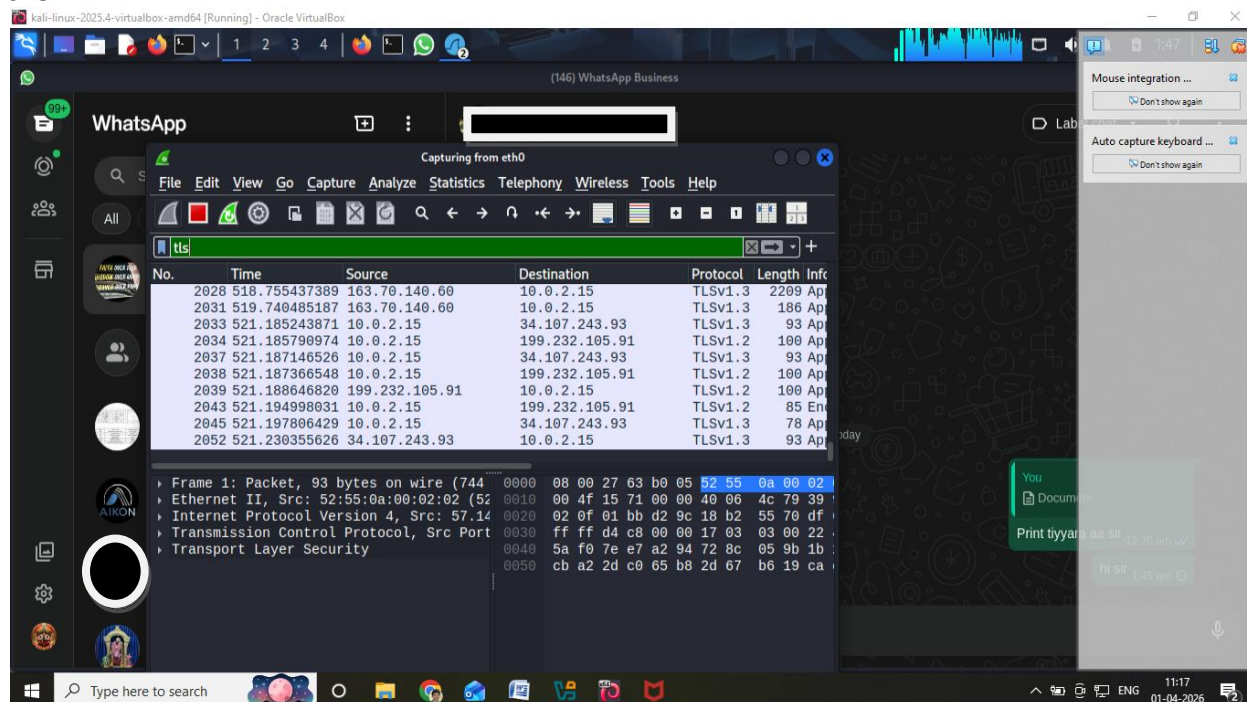


✓Step 4: Apply Filters

Use filters:

tls

dns

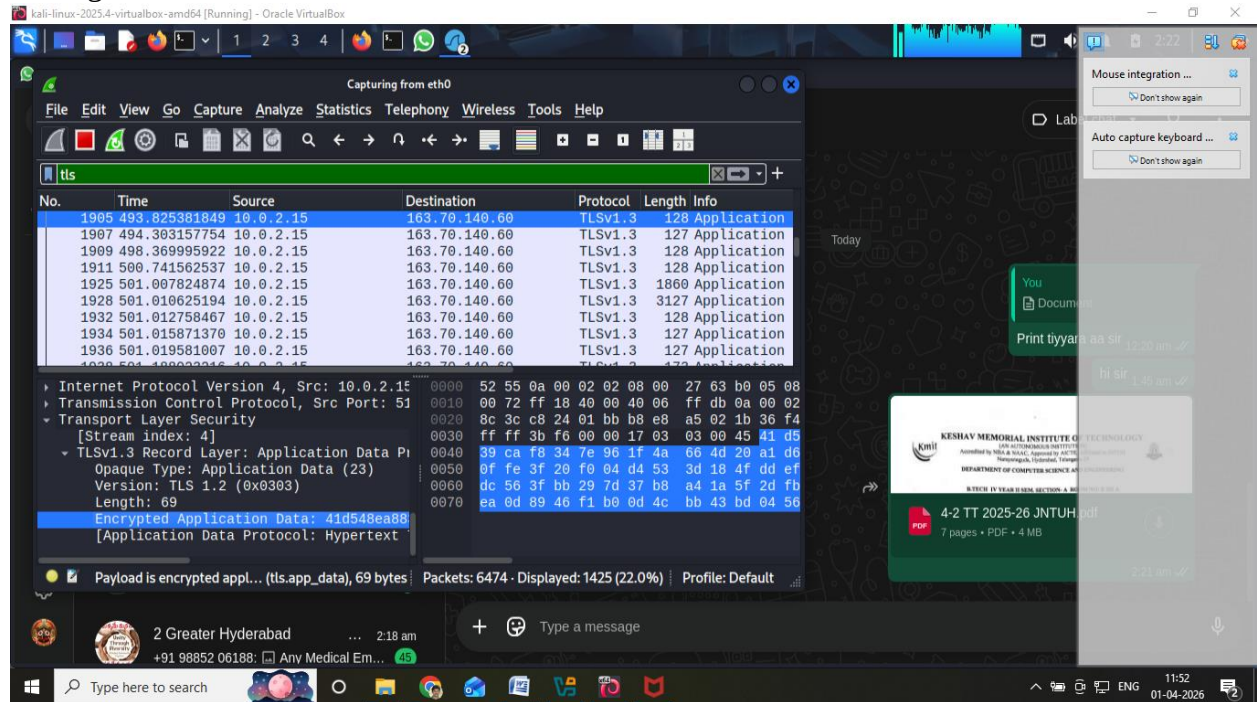


Observations

- Encrypted traffic (TLS packets)
- Domains:

- whatsapp.net
- facebook.com

Message content NOT visible



Findings

End-to-End Encryption

Secure communication

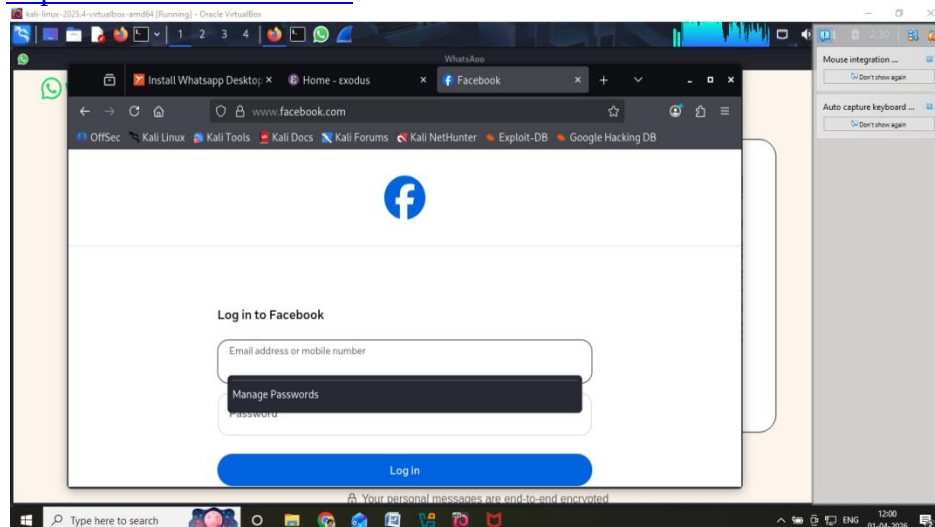
Metadata visible:

- IP address
- Time
- Server communication

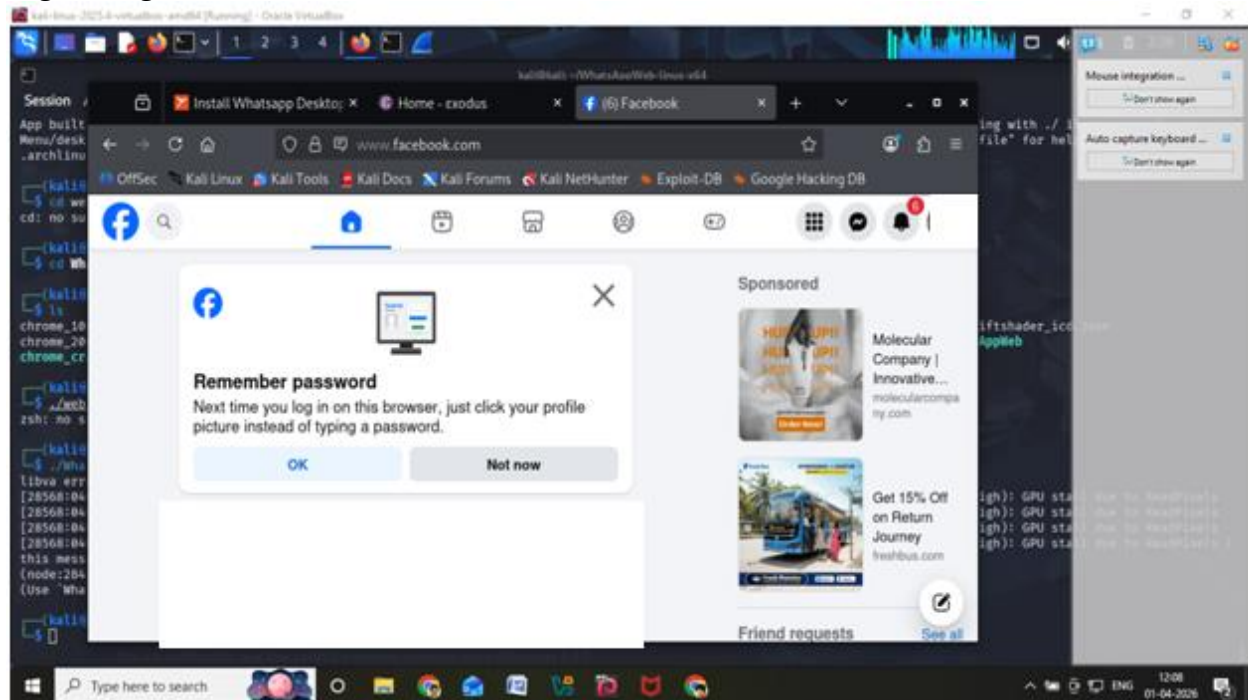
PART 2: Privacy Audit of Facebook Website

Step 1: Open Facebook

<https://www.facebook.com>

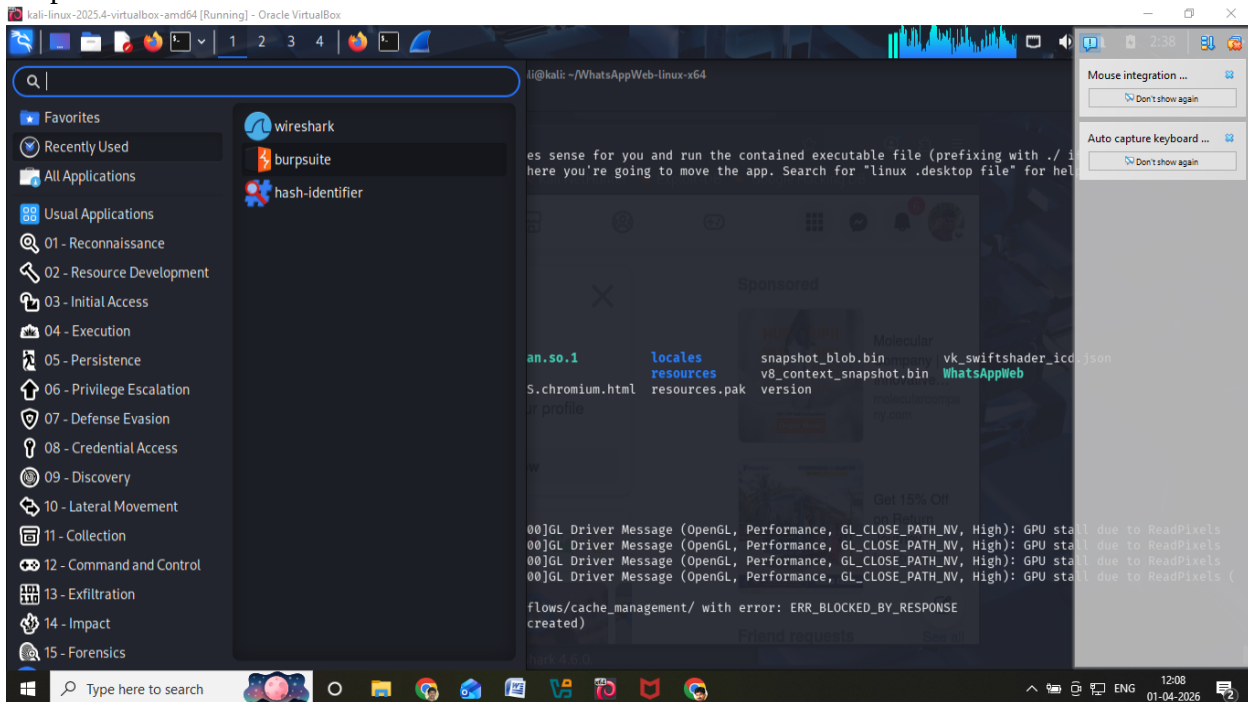


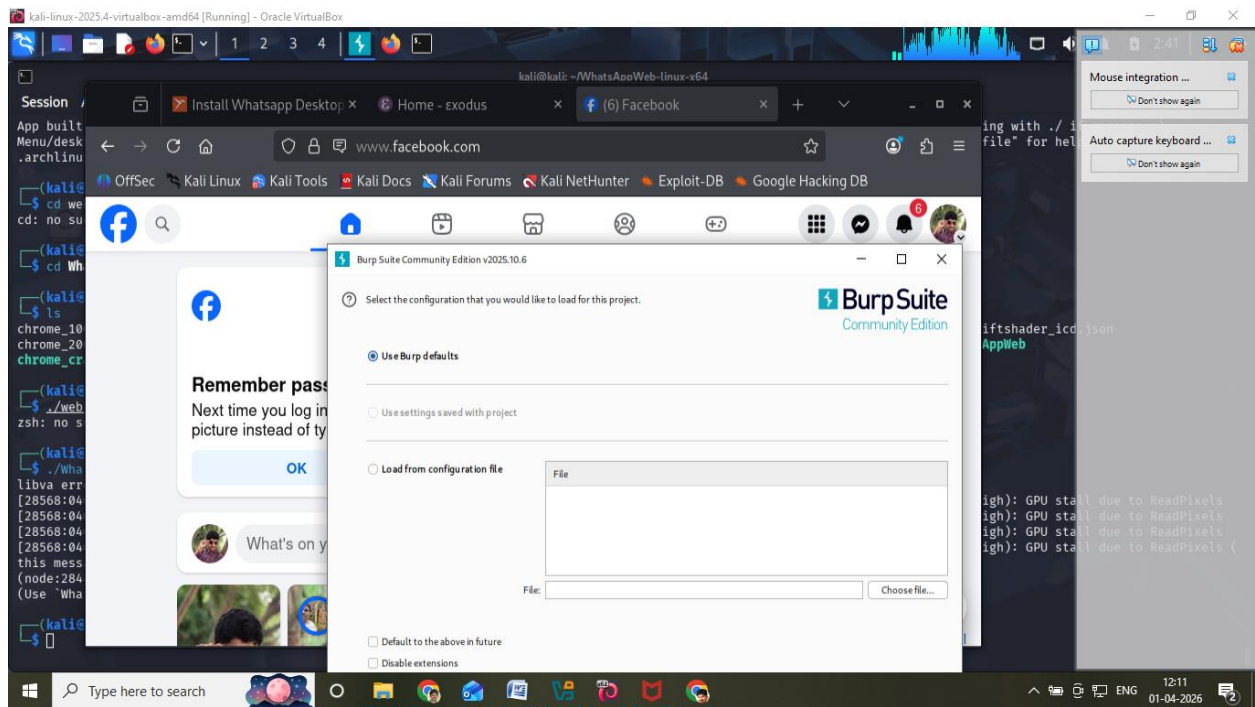
Login using test account



Step 2: Start Burp Suite

Burpsuite



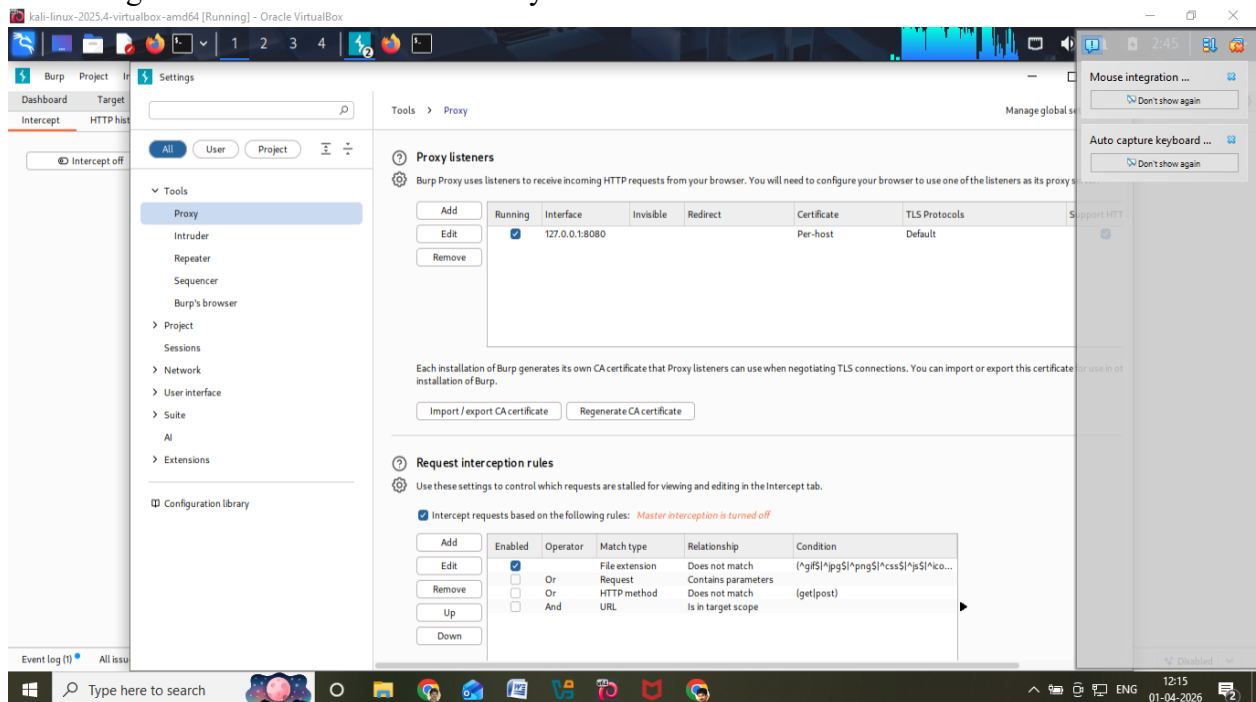


✓ Step 3: Configure Proxy

- IP: 127.0.0.1
- Port: 8080

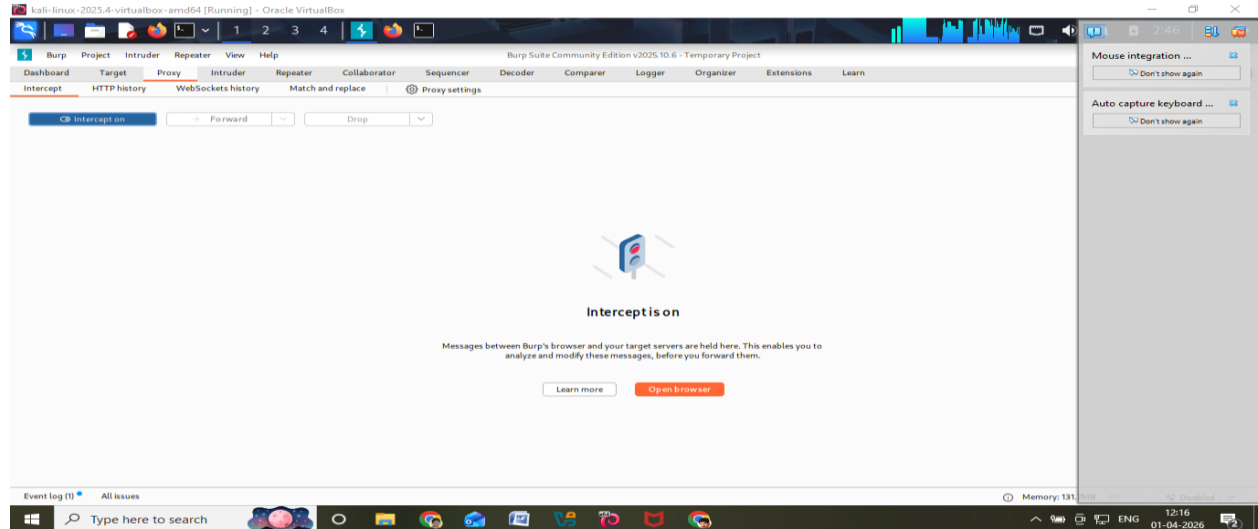
In browser:

- Settings → Network → Manual Proxy



✓ Step 4: Intercept Traffic

- Turn **Intercept ON**
- Reload Facebook



Observe:

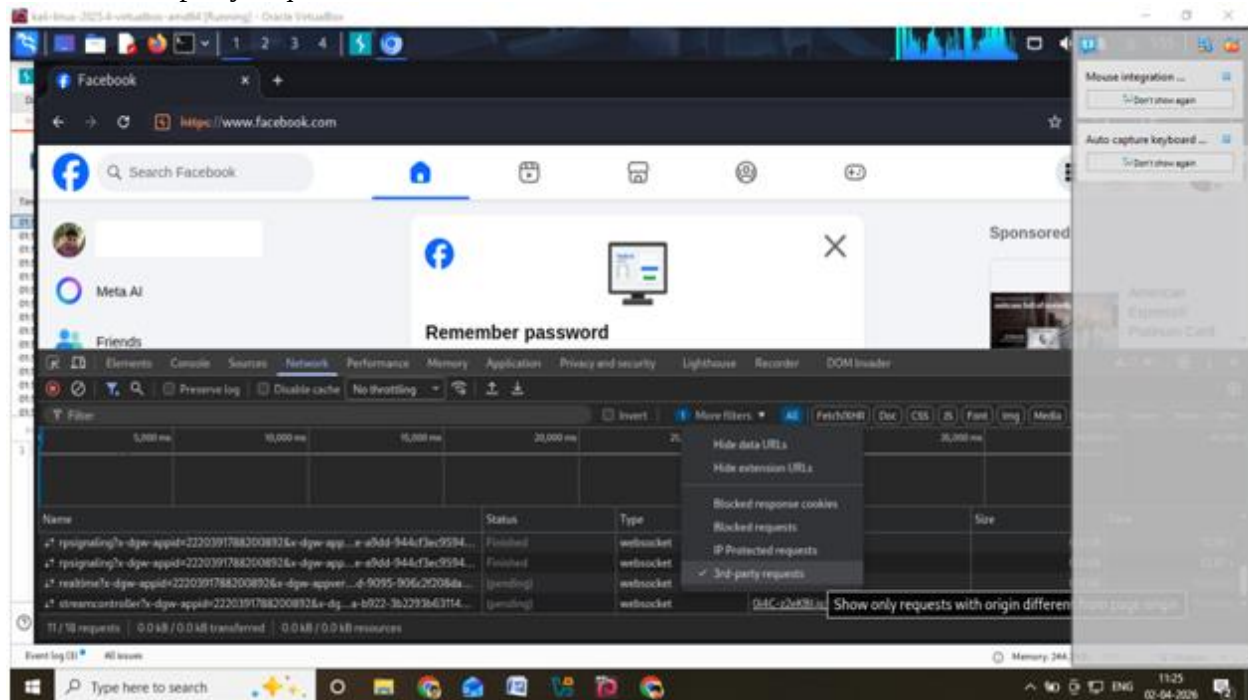
- Cookies (datr, fr)
- Session IDs
- Headers

Step 5: Analyze Tracking

Press **F12** → **Network** tab

Look for:

- Ads
- Analytics scripts
- Third-party requests



Findings

Heavy tracking

Uses cookies for personalization

Third-party integrations

PART 3: Data Breach Case Study

Case Study 1: Facebook Data Breach

- 533 million users affected
- Data leaked:
 - Phone numbers
 - Emails

Cause:

- API vulnerability

Case Study 2: WhatsApp Pegasus Attack

- Spyware attack via missed call
- Remote device access

Step 3: Check Email Breach

Go to:

□ <https://haveibeenpwned.com>

Enter email → check breach status

Analysis Points

- Type of attack
- Vulnerability
- Data exposed
- Prevention

Observations

- WhatsApp uses strong encryption
- Facebook performs extensive tracking
- Data breaches occur due to vulnerabilities

Result

Privacy audit was successfully performed using Kali Linux tools. WhatsApp ensures secure communication through encryption, while Facebook shows extensive tracking behavior. Data breach analysis highlighted risks of poor security practices.

Experiment No. 3: Testing IoT Device Security (Default Passwords & Open Ports)

Aim

To analyze security weaknesses in a simulated IoT device by identifying:

- Open ports
- Running services
- Service version disclosure
- Default/weak authentication
- Unencrypted communication

Tools Used

- Windows 10 (Host System)
- Docker Desktop
- OWASP Juice Shop
- Kali Linux
- Nmap

Experimental Setup

- A vulnerable web application (Juice Shop) was deployed using Docker on Windows.
- Kali Linux was used as the attacker machine.
- The Windows system IP address was scanned from Kali Linux to simulate IoT network testing.

Attack Scenario Description

In this experiment:

- Kali Linux acted as the attacker machine.
- Windows system hosting the Docker container acted as the IoT device.
- The vulnerable web application (Juice Shop) simulated the IoT dashboard.
- Network scanning was performed from Kali to the Windows IP address over LAN.

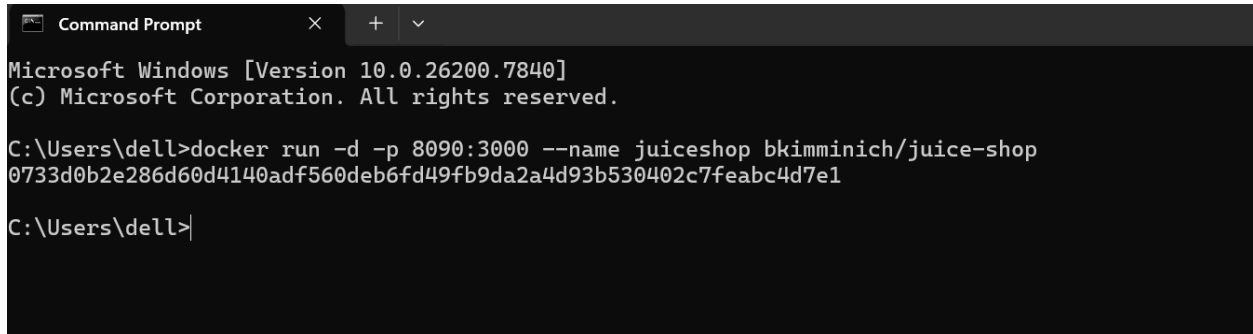
This simulates a real-world scenario where an attacker scans an IoT device connected to the same network.

Procedure

Step 1: Deploy Vulnerable IoT Simulation

The following command was used to run Juice Shop in Docker:

```
docker run -d -p 8090:3000 --name juiceshop bkimminich/juice-shop
```

A screenshot of a Windows Command Prompt window. The title bar reads "Command Prompt" with standard window controls. The text inside shows the Windows version "Microsoft Windows [Version 10.0.26200.7840]" and copyright information "(c) Microsoft Corporation. All rights reserved.". The user is at the C:\Users\dell directory. They have entered the command "docker run -d -p 8090:3000 --name juiceshop bkimminich/juice-shop" and the output is a long alphanumeric string: "0733d0b2e286d60d4140adf560deb6fd49fb9da2a4d93b530402c7feabc4d7e1". The prompt is now waiting for the next command at "C:\Users\dell>".

```
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\dell>docker run -d -p 8090:3000 --name juiceshop bkimminich/juice-shop
0733d0b2e286d60d4140adf560deb6fd49fb9da2a4d93b530402c7feabc4d7e1

C:\Users\dell>
```

Figure 1: Deployment of simulated IoT device using Docker

The application was accessed at:

<http://localhost:8090>

Step 2: Identify Host IP Address

On Windows: ipconfig


```

C:\Users\dell>ipconfig

Windows IP Configuration

Unknown adapter Local Area Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::5cd0:b754:6025:656e%10
    Autoconfiguration IPv4 Address. . . : 169.254.54.155
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : dlink
    Link-local IPv6 Address . . . . . : fe80::a080:af7:5ca0:94a7%17
    IPv4 Address. . . . . : 192.168.0.148
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.0.1

```

Figure 2: Host system IP address identification

Step 3: Perform Network Scan from Kali Linux

From Kali Linux terminal:

```
nmap -sV 192.168.0.148
```

This scan performed:

- Port scanning
- Service detection
- Version detection
- Banner grabbing

Step 6: Testing Default/Weak Credentials

The login functionality was tested using administrator credentials to check for default password vulnerability

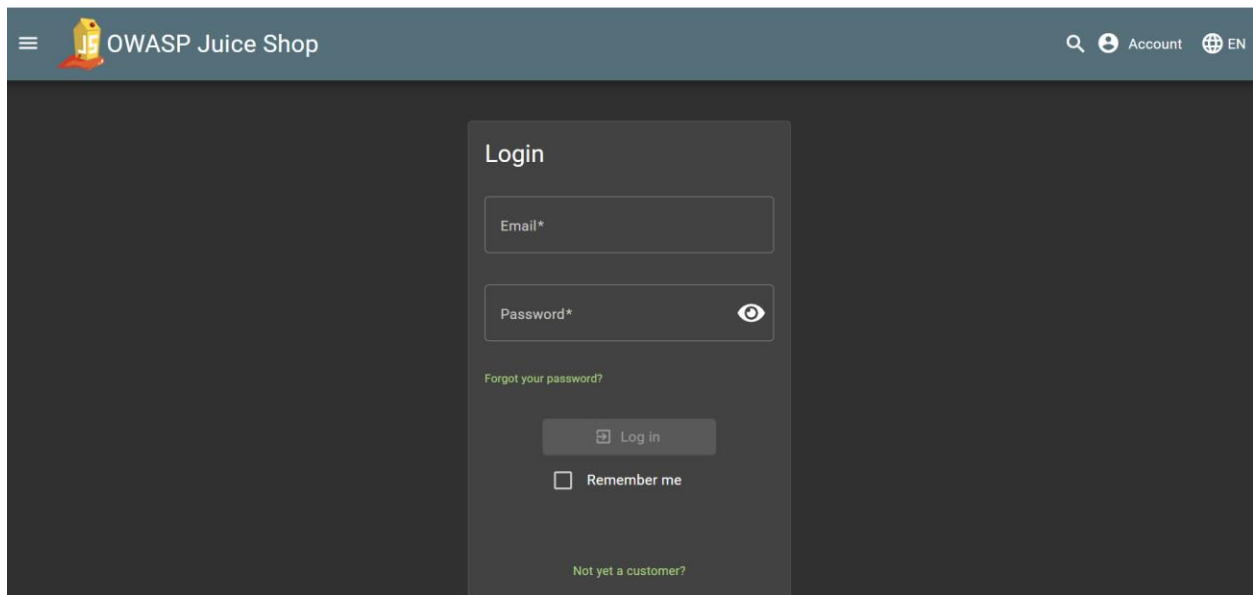


Figure 6: IoT device login interface

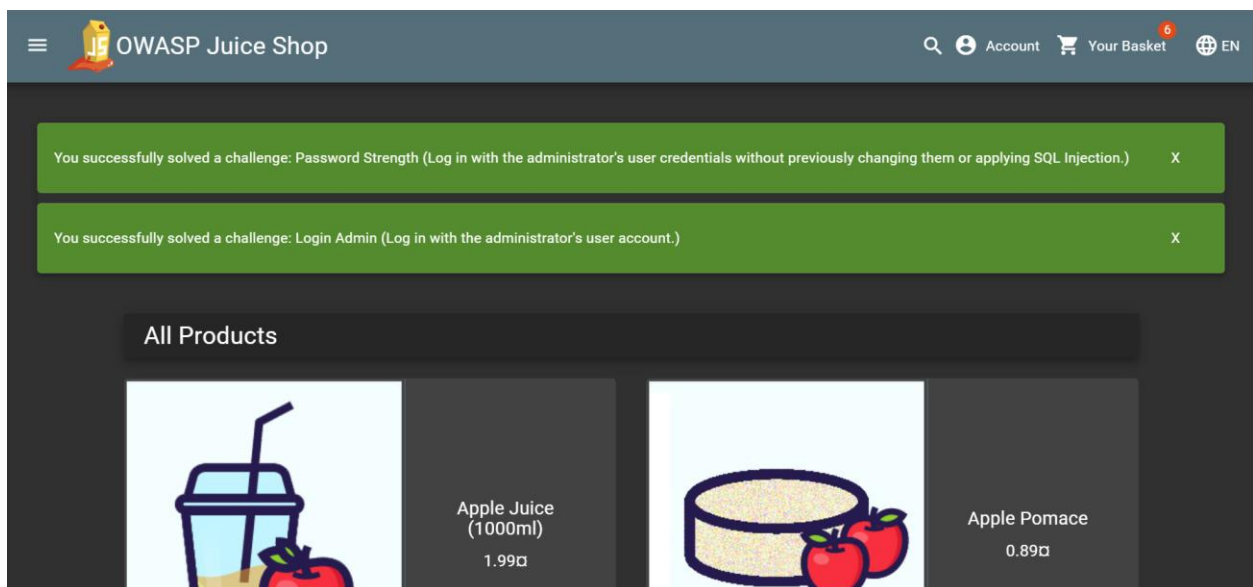


Figure 7: Successful login using default/weak credentials

Successful login using administrator credentials demonstrates the risk of default or weak passwords in IoT devices. If default credentials are not changed after deployment, attackers can gain unauthorized administrative access.

Step 7: Developer Tools → Network Tab

- Network tab showing HTTP requests
- Plain text traffic visible

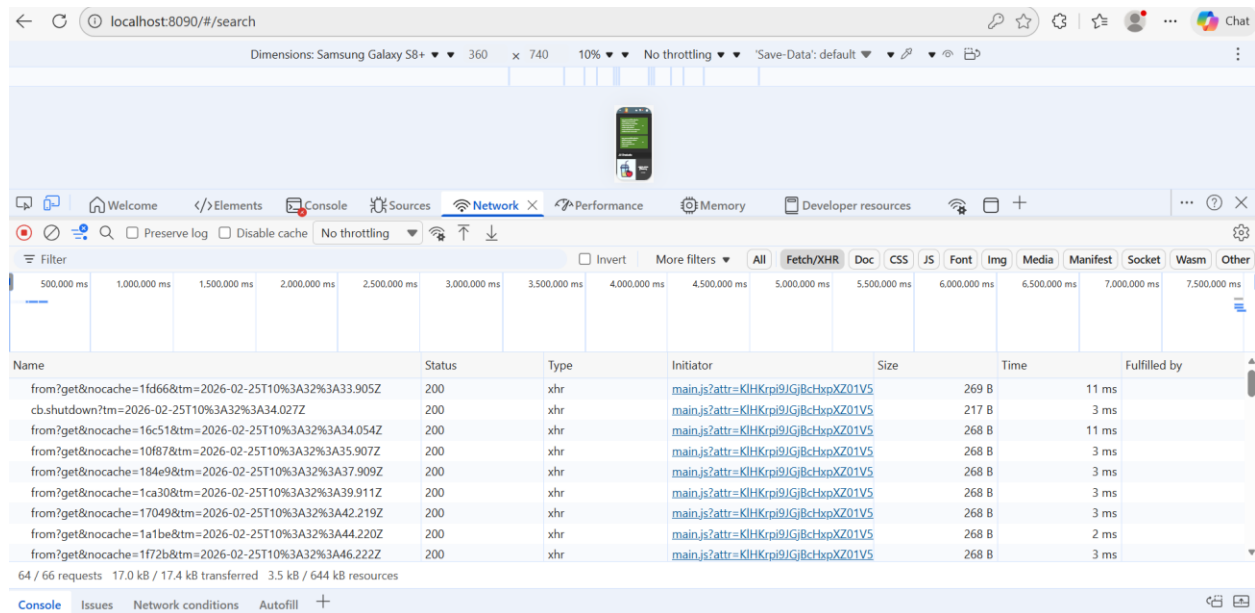


Figure 8: Unencrypted HTTP communication analysis

Observations

The scan revealed:

- Port 8090 open (HTTP service running).
- Service fingerprint identified as Juice Shop.
- Windows system ports (135, 139, 445) visible.
- Service version information exposed.
- MAC address detected.
- Web communication running over HTTP (not HTTPS).

Using Browser Developer Tools:

- Network requests visible in plain text.
- No SSL/TLS encryption detected.

Vulnerabilities Identified

1. Open Port Exposure

- Port 8090 accessible over network.
- Increases attack surface.

2. Service Enumeration

- Service version and fingerprint visible.
- Enables targeted exploit attempts.

3. Unencrypted Communication

- Application runs over HTTP.
- Data transmitted in plain text.

4. Weak Authentication Risk

- Login interface exposed.
- No strict account lockout mechanism.

5. Information Disclosure

- Banner reveals application identity.
- Server headers visible.

Conclusion

The experiment demonstrated how an attacker on the same network can identify open ports, detect running services, and test default credentials on an IoT device. The presence of exposed services and unencrypted communication highlights common IoT security risks. Proper configuration, strong authentication, and secure communication protocols are essential to protect IoT devices from exploitation.